Compiler Design, Computer Network & Operating System

Workbook

Computer Science Engineering
Information Technology

GATE

unacademy

Compiler Design, Computer Network & Operating System

Workbook

Computer Science Engineering

Information Technology

Printing of books passes through many stages - writing, composing, proof reading, printing etc. We try our level best to make the book error- free. If any mistake has inadvertently crept in, we regret it and would be deeply indebted to those who point it out. We do not take any legal responsibility.

Edition : CD.CN.OS-2024/2025

Sorting HAT Technologies Pvt. Ltd.

353, 12th 'A' cross, off double road, Opp. City Centre

Residency, Indira nagar, 1st stage Bangalore

Bangalore KA 560038 IN.

GATE Syllabus

Compiler Design : Lexical analysis, parsing, syntax-directed translation. Runtime environments. Intermediate code generation. Local optimisation, Data flow analyses: constant propagation, liveness analysis, common sub expression elimination.

Computer Network : Concept of layering: OSI and TCP/IP Protocol Stacks; Basics of packet, circuit and virtual circuit-switching; Data link layer: framing, error detection, Medium Access Control, Ethernet bridging; Routing protocols: shortest path, flooding, distance vector and link state routing; Fragmentation and IP addressing, IPv4, CIDR notation, Basics of IP support protocols (ARP, DHCP, ICMP), Network Address Translation (NAT); Transport layer: flow control and congestion control, UDP, TCP, sockets; Application layer protocols: DNS, SMTP, HTTP, FTP, Email.

Operating System : System calls, processes, threads, inter-process communication, concurrency and synchronization. Deadlock. CPU and I/O scheduling. Memory management and virtual memory. File systems.

Contents

Compiler Design :

Computer Network :

Operating System :

Compiler Design

1 Lexical & Syntax Analysis

Classroom Questions

Q.1 In a compiler the module that checks every character of the source text is called:

(A) The code generator

(B) The code optimizer

(C) The lexical analyzer

(D) The syntax analyzer

[GATE 1989 : IIT Kanpur]

Q.2 Match the following:

List - I

(a) Lexical Analysis

(b) Code Optimization

(c) Code Generation

(d) Abelian Group

List - II

(p) DAG's

(q) Syntax trees

(r) Push Down automata

(s) Finite automata

(A) a-s, b-p, c-q, d-r

(B) a-r, b-p, c-q, d-s

(C) a-s, b-q, c-p, d-r

(D) a-s, b-p, c-r, d-q

[GATE 1991 : IIT Madras]

Q.3 In some programming languages, an identifier is permitted to be a letter followed by any number of letters or digits. If L and D denotes the set of letters and digits respectively, which of the following expressions defines an identifier?

(A) $(L \cup D)^{+}$ (B) $L(L \cup D)^{*}$

(C) $(L.D)^{*}$ (D) $L(L.D)^{*}$

[GATE 1995 : IIT Kanpur]

Q.4 Type checking is normally done during

(A) Lexical analysis

(B) Syntax analysis

(C) Syntax directed translation

(D) Code optimization

[GATE 1998 : IIT Delhi]

Q.5 The number of tokens in the FORTRAN statement **DO 10 I = 1.25** is

(A) 3

(B) 4

(C) 5

(D) None of the above

[GATE 1999 : IIT Bombay]

Q.6 The number of tokens in the following C statement is

printf ("$i = \%d, \&i = \%x$", $i, \&i$);

(A) 33 (B) 26

(C) 10 (D) 21

[GATE 2000 : IIT Kharagpur]

Q.7 Consider line number 3 of the following C - program

```
int main ( ) {              /*Line 1 * /
int I, N;                   /*Line 2 * /
for (I = 0; I < N; I++);    /*Line 3 * /
}
```

Identify the compiler's response about this line while creating the object module **[GATE 2005 : IIT Bombay]**

(A) No compilation error

(B) Only a lexical error

(C) Only syntactic errors

(D) Both lexical and syntactic errors.

Q.8 Match all items in Group 1 with correct option from those given in Group 2

Group 1

P. Regular expression

Q. Pushdown automata

R. Dataflow analysis

S. Register allocation

Group 2

1. Syntax analysis
2. Code generation
3. Lexical analysis
4. Code optimization

Codes :

(A) $P-4, Q-1, R-2, S-3$

(B) $P-3, Q-1, R-4, S-2$

(C) $P-3, Q-4, R-1, S-2$

(D) $P-2, Q-1, R-4, S-3$

[GATE 2009 : IIT Roorkee]

Q.9 In a compiler, keywords of a language are recognized during
(A) parsing of the program
(B) the code generation
(C) the lexical analysis of the program
(D) dataflow analysis
[GATE 2011 : IIT Madras]

Q.10 Match the following :

List-I
(P) Lexical analysis
(Q) Top down parsing
(R) Semantic analysis
(S) Runtime environment

List-II
(i) Leftmost derivation
(ii) Type checking
(iii) Regular expressions
(iv) Activation records

(A) $P \leftrightarrow i, Q \leftrightarrow ii, R \leftrightarrow iv, S \leftrightarrow iii$
(B) $P \leftrightarrow iii, Q \leftrightarrow i, R \leftrightarrow ii, S \leftrightarrow iv$
(C) $P \leftrightarrow ii, Q \leftrightarrow iii, R \leftrightarrow i, S \leftrightarrow iv$
(D) $P \leftrightarrow iv, Q \leftrightarrow i, R \leftrightarrow ii, S \leftrightarrow iii$
[GATE 2016 : IISc Bangalore]

Q.11 Match the following according to input (from the left column) to the compiler phase (in the right column) that process is :

List-I
(P) Syntax tree
(Q) Character stream
(R) Intermediate representation
(S) Token stream

List-II
(i) Code generator
(ii) Syntax analyzer
(iii) Semantic analyzer
(iv) Lexical analyzer

(A) $P \rightarrow (ii), Q \rightarrow (iii), R \rightarrow (iv), S \rightarrow (i)$
(B) $P \rightarrow (ii), Q \rightarrow (i), R \rightarrow (iii), S \rightarrow (iv)$
(C) $P \rightarrow (iii), Q \rightarrow (iv), R \rightarrow (i), S \rightarrow (ii)$
(D) $P \rightarrow (i), Q \rightarrow (iv), R \rightarrow (ii), S \rightarrow (iii)$
[GATE 2017 : IIT Roorkee]

Q.12 A lexical analyzer uses the following patterns to recognize three tokens T1, T2, and T3 over the alphabet {a,b,c}.
T1: a?(b|c)*a
T2: b?(a|c)*b
T3: c?(b|a)*c
Note that ‘x?’ means 0 or 1 occurrence of the symbol x. Note also that the analyzer outputs the token that matches the longest possible prefix.
If the string bbaacabc is processes by the analyzer, which one of the following is the sequence of tokens it outputs?
(A) T1T2T3 (B) T1T1T3
(C) T2T1T3 (D) T3T3
[GATE 2018 : IIT Guwahati]

Q.13 Which one of the following statements is FALSE?
(A) Context-free grammar can be used to specify both lexical and syntax rules.
(B) Type checking is done before parsing.
(C) High-level language programs can be translated to different Intermediate Representations.
(D) Arguments to a function can be passed using the program stack.
[GATE 2018 : IIT Guwahati]

Q.14 The grammar $A \rightarrow AA|(A)|\varepsilon$ is not suitable for predictive parsing because the grammar is
(A) Ambiguous
(B) Left recursive
(C) Right recursive
(D) An operator grammar

Common Data for Questions 15 & 16

Consider the context-free grammar
$E \rightarrow E + E$
$E \rightarrow (E * E)$
$E \rightarrow id$
where E is the starting symbol, the set of terminals is {id, (,+,),*}, and the set of non-terminals is {E}.

Q.15 Which of the following terminal strings has more than one parse tree when parsed according to the above grammar?
(A) id + id + id + id
(B) id + (id* (id * id))
(C) (id* (id * id)) + id
(D) ((id * id + id) * id)
[GATE 2005 : IIT Bombay]

Q.16 For the terminal string with more than one parse tree obtained as solution to in above question, how many parse trees are possible?

(A) 5 (B) 4

(C) 3 (D) 2

[GATE 2005 : IIT Bombay]

Q.17 Which one of the following grammars generates the languages

$L=(a^i b^j | i \neq j)$?

(A) $S \rightarrow AC|CB$

$C \rightarrow aCb|a|b$

$A \rightarrow aA|\varepsilon$

$B \rightarrow Bb|\varepsilon$

(B) $S \rightarrow aS|Sb|a|b$

(C) $S \rightarrow AC|CB$

$C \rightarrow aCb|\varepsilon$

$A \rightarrow aA|\varepsilon$

$B \rightarrow Bb|\varepsilon$

(D) $S \rightarrow AC|CB$

$C \rightarrow aCb|\varepsilon$

$A \rightarrow aA|a$

$B \rightarrow Bb|b$

[GATE 2006 : IIT Kharagpur]

Q.18 In the correct grammar above, what is the length of the derivation (number of steps starting from S) to generates the strings $a^l b^m$ with $l \neq m$

(A) $\max(l,m)+2$ (B) $l+m+2$

(C) $l+m+3$ (D) $\max((l,m)+3$

[GATE 2006 : IIT Kharagpur]

Common Data for Questions 19 & 20

Consider the CFG with {S, A, B} as the non-terminal alphabet, {a, b} as the terminal alphabet, S as the start symbol and the following set of production rules.

$S \rightarrow bA$	$S \rightarrow aB$
$A \rightarrow a$	$B \rightarrow b$
$A \rightarrow aS$	$B \rightarrow bS$
$A \rightarrow bAA$	$B \rightarrow aBB$

Q.19 Which of the following strings is generated by the grammar?

(A) aaaabb (B) aabbbb

(C) aabbab (D) abbbba

[GATE 2007 : IIT Kanpur]

Q.20 How many derivation trees are there?

(A)1 (B)2

(C)3 (D) 4

[GATE 2007 : IIT Kanpur]

Q.21 What is the maximum number of reduce moves that can be taken by a bottom up parser for a grammar with no epsilon and unit production (i.e. of type $A \rightarrow \varepsilon$ and $A \rightarrow a$) to parse a string with n tokens?

(A) $n/2$ (B) $n-1$

(C) $2n-1$ (D) 2^n

[GATE 2013 : IIT Bombay]

Q.22 Consider the grammar defined by the following production rules, with two operators* and +

$S \rightarrow T*P$

$T \rightarrow U|T*U$

$P \rightarrow Q+P|Q$

$Q \rightarrow id$

$U \rightarrow id$

Which one of the following is TRUE?

(A) + is left associative, while * is right associative

(B) + is right associative, while * is left associative

(C) Both + and * are right associative

(D) Both + and * are left associative

[GATE 2014 : IIT Kharagpur]

Common Data for Questions 23 & 24

For the grammar below, a partial LL(1) parsing table is also presented along with the grammar. Entries that need to be filled are indicated as E1, E2 and E3. ε is the empty string, $ indicates end if input, and, | separates alternate right hand sides of productions.

$S \rightarrow aAbB|bAaB|\varepsilon$

$A \rightarrow S$

$B \rightarrow S$

	a	b	$
S	E1	E2	$S \to \varepsilon$
A	$A \to S$	$A \to S$	Error
B	$B \to S$	$B \to S$	E3

Q.23 The FIRST and FOLLOW sets for the non-terminals A and B are

(A) FIRST $(A) = \{a, b, \varepsilon\} = \text{FIRST(B)}$

FOLLOW(A) $= \{a, b\}$

FOLLOW(B) = $\{a, b, \$\}$

(B) FIRST (A) $= \{a, b, \$\}$

FIRST (B) $= \{a, b, \varepsilon\}$

FOLLOW(A) $= \{a, b\}$

$\text{FOLLOW(B)} = \{\$\}$

(C) FIRST (A) = $\{a, b, \varepsilon\} = \text{FIRST(B)}$

$\text{FOLLOW(A)} = \{a, b\}$

$\text{FOLLOW(B)} = \phi$

(D) FIRST (A) = $\{a, b\} = \text{FIRST(B)}$

FOLLOW(A) = {a, b}

FOLLOW(B) = {a, b}

[GATE 2012 : IIT Delhi]

Q.24 The appropriate entries for E1, E2 and E3 are

(A) $E1: S \to aAbB, A \to S$

$E2: S \to bAaB, B \to S$

$E3: B \to S$

(B) $E1: S \to aAbB, S \to \varepsilon$

$E2: S \to bAaB, S \to \varepsilon$

$E3: S \to \varepsilon$

(C) $E1: S \to aAbB, S \to \varepsilon$

$E2: S \to bAaB, S \to \varepsilon$

$E3: B \to S$

(D) $E1: A \to S, S \to \varepsilon$

$E2: B \to S, S \to \varepsilon$

[GATE 2012 : IIT Delhi]

Q.25 Consider the following grammar :

$P \to xQRS$

$Q \to yz | z$

$R \to w | \varepsilon$

$S \to y$

What is FOLLOW(Q)?

(A){R} (B){w} (C){w, y}
(D) $\{w, \notin\}$

[GATE 2017 : IIT Roorkee]

Q.26 Consider the grammar given below:

$S \to Aa$

$A \to BD$

$B \to b | \varepsilon$

$D \to d | \varepsilon$

Let a,b,d, and $ be indexed as follows:

a	b	d	$
3	2	1	0

Compute the FOLLOW set of the non-terminal B and write the index values for the symbols in the FOLLOW set in the descending order. (For example, if the FOLLOW set is {a, b, d,$}, then the answer should be 3210)

[GATE 2019 : IIT Madras]

Q.27 Consider the SLR (1) and LALR (1) parsing tables for a context free grammar. Which of the following statements is/ are true?

(A) The goto part of both tables may be different

(B) The shift entries are identical in both the tables

(C) The reduce entries in the tables may be different

(D) The error entries in the table may be different.

[GATE 1992 : IIT Delhi]

Q.28 Which of the following statements is true?

(A) SLR parser is more powerful than LALR.

(B) LALR parser is more powerful than Canonical LR parser.

(C) Canonical LR parser is more powerful than LALR parser.

(D) The parsers SLR, Canonical LR, and LALR have the same power.

[GATE 1998 : IIT Delhi]

Q.29 Which of the following statements is false?

(A) An unambiguous grammar has the same leftmost and rightmost derivation

(B) An LL(1) parser is a top-down parser

(C) LALR is more powerful than SLR (Simple LR)

(D) An ambiguous grammar can never be LR(k) for any k.

[GATE 2001 : IIT Kanpur]

Q.30 Which of the following suffices to convert an arbitrary CFG to an LL (1) grammar?

(A) Removing left recursion alone

(B) Factoring the grammar alone

(C) Removing left recursion and factoring the grammar

(D) None of the above

[GATE 2003 : IIT Madras]

Q.31 Assume that the SLR parser for a grammar G has n_1 states and the LALR parser for G has n_2 states. The relationship between n_1 and n_2 is

(A) n_1 is necessarily less than n_2

(B) n_1 is necessarily equal to n_2

(C) n_1 is necessarily greater than n_2

(D) None of the above

[GATE 2003 : IIT Madras]

Q.32 Consider the grammar shown below

$$S \to i\,E\,t\,S\,S' | a$$

$$S' \to e\,S | \varepsilon$$

$$E \to b$$

In the predictive parse table M, of this grammar, the entries $M[S',e]$ and $M[S',\$]$ respectively are

(A) $\{S' \to eS\}$ and $\{S' \to \varepsilon\}$

(B) $\{S' \to eS\}$ and $\{\ \}$

(C) $\{S' \to \varepsilon\}$ and $\{S' \to \varepsilon\}$

(D) $\{S' \to eS, S' \to \varepsilon\}$ and $\{S' \to \varepsilon\}$

[GATE 2003 : IIT Madras]

Q.33 Consider the grammar shown below :

$$S \to CC$$

$$C \to cC | d$$

The grammar is

(A) LL (1)

(B) SLR (1) but not LL (1)

(C) LALR (1) but not SLR (1)

(D) LR (1) but not LALR (1)

[GATE 2003 : IIT Madras]

Q.34 Which of the following grammar rules violate the requirement of an operator grammar? P, Q, R are non-terminals and r, s, t are terminals.

(i) $P \to QR$ (ii) $P \to QsR$

(iii) $P \to \varepsilon$ (iv) $P \to QtRr$

(A) (i) only

(B) (i) and (iii) only

(C) (ii) and (iii) only

(D) (iii) and (iv) only

[GATE 2004 : IIT Delhi]

Q.35 Consider the grammar

$$S \to (S) | a$$

Let the number of states in SLR (1), LR (1) and LALR (1) parsers for the grammar be n_1, n_2 and n_3 respectively. The following relationship holds good

(A) $n_1 < n_2 < n_3$ (B) $n_1 = n_3 < n_2$

(C) $n_1 = n_2 = n_3$ (D) $n_1 \geq n_3 \geq n_2$

[GATE 2005 : IIT Bombay]

Q.36 Consider the following grammar.

$$S \to S * E$$

$$S \to E$$

$$E \to F + E$$

$$E \to F$$

$$F \to id$$

Consider the following LR(0) items corresponding to the grammar above.

(i) $S \to S * \cdot E$

(ii) $E \to F. + E$

(iii) $E \to F. + E$

Given the items above, which two of them will appear in the same set in the canonical sets of items for the grammar?

(A) (i) and (ii)

(B) (ii) and (iii)

(C) (i) and (iii)

(D) None of these

[GATE 2006 : IIT Kharagpur]

Q.37 Consider the following grammar

$S \to FR$

$R \to *S | \varepsilon$

$F \to id$

In the predictive parser table, M, of the grammar the entries M[S, id] and M [R, $] respectively.

(A) $\{S \to FR\}$ and $\{R \to \varepsilon\}$

(B) $\{S \to FR\}$ and $\{\ \}$

(C) $\{S \to FR\}$ and $\{R \to *S\}$

(D) $\{F \to id\}$ and $\{R \to \varepsilon\}$

[GATE 2006 : IIT Kharagpur]

Q.38 Which one of the following is a top down parser?

(A) Recursive descent parser

(B) Operator precedence parser

(C) An LR (k) parser

(D) An LALR (k) parser

[GATE 2007 : IIT Kanpur]

Q.39 Consider the following two statements :

P : Every regular grammar is LL (1)

Q : Every regular set has LR(1) grammar.

Which of the following is TRUE?

(A) Both P and Q are true

(B) P is true and Q is false

(C) P is false and Q is true

(D) Both P and Q are false

[GATE 2007 : IIT Kanpur]

Q.40 Consider the grammar with non-terminals $N = \{S, C, S_1\}$, terminals $T = \{a, b, i, t, e\}$, with S as the start symbol, and the following set of rules

$S \to iCtS\,S_1 | a$

$S_1 \to eS | \varepsilon$

$C \to b$

The grammar is not LL(1) because :

(A) It is left recursive

(B) It is right recursive

(C) It is ambiguous

(D) It is not context free

[GATE 2007 : IIT Kanpur]

Q.41 Which of the following describes a handle (as applicable to LR parsing) appropriately?

(A) It is the position in a sentential form where the next shift or reduce operation will occur

(B) It is a non-terminal whose production will be used for reduction in the next step.

(C) It is production that may be used for reduction in a future step along with a position in the sentential form where the next shift or reduce operation will occur.

(D) It is the production p that will be used for reduction in the sentential form where the right hand side of the production may be found.

[GATE 2008 : IISc Bangalore]

Q.42 An LALR(1) parser for a grammar G can have shift reduce (S-R) conflicts if and only if

(A)The SLR (1) parser for G has S-R conflicts

(B)The LR (1) parser for G has S-R conflicts

(C)The LR (0) parser for G has S-R conflicts

(D)The LALR (1) parser for G has reduce-reduce conflicts

[GATE 2008 : IISc Bangalore]

Q.43 The grammar $S \to aSa | bS | c$ is

(A)LL(1) but not LR (1) (B)LR (1) but not LL(1)

(C)Both LL (1) and LR (1)

(D)Neither LL(1) nor LR(1)

[GATE 2010 : IIT Guwahati]

Q.44 Consider the following two sets of LR(1) items of an LR (1) grammar

$X \to c.X, c/d$ $\quad X \to c.X, \$$

$X \to .cX, c/d$ $\quad X \to .cX, \$$

$X \to .d, c/d$ $\quad X \to .d, \$$

Which of the following statements related to merging of the two sets in the corresponding LALR parser is/are FALSE?

1. Cannot be merged since look aheads are different.
2. Can be merged but will result in S-R conflict.
3. Can be merged but will result in R-R conflict.
4. Cannot be merged since goto on c will lead to two different sets.

(A) 1 only (B) 2 only
(C) 1 and 4 only (D) 1, 2, 3 and 4

[GATE 2013 : IIT Bombay]

Q.45 A canonical set of items is given below

$S \to L. > R$

$Q \to R.$

On input symbol <the set has

(A) A shift reduce conflict and a reduce-reduce conflict.
(B) A shift-reduce conflict but not a reduce-reduce conflict.
(C) A reduce –reduce conflict but not a shift-reduce conflict.
(D) Neither a shift-reduce nor a reduce conflict.

[GATE 2014 : IIT Kharagpur]

Q.46 Which one of the following is TRUE at any valid state in shift-reduce parsing?

(A) Viable prefixes appear only at the bottom of the stack and not inside
(B) Viable prefixes appear only at the top of the stack and not inside
(C) The stack contains only a set of viable prefixes
(D) The stack never contains viable prefixes

[GATE 2015 : IIT Kanpur]

Q.47 Match the following :

List-I

P. Lexical analysis
Q. Parsing
R. Register allocation
S. Expression evaluation

List-II

1. Graph coloring
2. DFA minimization
3. Post-order traversal
4. Production tree

Codes :

(A) $P-2, Q-3, R-1, S-4$
(B) $P-2, Q-1, R-4, S-3$
(C) $P-2, Q-4, R-1, S-3$
(D) $P-2, Q-3, R-4, S-1$

[GATE 2015 : IIT Kanpur]

Q.48 Among simple LR (SLR), canonical LR, and look – ahead LR (LALR), which of the following pairs identify the method that is very easy to implement and the method that is the most powerful, in that order

(A) SLR, LALR
(B) Canonical LR, LALR
(C) SLR, canonical LR
(D) LALR, canonical LR

[GATE 2015 : IIT Kanpur]

Q.49 Consider the following grammar G

$S \to F|H$

$F \to p|c$

$H \to d|c$

Where S, F, and H are non-terminal symbol, p, d, and c are terminal symbol. Which of the following statements (s) is/are correct?

S1 : LL(1) can parse all string that are generated using grammar G

S2 : LR(1) can parse all strings that are generated using grammar G

(A) Only S1
(B) Only S2
(C) Both S1 and S2
(D) Neither S1 nor S2

[GATE 2015 : IIT Kanpur]

Q.50 Which of the following statements about parser is/are CORRECT?

(i) Canonical LR is more powerful than SLR.
(ii) SLR is more powerful than LALR.
(iii) SLR is more powerful than Canonical LR.

(A) (i) only
(B) (ii) only
(C) (iii) only
(D) (ii) and (iii) only

[GATE 2017 : IIT Roorkee]

Q.51 Consider the augmented grammar given below:

S′ → S
S → ⟨L⟩ | id
L → L,S | S

Let I0 = CLOSURE ({[S′ → S]}). The number of items in the set GOTO (I0, <) is ______ **[GATE 2017 : IIT Roorkee]**

Q.52 Which one of following kinds of derivation is used by LR parsers?

(A) Leftmost
(B) Leftmost in reverse
(C) Rightmost
(D) Rightmost in reverse

[GATE 2019 : IIT Madras]

Q.53 Consider the following grammar.

S → aSB | d

B → b

The number of reduction steps taken by a bottom-up parser while accepting the string aaadbbb is ________

[GATE 2020 : IIT Delhi]

Practice Questions

Q.1 The number of tokens in the following C code is:

Print f ("i= %d, j= % d, & i= %x, i, j & i);

(A) 3 (B) 28
(C) 12 (D) 23

Q.2 A non left recursive and left factored grammar in which all non- empty rules defining the same non terminal have disjoint first sets, such grammar is called is___

(A) LL (1) (B) LR (0)
(C) SLR (1) (D) None of these

Q.3 Consider the grammar

$Z \to Z \uparrow Y$

$Y \to T^{\wedge}Y / id$

Which of the following is false

(A) $\uparrow$ is left associative
(B) $^{\wedge}$ is right associative
(C) $^{\wedge}$ has higher precedence than $\uparrow$
(D) Both $\uparrow$ and $^{\wedge}$ are left associative

Q.4 The # define..... Direction in C is handled on a C compiler

(A) By the lexical analysis
(B) By the syntax analysis
(C) By the semantic analysis
(D) By the code optimizer

Q.5 The grammar $S \to T \mid \varepsilon \mid a,\ T \to S \mid a$

(A) is ambiguous and hence not SLR (1)
(B) is unambiguous & LR (0)
(C) is not LALR (1) but is SLR (1)
(D) is Unambiguous and hence not SLR (0)

Q.6 Which one of the following is TRUE at any valid state in shift reduce parsing?

(A) Viable prefixes appear only at the bottom of the stack and not inside
(B) Viable prefixes appear only at the top of the stack and not inside
(C) The stack contains only a set of viable prefixes.
(D) The stack never contains viable prefixes.

Q.7 Compute the FOLLOW set of S for the following CFG:

$S \to SPQR$

$P \to pPt \mid \varepsilon$

$Q \to qQ \mid \varepsilon$

$R \to Rr \mid Qm \mid \varepsilon$

(A) $\{p, q, r, m, \$\}$ (B) $\{p, q, r, t, \$\}$
(C) $\{q, r, m, \$\}$ (D) $\{p, q, r, m, t, \$\}$

Q.8 Consider the following CFG:

$E \to A$

$A \to BC \mid DBC$

$B \to Bb \mid \in$

$C \to c \mid \in$

$D \to a \mid d$

What is the FIRST and FOLLOW set of nonterminal B?

(A) FIRST $= \{b, \in\}$, FOLLOW $= \{b, c\}$
(B) FIRST $= \{b\}$, FOLLOW $= \{b, c, \$\}$
(C) FIRST $= \{b\}$, FOLLOW $= \{b, \$\}$
(D) FIRST $= \{b, \in\}$, FOLLOW $= \{b, c, \$\}$

Q.9 Consider the following grammar:

$E \to E * F \mid F$

$F \to E + G \mid G$

$G \to id \mid \in$

In the above grammar:

(A) * is left associative, + is right associative

(B) + is left associative, + is right associative

(C) Both are left associative

(D) Both are right associative

Q.10 Consider the following statements about parsers:

S1: Top- down parser cant parse left recursive grammar.

S2: Keywords of a language are recognized during parsing of the program.

Choose the suitable option about above statements.

(A) S_1 is false, S_2 is true

(B) S_1 is true, S_2 is false

(C) Both statements are true

(D) Both statement are false

Q.11 Choose the False statement.

(A) No left recursive / ambiguous grammar can be LL (1)

(B) The class of grammars that can be parsed using LR methods is proper subset of the class of grammar that can be parsed by LL method

(C) LR parsing is non- backtracking method

(D) LR parsing can describe more languages than LL parsing

Q.12 Which of the following is True?

(A) Handle of a string is a sub string that matches left hand side of production

(B) RR conflicts never occur in LALR (1)

(C) SR conflicts occur in LALR (1)

(D) None of these

Q.13 The grammar

$S \to FA$

$S \to \varepsilon \mid \phi TA$

$F \to i$

Reflects that

(A) ϕ is left associative.

(B) ϕ is right associative

(C) Cannot deduce associative from the grammar

(D) None of the above.

Q.14 In the grammar $E \to E = E \mid E \mid E \mid i$ when we construct a SLR (1) machine

(A) No inadequate states parse

(B) Inadequate states can be resolved using the associativity and precedence of $= \& +$.

(C) Inadequate states cannot be resolved

(D) None of the above

Q.15 Consider the following statements:

(A) LL (k) grammars have one to one correspondence with DCFLs.

(B) LE (k) grammars have one to one correspondence with CFLs.

(A) A is true but B is False.

(B) A is false but B is True.

(C) Both are False

(D) Both are True.

Q.16 Consider grammar with start symbol ‘E”

$E \to TE'$

$E' \to +TE' \mid \in$

$T \to FT'$

$T' \to *FT' \mid \in$

$F \to (E) \mid id$

Which of the following statement is correct?

(i) ‘+’has more precedence than ‘*’

(ii) ‘*’ has more precedence than ‘+’

(iii) Both have same precedence

(iv) ‘ id’ has less precedence than ‘(‘

(A) i (B) ii

(C) iii (D) iv

Q.17 Construct the LALR (1) set of items for the grammar:

$S' \to S$

$S \to +SS \mid a$

Then, identify, in the list below, one of the LALR (1) sets of items for

(A) $[S \to a., \$]$

(B) $[S \to a., +a]$

(C) $[S \to +SS., \$ + a]$

(D) $[S \to +SS., \$]$

Q.18 Consider the following CFG, with S as start symbol:

$S \to aA \mid CB$

$A \to BaA \mid \in$

$B \to bB \mid Abc \mid \in$

$C \to B$

Which of the following is correct/

First $(S) = \{a, b, \in\}$ $(S) = \{a, \in\}$

(A) First $(B) = \{b, \in\}$

Follow $(C) = \{b, a, \$\}$

First $(S) = \{a, b, \$\}$

(B) First $(B) = \{b, \in\}$

Follow $(C) = \{a, b, \in\}$

(C) First $(B) = \{b, \in\}$

Follow $(C) = \{b, a, \$\}$

(D) First $(B) = \{a, \in\}$

Follow $(C) = \{b, a, \$\}$

Q.19 Consider an SLR (1) and LALR (1) tables, which of the following is true?

(A) Shift entries are different

(B) Reduce entries are same

(C) Goto entries are different

(D) Error entries are different

Q.20 Among simple LR (SLR), canonical LR, and Look- ahead LR (LALR), which of the following pairs identify the method that is very easy to implement and the method that is the most powerful, in that order?

(A) SLR, LALR

(B) Canonical LR, LALR

(C) SLR, canonical LR

(D) LALR, canonical LR

Q.21 $S \to aSAb \mid bSBc \Rightarrow \text{First}(S) = \{a, b\}$

$A \to +AB \mid \varepsilon \Rightarrow \text{First}(A) = \{+, \varepsilon\}$

$B \to *BC \mid \varepsilon \Rightarrow \text{First}(B) = \{*, \varepsilon\}$

$C \to aC + d \Rightarrow \text{First}(C) = \{a, d\}$

What is in the follow (S)?

(A) $\{a, b, c, +, \$\}$ (B) $\{a, c, +, *, \$\}$

(C) $\{b, c, +, *, \$\}$ (D) $\{a, b, d, *, \$\}$

Q.22 Consider the following two grammars.

$G_1 : A \to A1 \mid 0A \mid 01$

$G_2 : A \to 0A \mid 1$

Which of the following is True regarding above grammars?

(A) G_1 is LR (k)

(B) G_2 is LR (k)

(C) Both G_1 and G_2 is LR (k)

(D) None is LR (k)

Q.23 Consider the following grammar.

$S \to aB \mid aAb$

$A \to aAb \mid a$

$B \to aB \mid \varepsilon$

How many back tracks are required to generate the string aab from the above grammar?

(A) 1(B) 2

(C) 3 (D) 4

Q.24 $S \to aA * S$

$A \to +S \mid (S \mid \varepsilon$

Set {+,(} will be in the

(A) First (A) (B) First (E)

(C) Follow (E) (D) Follow (A)

Q.25 The grammar $S \to aSa \mid bS \mid \in$ is

(A) LL (1) but not LR (1)

(B) LR (1) but not LL (1)

(C) Both LL (1) and LR (1)

(D) Neither LL (1) nor LR (1)

Q.26 Consider the SDTS below

$E \to E + T \mid T$

$T \to id$

Choose the correct statement?

(A) + is left associative

(B) + is right associative

(C) The grammar is ambiguous

(D) Associativity cannot be associated with +.

Q.27 $S \to Sa \mid b$

Which of the following is True?

(A) There will be SR conflict during parsing

(B) There will be RR conflict during parsing

(C) There will be both conflict

(D) There will be no conflict

Q.28 $P \to P\alpha Q \mid Q$

$Q \to Q\beta R \mid R$

$R \to$ num

If $2\alpha3\alpha4\beta1\alpha2\beta1$ is evaluated to 18, then which of the following is the correct value for α and β

(A) +, * (B) +, −

(C) *, − (D) −,+

Q.29 Which of the following is operator grammar?

(A) $S \to AA$
$A \to a \mid \in$

(B) $S \to SAS$
$A \to a$

(C) $S \to AB$
$A \to aA \mid + \mid a$
$B \to aB \mid + \mid b$

(D) $S \to A + B$
$A \to aA \mid a$
$B \to bB \mid b$

Q.30 Consider the following statements:

S_1: A regular grammar is always linear but not all linear grammar is regular.

S_2: In LL grammar, the usage of production rule can be predicted exactly, by looking at a limited part of input.

Which of the above statements are true?

(A) S_1 is true and S_2 is false

(B) S_2 is true and S_1 is false

(C) Both are true

(D) Both are false

Q.31 The grammar which is equivalent to

$S \to SAa \mid Sa \mid a$

$A \to Ab \mid b$

After eliminating of left recursion is

(A) $S \to aS'$
$S' \to AaS' \mid aS'$
$A \to bA'$
$A' \to bA' \mid \in$

(B) $S \to AaS' \mid aS'$
$S' \to aS' \mid \in$
$A \to bA'$
$A' \to bA' \mid \in$

(C) $S \to aS'$
$S' \to AaS' \mid aS'$
$A \to bA'$
$A' \to bA' \mid \in$

(D) $S \to aS'$
$S' \to AaS' \mid aS'$
$A \to bA'$
$A' \to \in$

Q.32 Consider the following grammar:

$E \to TE'$
$E' \to +TE' \mid \in$
$T \to FT'$
$T' \to *FT' \mid \in$
$E \to (E) \mid id$

Which of the following is correct regarding the entry in the predictive parsing table (LL (1) parsing table) M for above grammar?

(A) $M[T,(] = T \to FT', M[T',+] = T' \to \in$

(B) $M[E,(] = E' \to \in, M[E',*] - E' \to \in$

(C) $M[E,(] = E \to TE', M[E,*] \to E \to \in$

(D) $M[T,id] = T \to FT', M[T,)] = T \to FT'$

Q.33 Consider the following augmented grammar with labels a and b

$S' \to S$
$a: S \to (S)S$
$b: S \to \varepsilon$

LR (0) sets are given as following for the above grammar.

(1)	(2)	(3)	(4)	(5)	(6)
$S' \to S$	$S' \to S$	$S \to (S.)S$	$S \to (S.)S$	$S \to (S.)S$	$S \to (S)S.$
$S \to S(S)$		$S \to .(S)S$		$S \to .(S)S$	
$S \to$		$S \to .$		$S \to .$	

If the following SLR (1) table is constructed as below then find the missing entries at E_1, E_2 and E_3 respectively

	(	)	\$	S
1	S_3	r_b	r_b	2
2	.		Accept	
3	S_3	E_1	E_2	4
4		S_5		
5	S_3	r_b	r_b	E_3
6		r_a	r_a	

(A) $r_a, r_b, 5$
(B) $r_a, r_a, 6$
(C) $r_b, r_a, 5$
(D) $r_b, r_b, 6$

Q.34 Assume $x, -, +$ and $/$ are operators. Precedence and associativity given for those operators as following:

1. $\times$ has highest precedence among all operators and it is left associative
2. $-,+$ and $/$ are having equal precedence and they are right associative.

Using $\times$as Multiplication, $-$ as subtraction, $+$as Addition and $/$ is Division.

The output of the following expression: $10\times10-5+15-5\times10/5$ is _____,

Q.35 Consider the following grammar

$S \rightarrow ABA$

$A \rightarrow Bc \mid dA \mid \in$

$B \rightarrow eA$

How many entries have multiple productions in LL (1) table?

Q.36 Find the number of tokens in the following C code using lexical analyzer of compiler Main ( )

{

Int * a, b;

$B = 10;$

$A = \&b;$

Print (" % d % d", b, *a);

$B = */*$ pointer */ b;

}

Q.37 Consider the grammar given below

$S \rightarrow E\#$

$E \rightarrow .T \mid E;T$

$T \rightarrow Ta \mid \varepsilon$

Number of inadequate states in DFA with LR (0) items is

Q.38 Consider the following augmented grammar.

$S \rightarrow aAb \mid eb$

$A \rightarrow e \mid f$

The number of states in LR (0) construction are_____.

Q.39 Consider the following grammar:

$S \rightarrow Aa \mid B$

$B \rightarrow a \mid bC$

$C \rightarrow a \mid \in$

The number of productions in simplified CFG is______.

Q.40 Consider the following grammar which is not LL (1) because LL (1) table contain multiple entry for same production.

$S \rightarrow .aAbB \mid bAaB \mid \in$

$A \rightarrow S$

$B \rightarrow S$

The number of entries have multiple production in LL (1) table are_____.

Q.41 Let G to a grammar with the following productions:

$E \rightarrow E + T \mid T$

$T \rightarrow T * F \mid F$

$T \rightarrow (E) + T - F$

$F \rightarrow id$

If LR (1) parser is used to construct the DFA using the above production, then how many look a-heads are present for an item $T \rightarrow .T * F$ in the initial state______.

Q.42 Find the number of tokens in the following C code using lexical analyzer of compiler.

Main ( )

{

/* int a = 10; */

Int * u, * v, s;

u = & s;

v = & s;

printf (" %d %d", s, *u);

//code ended

}

Q.43 Consider the following program:

Main ( )

{ int x = 10 ;

It (x < 20;

Else

y = 20 ;

}

When lexical analyzer scanning the above program, how many lexical errors can be produced?

Q.44 How many DFA states are constructed for the following augmented grammar using LR (0) parser?

$S' \rightarrow \cdot S\$$

$S' \rightarrow x \mid (A)$

$A \rightarrow A, S \mid S$

Where S is the start symbol and $S' \rightarrow S\$$is augmented production.

Q.45 Let G be the following grammar:

$S \rightarrow aABbCD$

$A \rightarrow ASd \mid \in$

$B \rightarrow SAc \mid hC \mid \in$

$C \rightarrow Sf \mid Cg$

$C \rightarrow \cdot BD \mid \in$

How many number of productions will be in G after elimination of all null productions only?

{Consider $S \rightarrow a \mid b$ as 2 productions (i) $S \rightarrow a, \text{(ii)} S \rightarrow b\}$.

❑❑❑

2 Semantic Analyzer

Classroom Questions

Q.1 Which data structure in a compiler is used for maintaining information about variables and their attributes?

(A) Abstract syntax tree
(B) Symbol table
(C) Semantic stack
(D) Parse table

[GATE 2010 : IIT Guwahati]

Q.2 A shift reduce parser carries out the actions specified within braces immediately after reducing with the corresponding rule of grammar.

$S \rightarrow xxW\{\text{print"1"}\}$

$S \rightarrow y\{\text{print"2"}\}$

$W \rightarrow Sz\{\text{print"3"}\}$

What is the translation of “xxxxyzz” using the syntax directed translation scheme described by the above rules?

(A) 23131 (B)11233
(C) 11231 (D) 33211

[GATE 1995 : IIT Kanpur]

Q.3 Consider the productions A → PQ and A → XY. Each of the five non-terminals A, P,Q,X, and Y has two attributes: s is a synthesized attribute, and i is an inherited attribute. Consider the following rules.

Rule 1 : P.i=A.i+2, Q.i=P.i+A.i, and A.s=P.s+Q.s

Rule 2 : X.i=A.i+Y.s and Y.i=X.s+A.i

Which one of the following is TRUE?

(A) Both Rule 1 and Rule 2 are L-attributed
(B) Only Rule 1 is L-attributed
(C) only Rule 2 is L-attributed
(D) Neither Rule 1 nor Rule 2 is L-attributed

[GATE 2020 : IIT Delhi]

Q.4 In the following grammar

X :: = X ⊕ Y / Y
Y :: = Z * Y / Z
Z :: = id

Which of the following is true? a. ‘⊕’ is left associative while ‘*’ is right associative b. Both ‘⊕’ and ‘*’ are left associative c. ‘⊕’ is right associative while ‘*’ is left associative d. None of the above

(A) a (B) b
(C) c (D) d

[GATE 1997 : IIT Madras]

Q.5 In a bottom-up evaluation of a syntax directed definition, inherited attributes can

(A) Always be evaluated.
(B) Be evaluated only if the definition is L-attributed.
(C) Be evaluated only if the definition has synthesized attributes.
(D) Never be evaluated.

[GATE 2003 : IIT Madras]

Q.6 Consider the translation scheme shown below

S → T R
R → + T {print ('+');} R | ε
T → num {print (num.val);}

Here num is a token that represents an integer and num.val represents the corresponding integer value. For an input string ‘9 + 5 + 2’, this translation scheme will print

(A) 9 + 5 + 2 (B) 9 5 + 2 +
(C) 9 5 2 + + (D) + + 9 5 2

[GATE 2003 : IIT Madras]

Q.7 Consider the syntax directed definition shown below.

S → id : = E {gen (id.place = E.place;);}
E → E1 + E2 {t = newtemp ( ); gen (t = El.place + E2.place;); E.place = t}
E → id {E.place = id.place;}

Here, gen is a function that generates the output code, and newtemp is a function that returns the name of a new temporary variable on every call. Assume that ti’s are the temporary variable names generated by newtemp.

For the statement 'X: = Y + Z', the 3-address code sequence generated by this definition is
(A) X = Y + Z
(B) t1 = Y + Z; X = t1
(C) t1 =Y; t2 = t1 + Z; X = t2
(D) t1 = Y; t2 = Z; t3 = t1 + t2; X = t3

[GATE 2003 : IIT Madras]

Q.8 Consider the grammar with the following translation rules and E as the start symbol.

E → E1 # T { E.value = E1.value * T.value }
| T { E.value = T.value }
T → T1 & F { T.value = T1.value + F.value }
| F { T.value = F.value }
F → num { F.value = num.value }

Compute E.value for the root of the parse tree for the expression: 2 # 3 & 5 # 6 & 4.

(A) 200 (B) 180
(C) 160 (D) 40

[GATE 2004 : IIT Delhi]

Q.9 Consider the grammar $E \rightarrow E+n | E \times n | n$ for a sentence $n+n \times n$, then handles in the right sentential form of the reduction are

(A) $n, E+n \, and \, E+n \times n$
(B) $n, E+n \, and \, E+E \times n$
(C) $n, n+n \, and \, n+n \times n$
(D) $n, E+n \, and \, E \times n$

[GATE 2005 : IIT Bombay]

Q.10 Consider the following expression grammar. The semantic rules for expression calculation are stated next to each grammar production.

E → number E.val = number. val
| E '+' E E(1).val = E(2).val + E(3).val
| E '×' E E(1).val = E(2).val × E(3).val

The above grammar and the semantic rules are fed to a yacc tool (which is an LALR (1) parser generator) for parsing and evaluating arithmetic expressions. Which one of the following is true about the action of yacc for the given grammar?

(A) It detects recursion and eliminates recursion
(B) It detects reduce-reduce conflict, and resolves
(C) It detects shift-reduce conflict, and resolves the conflict in favor of a shift over a reduce action
(D) It detects shift-reduce conflict, and resolves the conflict in favor of a reduce over a shift action

[GATE 2005 : IIT Bombay]

Q.11 Consider the following expression grammar. The semantic rules for expression calculation are stated next to each grammar production.

E → number E.val = number. val
| E '+' E E(1).val = E(2).val + E(3).val
| E '×' E E(1).val = E(2).val × E(3).val

Assume the conflicts in Part (a) of this question are resolved and an LALR(1) parser is generated for parsing arithmetic expressions as per the given grammar. Consider an expression 3 × 2 + 1. What precedence and associativity properties does the generated parser realize?

(A) Equal precedence and left associativity; expression is evaluated to 7
(B) Equal precedence and right associativity; expression is evaluated to 9
(C) Precedence of '×' is higher than that of '+', and both operators are left associative; expression is evaluated to 7
(D) Precedence of '+' is higher than that of '×', and both operators are left associative; expression is evaluated to 9

[GATE 2005 : IIT Bombay]

Q.12 Consider the following translation scheme.

$$S \rightarrow ER$$
$$R \rightarrow E\{\text{print}('*');\}R | \varepsilon$$
$$E \rightarrow F+E\{\text{print}('+');\} | F$$
$$F \rightarrow (S) | id\,\{\text{print}\,(id.value);\}$$

Here id is a token that represents an integer and id.value represents the corresponding integer value. For an input '2*3+4' this translation scheme prints

(A) 2*3+4 (B) 2*+3 4

(C) 2 3*4+ (D) 2 3 4+*

[GATE 2006 : IIT Kharagpur]

Q.13 One of the purpose of using intermediate code in compilers is to

(A) Make parsing and semantic analysis simpler.

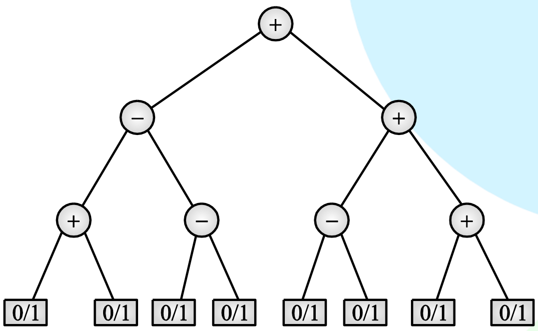
(B) Improve error recovery and error reporting.

(C) Increase the chances of reusing the machine-independent code optimizer in other compilers.

(D) Improve the register allocation.

[GATE 2014 : IIT Kharagpur]

Q.14 Consider the expression tree shown. Each leaf represents a numerical value, which can either be 0 or 1. Over all possible choices of the values at the leaves, the maximum possible value of the expression represented by the tree is ___.



[GATE 2014 : IIT Kharagpur]

Q.15 Consider the following Syntax Directed Translation Scheme (SDTS), with non-terminals {S,A} and terminals {a,b}.

$S \to aA\,\{\text{print}\,1\}$

$S \to a \qquad \{\text{print}\,2\}$

$A \to Sb\,\{\text{print}\,3\}$

Using the above SDTS, the output printed by a bottom up parser, for the input aab is :

(A) 1 3 2 (B) 2 2 3

(C) 2 3 1 (D) Syntax error

[GATE 2016 : IISc Bangalore]

Q.16 The attributes of three arithmetic operators in some programming language are given below.

Operator	Precedence	Associativity	Arity
+	High	Left	Binary
–	Medium	Right	Binary
*	Low	Left	Binary

The value of the expression 2 – 5 + 1 – 7 * 3 in this language is __________ ?

[GATE 2016 : IISc Bangalore]

Q.17 Which one of the following statements is FALSE?

(A) Context-free grammar can be used to specify both lexical and syntax rules.

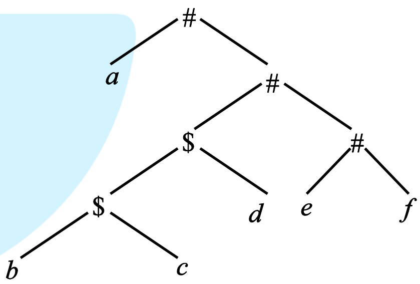
(B) Type checking is done before parsing.

(C) High-level language programs can be translated to different Intermediate Representations.

(D) Arguments to a function can be passed using the program stack.

[GATE 2018 : IIT Guwahati]

Q.18 Consider the following parse tree for the expression a # b $ c $ d # e # f, involving two binary operators $ and #.



Which one of the following is correct for the given parse tree?

(A) $ has higher precedence and is left associative; # is right associative

(B) # has higher precedence and is left associative; $ is right associative

(C) $ has higher precedence and is left associative; # is left associative

(D) # has higher precedence and is right associative; $ is left associative

[GATE 2018 : IIT Guwahati]

Q.19 Consider the following grammar and the semantic actions to support the inherited type declaration attributes. Let $X_1, X_2, X_3, X_4, X_5,$ and X_6 be the placeholders for the non-terminals D, T, L or L_1 in the following table :

Production rule	Semantic action
D→T L	$X_1 \cdot \text{type} = X_2 \cdot \text{type}$
T→int	$\text{T} \cdot \text{type} = \text{int}$
T→float	$\text{T} \cdot \text{type} = \text{float}$

$L \to L_1, id$	$X_3 \cdot type = X_4 \cdot type$ addType(id.entry, $X_5 \cdot type$)
$L \to id$	addType(id.entry, $X_6 \cdot type$)

Which one of the following are the appropriate choices for X_1, X_2, X_3 and X_4?

(A) $X_1 = L,\ X_2 = T,\ X_3 = L_1,\ X_4 = L$
(B) $X_1 = T,\ X_2 = L,\ X_3 = L_1,\ X_4 = T$
(C) $X_1 = L,\ X_2 = L,\ X_3 = L_1,\ X_4 = T$
(D) $X_1 = T,\ X_2 = L,\ X_3 = T,\ X_4 = L_1$

[GATE 2019 : IIT Madras]

Practice Questions

Q.1 Consider the following SDT.
$A \to BC*$
(I) $B.i = f(A,i)$
(II) $B.i = f(A,S)$
(III) $A.S = f(B.s)$
Which of the above is violating L-attribute definition?
(A) I only (B) II only
(C) I, II (D) I, II, III

Q.2 If attribute can be evaluated in depth-first order then definition is
(A) S-attributed
(B) L-attributed
(C) Both (A) and (B)
(D) None of these

Q.3 Let *G* be the grammar with following transition
$S \to a\{print"0"\}A$
$A \to b\{print"1"\}B$
$A \to c\{print"2"\}$
$A \to \varepsilon\{print"-"\}$
$B \to d\{print"1"\}A$
$B \to \varepsilon\{print"0"\}$
What is the output produced for the input "*abdbdc*" using the bottom up parsing with above translations?
(A) 211-10 (B) 211110
(C) 111-20 (D) 111012

Q.4 Let "*G*" be a grammar with the following translations :
$S \to p\{print"G"\}P$
$P \to q\{print"A"\}Q$
$P \to r\{print"T"\}$
$P \to \varepsilon\{print"E"\}$
$Q \to S\{print"A"\}P$
$Q \to \varepsilon\{print"O"\}$
What is the output produced for the input "*pqsqsr*" using the bottom up parsing with above translations?
(A) *TAAAG* (B) *TAAAAG*
(C) *GAAA* (D) *AAAGAT*

Q.5 Consider the syntax directed translation scheme below for the grammar :
$N \to L$
$L \to L + B \mid B$
$B \to 0 \mid 1$
The Schema is
$N \to L$ $\{N.val = L.val\}$
$N \to L + B$
$\{L.val = L.val * 2 + B.val\}$
$L \to B$ $\{L.val = B.val\}$
$B \to 0$ $\{B.val = 0\}$
$B \to 1$ $\{B.val = 1\}$
The value of *N*.val is
(A) The number of bits in the input string
(B) The value of the non decimal input in binary form.
(C) The value of the binary input in decimal form
(D) None of the above.

Q.6 $X \to YZ$
$Y \to Y + Z\{print('+');\}$
$T\{Y.val = T.val\}$
$Z \to *Y\{print('*');\}Z$
$T\{Z.val = T.val\}\ \varepsilon$
$T \to num\{print(num.val);\}$
For $2+3*2$, the above translation scheme prints
(A) $2+3*2$ (B) $23+2*$
(C) $232*2+$ (D) $23*2+$

Q.7 A shift reduce parser carries out the actions specified within braces immediately after reducing with the corresponding rule of grammar.

$S \rightarrow xx$ W {print "1"}

$S \rightarrow y$ W {print "2"}

$W \rightarrow Sz$ {print "3"}

What is the translation of *xxxxyzz* using the syntax-directed translation scheme describe by the above rules?

(A) 23131 (B) 11233

(C) 11231 (D) 33211

Q.8 Match the following errors corresponding to their phase :

Group A

1. Unbalanced parenthesis
2. Appearance of illegal characters
3. Unbalanced parenthesis

Group B

A. Syntactic error

B. Semantic error

C. Lexical error

(A) 1$\rightarrow$A, 2$\rightarrow$C, 3$\rightarrow$B

(B) 1$\rightarrow$B, 2$\rightarrow$C, 3$\rightarrow$A

(C) 1$\rightarrow$A, 2$\rightarrow$B, 3$\rightarrow$C

(D) 1$\rightarrow$B, 2$\rightarrow$C, 3$\rightarrow$A

Q.9 In a bottom-up evaluation of a syntax directed definition, inherited attribute can

(A) Always be evaluated

(B) Be evaluated only if definition in L-attribute

(C) Never be evaluated

(D) Be evaluated only if the definition has synthesized attributes

Q.10 Type checking is normally done using

(A) Lexical analysis

(B) Syntax analysis

(C) Syntax directed translation

(D) Code optimization

Q.11 Consider the translation scheme given below :

$S \rightarrow T - R\{\text{print}('-')\} \mid R$

$R \rightarrow +T\{\text{print}('+') \mid R \mid F$

$F \rightarrow \text{id}\{\text{print}(\text{id.value})\} \mid \in$

For an input scheme $10-5+4$, this scheme will print

(A) $10\ 5+4-$ (B) $10\ 54-+$

(C) $105-4+$ (D) $1054+-$

Q.12 Consider following SDT

$S \rightarrow A$ sign // $S.\text{val} = A.\text{val};$

Sign $\rightarrow +$ // sign.sign $=1$

Sign $\rightarrow -$ // sign.sign $=0$

$A \rightarrow n$ // A.val = value (n)

$A \rightarrow A1, n$ // $A1$.sign = A.sign;

if $(A\text{sign} = 1)$ then

A.val $= \min(A1.\text{val}, \text{value}(n));$

else

A.val $= \max(A1.\text{val}, \text{value}(n));$

A partial modified grammar and actions exclusively using synthesized attributes is given below :

$S \rightarrow B+$ // $S.\text{val} = B.\text{val};$ print (B.val)

$S \rightarrow C-$ // $S.\text{val} = C.\text{val};$ print (C.val)

$B \rightarrow n$ // $B.\text{val} = \text{value}(n)$

$B \rightarrow B1.n$ // $B.\text{val} = B1$

$C \rightarrow n$ // $C.\text{val} = \text{value}(n)$

$C \rightarrow C1.n$ // $C.\text{val} = B2$

What is the value of $B1$ and $B2$ to complete the grammar respectively?

(A) Max ($B1$.val, value (n)), Min ($C1$.val, value (n))

(B) Min ($B1$.val, value (n)), Max ($C1$.val, value (n))

(C) Max ($B1$.val, value (n)), Max ($C1$.val, value (n))

(D) Min ($B1$.val, value (n)), Min ($C1$.val, value (n))

Q.13 Consider the following SDT :

$E \rightarrow E + E \ \{E.\text{val} = E_1.\text{val} + E_2.\text{val}\}$

$E \rightarrow E \times E \ \{E.\text{val} = E_1.\text{val} - E_2.\text{val}\}$

$E \rightarrow (E) \ \{E.\text{val} = E_1.\text{val}\}$

$E \rightarrow \text{id} \ \{E.\text{val} = \text{id.lex}\}$

The value of the attribute computed at root when the expression $[(3\times3)+(3\times5)-6]+7$ is evaluated using the above SDT are ________.

Q.14 A shift reduce parser carries out the action specified within braces immediately after reducing with the corresponding rule of grammar :

$S \rightarrow xxW$ {print "1"}

$S \rightarrow y$ {print "2"}

$W \rightarrow Sz$ {print "3"}

What is the translation of *xxxxyzz* using the syntax directed translation scheme described by the above rules?

Q.15 Consider the following SDT

$S \rightarrow S * S_1 \mid S_2 \quad (S.\text{val} = S_1.\text{val} + S.\text{val})$

$S \rightarrow S_2 \quad (S_1.\text{val} = S_2.\text{val})$

$S_2 \rightarrow S_2 \# S_3 \quad (S_2.\text{val} = S_2.\text{val} - S_3.\text{val})$

$S_2 \rightarrow S_3 \quad (S_2.\text{val} = S_3.\text{val})$

$S_3 \rightarrow \text{id} \quad (S_3.\text{val} = \text{id})$

Evaluate the expression

$15 \# 12 * 5 \# 25 \# 30 * 60$

❑❑❑

3 Intermediate Code Generation & Optimization

Classroom Questions

Q.1 The pass number for each of the following activities
(i) Object code generation
(ii) Literals added to literal table
(iii) Listing printed
(iv) Address resolution of local symbols that occur in a two assembler respectively are
(A) 1, 2, 1, 2 (B) 2, 1, 2, 1
(C) 2, 1, 1, 2 (D) 1, 2, 2, 2
[GATE 1996 : IISc Bangalore]

Q.2 The process of assigning load addresses to the various parts of the program and adjusting the code and data in the program to reflect the assigned addresses is called
(A) Assembly
(B) Parsing
(C) Relocation
(D) Symbol resolution
[GATE 2001 : IIT Kanpur]

Q.3 Match the following :
Group – I
(A) Pointer data type
(B) Activation record
(C) Repeat-until
(D) Coercion
Group – II
(P) Type conversion
(Q) Dynamic data structure
(R) Recursion
(S) Non-deterministic loop
(A) a - p, b - r, c - s, d - q
(B) a - q, b - r, c - s, d - p
(C) a - q, b - s, c - r, d - p
(D) a - r, b - q, c - s, d – p
[GATE 1990 : IISc Bangalore]

Q.4 Generation of intermediate code based on an abstract machine model is useful in compilers because
(A) It makes implementation of lexical analysis and syntax analysis easier.
(B) Syntax directed translations can be written for intermediate code generation.
(C) It enhances the portability of the front end of the compiler.
(D) It is not possible to generate code for real machines directly from high level language programs.
[GATE 1994 : IIT Kharagpur]

Q.5 A linker is given object modules for a set of programs that were compiled separately. What information need not be included in an object module?
(A) Object code.
(B) Relocation bits.
(C) Names and locations of all external symbol defined in the object module.
(D) Absolute addresses of internal symbols.
[GATE 1995 : IIT Kanpur]

Q.6 Consider the grammar rule $E \to E_1 - E_2$ for arithmetic expressions. The code generated is targeted to a CPU having a single user register. The subtraction operation requires the first operand to be in the register. If E_1 and E_2 do not have any common sub-expression, in order to get the shortest possible code.
(A) E_1 should be evaluated first
(B) E_2 should be evaluated first
(C) Evaluation of E_1 and E_2 should necessary be interleaved
(D) Order of evaluation of E_1 and E_2 is of no consequence.
[GATE 2004 : IIT Delhi]

Q.7 Consider the following C code segment.
for $(i = 0; i < n; i++)$
{
for $(j = 0; j < n; j++)$

{

if $(i\%2)$

{

$X += (4 * j + 5 * i);$

$Y += (7 + 4 * j);$

}

}

}

Which one of the following is false?

(A) The code contains loop-invariant computation

(B) There is scope of common sub expression elimination is this code

(C) There is scope of strength reduction in this code.

(D) There is scope of dead code elimination in this code.

[GATE 2006 : IIT Kharagpur]

Q.8 In a simplified computer the instructions are:

OP $R_j R_i$ – Performs R_j OP R_i and stores the result in register R_i

OP m, R_i – Performs val OP R_i and stores the result in R_i. Val denotes the content of memory location m.

MOV m, R_i – Moves the content of memory location m to register R_i.

MOV R_i, m – Moves the content of registers, and to memory location m

The computer has only two registers, and OP is either ADD or SUB. Consider the following basic block:

t1 = a + b

t2 = c + d

t3 = e – t2

t4 = t1 – t3

Assume that all operands are initially in memory. The final value of the computation should be in memory. What is the minimum number of MOV instruction in the code generated for this basic block?

(A) 2 (B) 3

(C) 5 (D) 6

[GATE 2007 : IIT Kanpur]

Q.9 Some code optimization are carried out on the intermediate code because

(A) They enhance the portability of the complier to other target processors.

(B) Program analysis is more accurate on intermediate code than on machine code.

(C) The information from dataflow analysis cannot otherwise be used for optimization.

(D) The information from the front end cannot otherwise be used for optimization.

[GATE 2008 : IISc Bangalore]

Q.10 Which languages necessarily need heap allocation in the runtime environment?

(A) Those that support recursion.

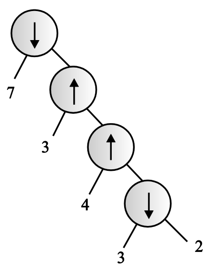
(B) Those that use dynamic scoping

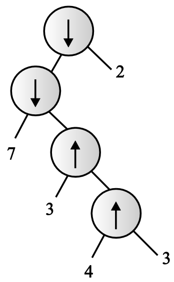
(C) Those that allow dynamic data structures

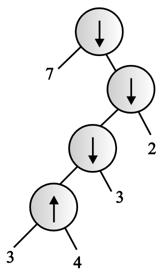
(D) Those that use global variables.

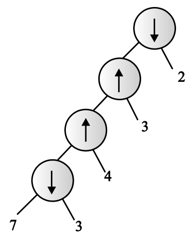
[GATE 2010 : IIT Guwahati]

Q.11 Consider two binary operators '$\uparrow$' and '$\downarrow$' with the precedence of operator $\downarrow$ being lower than that of the operator $\uparrow$. Operator $\uparrow$ is right associative while operator $\downarrow$, is left associative. Which one of the following represents the parse tree for expression $(7 \downarrow 3 \uparrow 4 \uparrow 3 \downarrow 2)$?

(A)


(B)


(C)


(D)


[GATE 2011 : IIT Madras]

Q.12 Which one of the following is FALSE?

(A) A basic block is a sequence of instructions where control enters the sequence at the beginning and exists at the end.

(B) Available expression analysis can be used for common sub-expression elimination.

(C) Live variable analysis can be used for dead code elimination.

(D) $x=4*5 \Rightarrow x=20$ is an example of common sub-expression elimination.

[GATE 2014 : IIT Kharagpur]

Q.13 Which one of the following is NOT performed during compilation?

(A) Dynamic memory allocation

(B) Type checking

(C) Symbol table management

(D) Inline expansion

[GATE 2014 : IIT Kharagpur]

Q.14 For a C program accessing $X[i][j][k]$, the following intermediate code is generated by a complier. Assume that the size of an integer is 32 bits and the size of character is bits.

$$t0 = i*1024$$
$$t1 = j*32$$
$$t2 = k*4$$
$$t3 = t1+t0$$
$$t4 = t3+t2$$
$$t5 = X[t4]$$

Which one of the following statements about the source code for the C program is CORRECT?

(A) X is declared as "int $X[32][32][8]$"

(B) X is declared as "int $X[4][1024][32]$"

(C) X is declared as "*char* $X[4][32][8]$"

(D) X is declared as "*char* $X[32][16][2]$"

[GATE 2014 : IIT Kharagpur]

Q.15 Which of the following statements are CORRECT?

1. Static allocation of all data areas by a compiler makes it impossible to implement recursion.
2. Automatic garbage collection is essential to implement recursion.
3. Dynamic allocation of activation records is essential to implement recursion.
4. Both heap and stack are essential to implement recursion.

(A) 1 and 2 only (B) 2 and 3 only

(C) 3 and 4 only (D) 1 and 3 only

[GATE 2014 : IIT Kharagpur]

Q.16 Consider the basic block given below.

$$a = b+c$$
$$c = a+d$$
$$d = b+c$$
$$e = d-b$$
$$a = e+b$$

The minimum number of nodes and edges present in the DAG representation of the above basic block respectively are

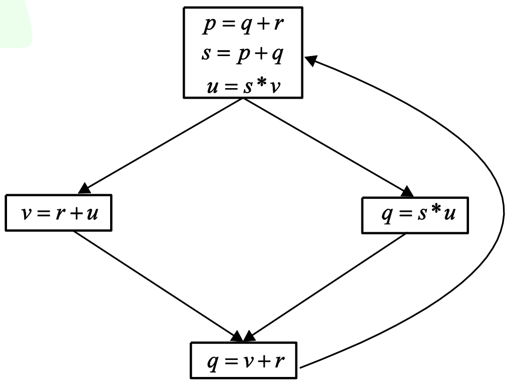
(A) 6 and 6 (B) 8 and 10

(C) 9 and 12 (D) 4 and 4

[GATE 2014 : IIT Kharagpur]

Q.17 A variable x is said to be live at a statement S_i in a program if the following three conditions hold simultaneously:

1. There exists a statement S_j that uses x
2. There is a path from S_i to S_j in the flow graph corresponding to the program
3. The path has no intervening assignment to x including at S_i and S_j



The variables which are live both at the statement in basic block 2 and at the statement in basic block 3 of the above control flow graph are

(A) p, s, u (B) r, s, u

(C) r, u (D) q, v

[GATE 2015 : IIT Kanpur]

Q.18 The least number of temporary variables required to create a three-address code in static single assignment form for the expression q + r/3 + s – t * 5 + u * v/w is _____.

[GATE 2015 : IIT Kanpur]

Q.19 In the context of abstract-syntax-tree (AST) and control-flow-graph (CFG), which one of the following is TRUE?

(A) In both AST and CFG, let node N_2 be the successor of node N_1. In the input program, the code corresponding to N_2 is present after the code corresponding to N_1

(B) For any input program, neither AST nor CFG will contain a cycle

(C) The maximum number of successors of a node in an AST and a CFG depends on the input program

(D) Each node in AST and CFG corresponds to at most one statement in the input program

[GATE 2015 : IIT Kanpur]

Q.20 Consider the intermediate code given below.

(1) $i = 1$

(2) $j = 1$

(3) $t_1 = 5*i$

(4) $t_2 = t_1 + j$

(5) $t_3 = 4*t_2$

(6) $t_4 = t_3$

(7) $a[t_4] = -1$

(8) $j = j+1$

(9) if $j <= 5\, goto(3)$

(10) $i = i+1$

(11) If $i < 5\, goto(2)$

The number of nodes and edges in the control-flow-graph constructed for the above code, respectively, are

(A) 5 and 7 (B) 6 and 7

(C) 5 and 5 (D) 7 and 8

[GATE 2015 : IIT Kanpur]

Q.21 A student wrote two context-free grammars G1 and G2 for generating a single C-Like array declaration. The dimension of the array is at least one. For example,

$\text{int } a[10][3];$

The grammars use D as the start symbol, and use six terminal symbols int; id[ ] num.

Grammar G1	**Grammar G2**
$D \to \text{int } L;$	$D \to \text{int } L;$
$L \to id[E$	$L \to id\ E$
$E \to num]$	$E \to E[num]$
$E \to num][E$	$E \to [num]$

Which of the grammar correctly generate the declaration mentioned above?

(A) Both G1 and G2

(B) Only G1

(C) Only G2

(D) Neither G1 nor G2

[GATE 2016 : IISc Bangalore]

Q.22 Consider the following intermediate program in three address code

$p = a - b$

$q = p * c$

$p = u * v$

$q = p + q$

Which one of the following corresponds to a static single assignment form of the above code?

(A) $p_1 = a - b$
$q_1 = p_1 * c$
$p_1 = u * v$
$q_1 = p_1 + q_1$

(B) $p_3 = a - b$
$q_4 = p_3 * c$
$p_4 = u * v$
$q_5 = p_4 + q_4$

(C) $p_1 = a - b$
$q_1 = p_2 * c$
$p_3 = u * v$
$q_2 = p_4 + q_3$

(D) $p_1 = a - b$
$q_1 = p * c$
$p_2 = u * v$
$q_2 = p + q$

[GATE 2017 : IIT Roorkee]

Q.23 Consider the following grammar :

$stmt \to if$ expr then expr else expr; stmt| 0

expr$\to$ term relop term |term

term $\to$id |number

id $\to a|b|c$

Number $\to$[0-9]

Where relop is relational operator (e.g., <,>,..), 0 refers to the empty statement, and if, then, else are terminals. Consider a program P following the

above grammar containing then if terminals. The number of control flow paths in P is _____.
For example the program

If e_1 then e_2 else e_3 has 2 control flow paths, $e_1 \to e_2$ and $e_1 \to e_3$

[GATE 2017 : IIT Roorkee]

Practice Questions

Q.1 Consider the following statements regarding run-time environment
Which of the above statement is true?
(A) The storage used for heap section can grow at run time but not stack section.
(B) Only control links and access links are saved or restored when a function call or return happen at runtime.
(C) Control links are used in the activation record to access the non-local data.
(D) Temporary variables are one of the contents of an activation record.

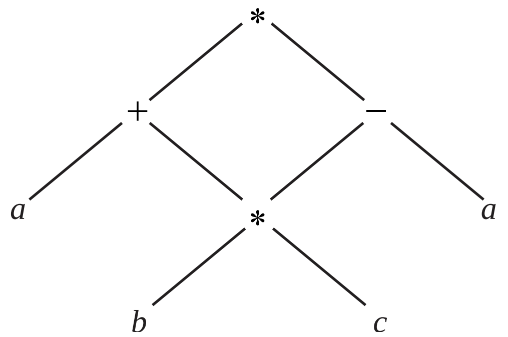
Q.2 Which of the following is the postfix form of the following expression?
$-b+c*d/e$
(A) $-b\ cde*/+$ (B) $b-cd*e/+$
(C) $b\ cd*e/+-$ (D) $b\ cde*/+-$

Q.3 Number of internal nodes in the *DAG* representation of the following expression are :
$\left((a*a)*(a*a)*\left((a*a)*(a*a)\right)\right)$
(A) 2 (B) 3
(C) 4 (D) 5

Q.4 Post fix conversion for the given expression is
$(x+y)*(p-q)/(m+n)$
(A) $xy+pq-*mn+/$
(B) $xypq+-*mn+/$
(C) $xy+pq*-mn+/$
(D) $xy+-*mn/+$

Q.5 Arrays with dynamic bounds are unpopular in HLLs as
(A) The checking of bounds is prohibitively expensive.
(B) Dynamic storage allocation is more expensive than static storage allocation.
(C) More efficient code can be generated.
(D) All of the above

Q.6 The equivalent expression for the DAG is



(A) $\left((a+b)*c)*(b*(c-a)\right)$
(B) $a+(b*c-a)$
(C) $(a+(b*c))*((b*c)-a)$
(D) $a*(a+b*c)-a$

Q.7 Consider the C program given below :
```
main ( )
{
    a = a + b;
    c = a × c;
    d = c − d;
    a = c/d;
    printf ("%d", a);
}
```
What will be the minimum number of nodes and edges present in the DAG representation of the output of above C program?
(A) 6 and 6 (B) 7 and 6
(C) 6 and 7 (D) 8 and 8

Q.8 What will be the output of the following program for call by name and copy restore parameter passing mechanism respectively?
```
void func (int a, int b)
{
    a = a + b;
    b = a + b;
    a = b − a;
    b = a − b
}
```

```
main ( )
{
    int j, i = 4;
    int array [10] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 0}
    func (i, array [i]);
    for (j = 0; j < 10; j++)
    printf ("%d", array [j]);
}
```

(A) call by name : $-1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 9$
copy restore : $1\ 2\ 3\ 4\ -9\ 6\ 7\ 8\ 9\ 0$

(B) call by name : $-1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 9$
copy restore : $1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 0$

(C) call by name : $-1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 9$
copy restore : $1\ 2\ 3\ 4\ -8\ 6\ 7\ 8\ 9\ 0$

(D) call by name : $-1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 9$
copy restore : $1\ 2\ 3\ 4\ -8\ 6\ 7\ 8\ 9\ 0$

Q.9 Which of the following parameter is not included in the activation record of recursive function call?

(A) Local variables (B) Return value
(C) Global variable (D) Access links

Q.10 Consider the following three address code table :

Location	Operator	Operand 1	Operand 2
(1)	*	*a*	*b*
(2)	Uminus	(1)	
(3)	+	c	*d*
(4)	+	(2)	(3)
(5)	+	*a*	*b*
(6)	+	(5)	(3)
(7)	–	(4)	(6)

Which of the following expression represents the above three address code (triple representation)?

(A) $-(a*b)-(c+d)+(a+b+c+d)$

(B) $+(a*b)-(c+d)-(a+b+c+d)$

(C) $-(a*b)+(c+d)-(a+b+c+d)$

(D) $-(a*b)+(c+d)+(a+b-(c+d))$

Q.11 Consider the following statements :

S_1 : Static allocation can not support recursive function.

S_2 : Stack allocation can support pointers but cannot deallocate storage at run-time.

S_3 : Heap allocation can support pointers and it can allocate or deallocate storage at run-time.

Which of the above statements are true?

(A) S_1 and S_2 (B) S_2 and S_3
(C) S_3 and S_1 (D) S_1, S_2 and S_3

Q.12 What is the minimum number of extra registers required to swap two numbers where two numbers are stored in two different registers?

Q.13 A directed acyclic graph represents one form of intermediate representation. The number of non-terminal nodes in DAG of $a=(b+c)*(b+c)$ expression is

Q.14 Consider the following expression :
The number

$$\left((x+y)-\left((x+y)*(x+y)\right)\right)$$
$$+\left((x+y)*(x+y)\right)$$
$$+\left((x+y)\div(x+y)\right)$$

r of nodes to represent the DAG for the above expression is ________

Q.15 Consider the following expression

$x=a*b-c*d+e$

For generating target code how many register will be required apart from accumulator *A*?

Q.16 Consider two binary operators * and ^ with precedence of operator * being lower than of ^. Operator * is left associative while operator ^ is right associative. The value of 3*2^3*4 is ________

Q.17 Given the 3-address code for a basic block

Num	Instruction	Meaning
1.	$Ld\ a, T_1$	$T_1 \leftarrow a$
2.	$Ld\ b, T_2$	$T_2 \leftarrow b$
3.	$Ld\ c, T_3$	$T_3 \leftarrow c$
4.	$Ld\ d, T_4$	$T_4 \leftarrow d$
5.	Add T_1, T_2, T_5	$T_5 \leftarrow T_1+T_2$
6.	Add T_5, T_3, T_6	$T_6 \leftarrow T_3+T_5$
7.	Add T_6, T_4, T_7	$T_7 \leftarrow T_6+T_4$
8.	$ST\ T_7, a$	$a \leftarrow T_7$

How many registers are needed to allocate this basic block with no spills?

Q.18 Find the minimum number of temporary variables created in a 3-address code for the following expression

$a+b*c+d-e-a+b*c$

Consider following grammar for precedence and associativity.

Q.19 Given the 3-address code for a basic block:

Number	Instruction	Meaning
1	$Ld\, a, T_1$	$T_1 \leftarrow a$
2	$Ld\, b, T_2$	$T_2 \leftarrow b$
3	$Ld\, c, T_3$	$T_3 \leftarrow c$
4	$Ld\, d, T_4$	$T_4 \leftarrow d$
5	Add T_1, T_2, T_5	$T_5 \leftarrow T_1 + T_2$
6	Add T_1, T_2, T_5	$T_5 \leftarrow T_1 + T_2$
7	Add T_1, T_2, T_5	$T_5 \leftarrow T_1 + T_2$
8	$ST\ T_7, a$	$a \leftarrow T_7$

How many registers are needed to allocate this basic block with no spills?

$S \rightarrow ES$

$E \rightarrow E - F \mid F$

$F \rightarrow F + G \mid T$

$G \rightarrow G * H \mid H$

$H \rightarrow id \mid \varepsilon$

Q.20 Consider the intermediate code given below :

1. $a=10$
2. $b=15$
3. $a=a+b$
4. $b=a-b$
5. $a=a-b$
6. if $(a==b)$ go to (3)

The number of nodes and edges in the control-flow graph constructed for the above code, respectively are X and Y. The value of $X+Y$ is _______.

❑❑❑

Computer Network

1 Basics Concept of Network

Classroom Questions

Q.1 Consider that 15 machines need to be connected in a LAN using 8-port Ethernet switches. Assume that these switches do not have any separate uplink ports. The minimum number of switches needed is ______.

(A) 3 (B) 7

(C) 1 (D) 5

[GATE 2019, IIT Madras]

Q.2 In the following pairs of OSI protocol layer/sub-layer and its functionality, the INCORRECT pair is

(A) Network layer and Routing

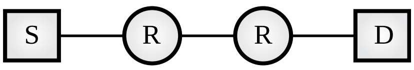
(B) Data Link Layer and Bit synchronization

(C) Transport layer and End-to-end process communication

(D) Medium Access Control sub-layer and Channel sharing

[GATE 2014, IIT Kharagpur]

Q.3 Assume that source S and destination D are connected through two intermediate routers labeled R. Determine how many times each packet has to visit the network layer and the data link layer during a transmission from S to D.



(A) Network layer – 4 times and Data link layer – 4 times

(B) Network layer – 4 times and Data link layer – 3 times

(C) Network layer – 4 times and Data link layer – 6 times

(D) Network layer – 2 times and Data link layer – 6 times

[GATE 2013, IIT Bombay]

Q.4 The protocol data unit (PDU) for the application layer in the Internet stack is:

(A) Segment (B) Datagram

(C) Message (D) Frame

[GATE 2012, IIT Delhi]

Q.5 In Ethernet when Manchester encoding is used, the bit rate is :

(A) Half the baud rate

(B) Twice the baud rate

(C) Same as the baud rate

(D) None of the above

[GATE 2007, IIT Kanpur]

Q.6 Which of the following statements is FALSE regarding a bridge?

(A) Bridge is a layer 2 device

(B) Bridge reduces collision domain

(C) Bridge is used to connect two or more LAN segments

(D) Bridge reduces broadcast domain

[GATE 2005, IIT Bombay]

Q.7 Choose the best matching between Group-1 and Group-2 :

Group-1	Group-2
P. Data link layer	1. Ensures reliable transport of data over a physical point-to-point link
Q. Network layer	2. Encodes/decodes data for physical transmission
R.Transport layer	3. Allows end-to-end communication between two processes
	4. Routes data from one network node to the next

(A) P-1, Q-4, R-3 (B) P-2, Q-4, R-1

(C) P-2, Q-3, R-1 (D) P-1, Q-3, R-2

[GATE 2004, IIT Delhi]

Q.8 Which of the following is NOT true with respect to a transparent bridge and a router?

(A) Both bridge and router selectively forward data packets

(B) A bridge uses IP addresses while a router uses MAC addresses

(C) A bridge builds up its routing table by inspecting incoming packets
(D) A router can connect between a LAN and a WAN

[GATE 2004, IIT Delhi]

Q.9 Which of the following functionality must be implemented by a transport protocol over and above the network protocol?
(A) Recovery from packet losses
(B) Detection of duplicate packets
(C) Packet delivery in the correct order
(D) End to end connectivity

[GATE 2003, IIT Madras]

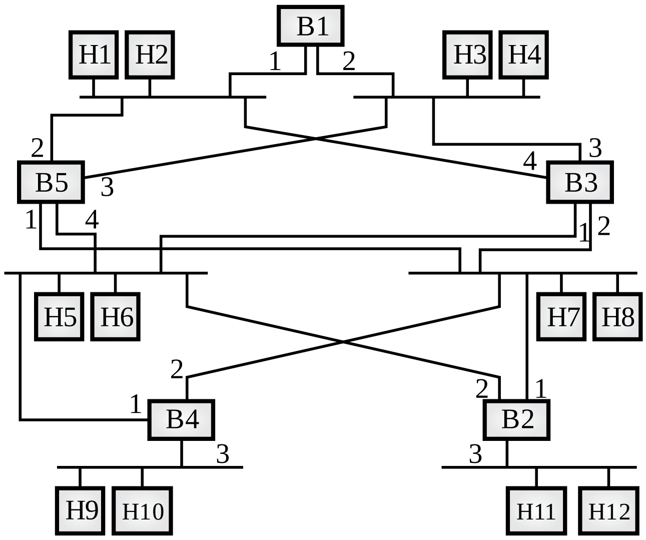
Q.10 In a network of LANs connected by bridges, packets are sent from one LAN to another through intermediate bridges. Since more than one path may exist between two LANs, packets may have to be routed through multiple bridges. Why is the spanning tree algorithm used for bridge-routing?
(A) For shortest path routing between LANs
(B) For avoiding loops in the routing paths
(C) For fault tolerance
(D) For minimizing collisions

[GATE 2003, IIT Madras]

Common Data Question 11 & 12

Consider the diagram shown below where a number of LANs are connected by (transparent) bridges. In order to avoid packets looping through circuits in the graph, the bridges organize themselves in a spanning tree. First, the root bridge is identified as the bridge with the least serial number. Next, the root sends out (one or more) data units to enable the setting up of the spanning tree of shortest paths from the root bridge to each bridge.

Each bridge identifies a port (the root port) through which it will forward frames to the root bridge. Port conflicts are always resolved in favour of the port with the lower index value. When there is a possibility of multiple bridges forwarding to the same LAN (but not through the root port), ties are broken as follows: bridges closest to the root get preference and between such bridges, the one with the lowest serial number is preferred.



Q.11 For the given connection of LANs by bridges, which one of the following choices represents the depth first traversal of the spanning tree of bridges?
(A) B1, B5, B3, B4, B2
(B) B1, B3, B5, B2, B4
(C) B1, B5, B2, B3, B4
(D) B1, B3, B4, B5, B2

[GATE 2006, IIT Kharagpur]

Q.12 Consider the spanning tree B1, B5, B3, B4, B2 for the given connection of LANs by bridges that represents the depth first traversal of the spanning tree of bridges. Let host H1 send out a broadcast ping packet. Which of the following options represents the correct forwarding table on B3?

(A)

Hosts	Port
H1, H2, H3, H4	3
H5, H6, H9, H10	1
H7, H8, H11, H12	2

(B)

Hosts	Port
H1, H2	4
H3, H4	3
H5, H6	1
H7, H8, H9, H10, H11, H12	2

(C)

Hosts	Port
H3, H4	3
H5, H6, H9, H10	1
H1, H2	4
H7, H8, H11, H12	2

(D)

Hosts	Port
H1, H2, H3, H4	3
H5, H7, H9, H10	1
H7, H8, H11, H12	4

[GATE 2006, IIT Kharagpur]

Practice Questions

Q.1 Which of the following option is NOT TRUE about protocol?

(A) It is a set of rules followed by each node in network.

(B) It is implemented inside header of data packet.

(C) It is a program written at different layers to do specific task.

(D) It is used to design OSI model.

Q.2 Which one of the following addressing scheme is implemented at Application Layer?

(A) MAC-address

(B) IP-address

(C) Port-address

(D) Specific address

Q.3 Which of the following options is/are TRUE about header?

(A) All the protocols are implemented inside header.

(B) Header size can not be greater than size of data.

(C) Some of the layer may not include header.

(D) No packet can travel into the network without header.

Q.4 Which one of the protocol is NOT associated with physical Layer?

(A) Data rate

(B) Bit synchronization

(C) Circuit switching

(D) Network topology

(E) Access control

Q.5 Suppose a connecting device has four ports. What is the minimum number of devices required to connect 10 computers.

Q.6 Which of the following options is/are TRUE?

(A) MAC-address is used to identify a particular host inside network.

(B) IP-address is used to identify a host.

(C) Port-address is used to identify a process inside host.

(D) Specific-address is used to find an application inside host.

Q.7 Which of the following options is/are TRUE?

(A) The packet is known by Frame in Data Link Layer.

(B) The packet is known by Datagram at application Layer.

(C) The packet is known by segment at Transport Layer.

(D) The name of packet is different at different Layer due to different header size.

Q.8 Which one of the following is NOT TRUE?

(A) Circuit switching takes place at physical Layer and uses physical address.

(B) Datagram switching takes place at network Layer and resources are allocated on demand.

(C) In circuit switching all the packets follow same path.

(D) In datagram switching, packets might not follow same path.

Q.9 Which of the following address is NOT contained by header?

(A) MAC-address (B) IP-address

(C) Port-address (D) Specific address

Q.10 Which layers of the OSI reference model are host-to-host layers?

(A) Transport, session, presentation, application

(B) Session, presentation, application

(C) Datalink, transport, presentation, application

(D) Physical, datalink, network, transport

Q.11 The following are names of data units in each layer. Mark the correct choice

(A) Packet: Data Link Layer

(B) Message: Transport Layer

(C) Bits: Physical Layer

(D) None of these

Q.12 Which of the following protocols is NOT part of Networks Layer?

(A) DHCP (B) ARP

(C) Go-back-N (D) BGP

Q.13 Which of the following is NOT TRUE about protocols?

(A) It is used for flow control.

(B) It is used to forward data packet.

(C) It is used to design OSI model.

(D) It is used inside header of data packet.

Q.14 Which of the following address scheme is NOT represented by bits?

(A) MAC-Address

(B) IP-Address

(C) Port-Address

(D) Specific Address

Q.15 The Header part of packet in different layer does NOT contains

(A) MAC-Address

(B) IP-Address

(C) Port Address

(D) Specific Address

❑❑❑

2 Data Link Layer

Classroom Questions

Q.1 Consider the sliding window flow-control protocol operating between a sender and a receiver over a full-duplex error-free link. Assume the following:

- The time taken for processing the data frame by the receiver is negligible.
- The time taken for processing the acknowledgement frame by the sender is negligible.
- The sender has infinite number of frames available for transmission.
- The size of the data frame is 2000 bits and the size of the acknowledgement frame is 10 bits.
- The link data rate in each direction is 1 Mbps (= 10^6 bits per second).
- One way propagation delay of the link is 100 milliseconds.

The minimum value of the sender's window size in terms of the number of frames, (rounded to the nearest integer) needed to achieve a link utilization of 50% is ________.

[GATE 2021, IIT Bombay]

Q.2 Consider the cyclic redundancy check (CRC) based error detecting scheme having the generator polynomial X^3+X+1. Suppose the message $m_4m_3m_2m_1m_0=11000$ is to be transmitted. Check bits $c_2c_1c_0$ are appended at end of the message by the transmitter using the above CRC scheme. The transmitted bit string is denoted by $m_4m_3m_2m_1m_0c_2c_1c_0$. The value of the check bit sequence $c_2c_1c_0$ is

(A) 101 (B) 110
(C) 100 (D) 111

[GATE 2021, IIT Bombay]

Q.3 Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 1,000 bits. The channel transmission rate is 1 Mbps (=10^6 bits per second). The aggregate number of transmissions across all the nodes (including new frame transmissions and retransmitted frames due to collisions) is modelled as a Poisson process with a rate of 1,000 frames per second. Throughput is defined as the average number of frames successfully transmitted per second. The throughput of the network (rounded to the nearest integer) is ________.

[GATE 2021, IIT Bombay]

Q.4 Consider a simple communication system where multiple nodes are connected by a shared broadcast medium (like Ethernet or wireless). The nodes in the system use the following carrier-sense based medium access protocol. A node that receives a packet to transmit will carrier-sense the medium for 5 units of time. If the node does not detect any other transmission, it starts transmitting its packet in the next time unit. If the node detects another transmission, it waits until this other transmission finishes, and then begins to carrier-sense for 5 time units again. Once they start to transmit, nodes do not perform any collision detection and continue transmission even if a collision occurs. All transmissions last for 20 units of time. Assume that the transmission signal travels at the speed of 10 meters per unit time in the medium.

Assume that the system has two nodes P and Q, located at a distance d meters from each other. P start transmitting a packet at time t = 0 after successfully completing its carrier-sense phase.

Node Q has a packet to transmit at time t = 0 and begins to carrier-sense the medium.

The maximum distance d (in meters, rounded to the closest integer) that allows Q to successfully avoid a collision between its proposed transmission and P's ongoing transmission is _______.

[GATE 2018, IIT Guwahati]

Q.5 A computer network uses polynomials over GF(2) for error checking with 8 bits as information bits and uses x^3+x+1 as the generator polynomial to generate the check bits. In this network, the message 01011011 is transmitted as :

(A) 01011011010 (B) 01011011011
(C) 01011011101 (D) 01011011100

[GATE 2017, IIT Roorkee]

Q.6 The values of parameters for the Stop-and-Wait ARQ protocol are as given below :

- Bit rate of the transmission channel = 1 Mbps.
- Propagation delay from sender to receiver = 0.75 ms.
- Time to process a frame = 0.25 ms.
- Number of bytes in the information frame = 1980.
- Number of bytes in the acknowledge frame = 20.
- Number of overhead bytes in the information frame = 20.

Assume there are no transmission errors. Then, the transmission efficiency (expressed in percentage) of the Stop-and-Wait ARQ protocol for the above parameters is _______ (correct to 2 decimal places).

(A) 89.33% (B) 89.34%
(C) 89.35% (D) 89.36%

[GATE 2017, IIT Roorkee]

Q.7 Consider two hosts X and Y, connected by a single direct link of rate 10^6 bits/sec. The distance between the two hosts is 10,000 km and the propagation speed along the link is 2×10^8 m/sec. Host X sends a file of 50,000 bytes as one large message to host Y continuously. Let the transmission and propagation delays be p milliseconds and q milliseconds respectively. Then the value of p and q are

(A) p = 50 and q = 100
(B) p = 50 and q = 400
(C) p = 100 and q = 50
(D) p = 400 and q = 50

[GATE 2017, IIT Roorkee]

Q.8 A sender uses the Stop-and-Wait ARQ protocol for reliable transmission of frames. Frames are of size 1000 bytes and the transmission rate at the sender is 80 Kbps (1 Kbps = 1000 bits/second). Size of an acknowledgment is 100 bytes and the transmission rate at the receiver is 8 Kbps. The one-way propagation delay is 100 milliseconds. Assuming no frame is lost, the sender throughput is ______ bytes/second.

(A) 2500 (B) 2501
(C) 2502 (D) 2503

[GATE 2016, IISc Bangalore]

Q.9 A network has a data transmission bandwidth of 20×10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another node is 40 microseconds. The minimum size of a frame in the network is _______ bytes.

(A) 200 (B) 201
(C) 202 (D) 203

[GATE 2016, IISc Bangalore]

Q.10 Consider a 128×10^3 bits/second satellite communication link with one way propagation delay of 150 milliseconds. Selective retransmission (repeat) protocol is used on this link to send data with a frame size of 1 kilobyte. Neglect the transmission time of acknowledgement. The minimum number of bits required for the sequence number field to achieve 100% utilization is ______.

(A) 4 (B) 5
(C) 6 (D) 7

[GATE 2016, IISc Bangalore]

Q.11 Consider a LAN with four nodes S_1, S_2, S_3 and S_4. Time is divided into fixed-size slots, and a node can being its transmission only at the beginning of a slot. A collision is said to have occurred if more than one node transmit in the same slot. The probability of generation of a frame in a time slot by S_1, S_2, S_3 and S_4 are

$0.1, 0.2, 0.3$ and 0.4 respectively. The probability of sending a frame in the first slot without any collision by any of these four stations is _______.

(A) 0.4404 (B) 0.463

(C) 0.464 (D) 0.465

[GATE 2015, IIT Kanpur]

Q.12 Suppose that the stop-and-wait protocol is used on a link with a bit rate of 64 kilobits per second and 20 milliseconds propagation delay. Assume that the transmission time for the acknowledgment and the processing time at nodes are negligible. Then the minimum frame size in bytes to achieve a link utilization of atleast 50 % is ______.

(A) 320 (B) 321

(C) 322 (D) 323

[GATE 2015, IIT Kanpur]

Q.13 A link has transmission speed of 10^6 bits/sec. It uses data packets of size 1000 bytes each. Assume that the acknowledgment has negligible transmission delay and that its propagation delay is the same as the data propagation delay. Also, assume that the processing delays at nodes are negligible. The efficiency of the stop-and-wait protocol in this setup is exactly 25%. The value of the one way propagation delay (in milliseconds) is ______.

(A) 12 (B) 13

(C) 14 (D) 15

[GATE 2015, IIT Kanpur]

Q.14 Consider a CSMA/CD network that transmits data at a rate of 100 Mbps (10^8 bits per second) over a 1 km (kilometer) cable with no repeaters. If the minimum frame size required for this network is 1250 bytes, what is the signal speed (km/sec) in the cable?

(A) 8000 (B) 10000

(C) 16000 (D) 20000

[GATE 2015, IIT Kanpur]

Q.15 Consider a network connecting two systems located 8000 Km apart. The bandwidth of the network is 500×10^6 bits per second. The propagation speed of the media is 4×10^6 meters per second. It needs to design a Go-Back-N sliding window protocol for this network. The average packet size is 10^7 bits. The network is to be used to its full capacity. Assume that processing delays at nodes are negligible. Then, the minimum size in bits of the sequence number field has to be ______.

(A) 8 (B) 7

(C) 6 (D) 5

[GATE 2015, IIT Kanpur]

Q.16 Consider a selective repeat sliding window protocol that uses a frame size of 1 KB to send data on a 1.5 Mbps link with a one-way latency of 50 msec. To achieve a link utilization of 60%, the minimum number of bits required to represent the sequence number field is _________.

(A) 5 (B) 6

(C) 7 (D) 8

[GATE 2014, IIT Kharagpur]

Q.17 A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the output bit-string after stuffing is 01111100101, then the input bit-string is :

(A) 0111110100 (B) 0111110101

(C) 0111111101 (D) 0111111111

[GATE 2014, IIT Kharagpur]

Q.18 Determine the maximum length of the cable (in km) for transmitting data at a rate of 500 Mbps in an Ethernet LAN with frames of size 10,000 bits. Assume the signal speed in the cable to be 2,00,000 km/s.

(A) 1 (B) 2

(C) 2.5 (D) 5

[GATE 2013, IIT Bombay]

Q.19 Let G(x) be the generator polynomial used for CRC checking. What is the condition that should be satisfied by G(x) to detect odd number of bits in error?

(A) G(x) contains more than two terms

(B) G(x) does not divide $1+x^k$, for any k not exceeding the frame length

(C) 1+x is a factor of G(x)

(D) G(x) has an odd number of terms.

[GATE 2009, IIT Roorkee]

Common Data Question 20 & 21

Frames of 1000 bits are sent over a 10^6 bps duplex link between two hosts. The propagation time is 25 ms. Frames are to be transmitted into this link to maximally pack them in transit (within the link).

Q.20 What is the minimum number of bits (I) that will be required to represent the sequence numbers distinctly? Assume that no time gap needs to be given between transmission of two frames.

(A) I = 2 (B) I = 3

(C) I = 4 (D) I = 5

[GATE 2009, IIT Roorkee]

Q.21 Let I be the minimum number of bits (I) that will be required to represent the sequence numbers distinctly assuming that no time gap needs to be given between transmission of two frames.

Suppose that the sliding window protocol is used with the sender window size of 2^I, where I is the numbers of bits as mentioned earlier and acknowledgements are always piggy backed. After sending 2^I frames, what is the minimum time the sender will have to wait before starting transmission of the next frame? (Identify the closest choice ignoring the frame processing time)

(A) 16 ms (B) 18 ms

(C) 20 ms (D) 22 ms

[GATE 2009, IIT Roorkee]

Q.22 A 1 Mbps satellite link connects two ground stations. The altitude of the satellite is 36,504 km and speed of the signal is 3×10^8 m/s. What should be the packet size for a channel utilization of 25% for a satellite link using go-back-127 sliding window proto-col? Assume that the acknowledgment packets are negligible in size and that there are no errors during communication.

(A) 120 bytes (B) 60 bytes

(C) 240 bytes (D) 90 bytes

[GATE 2008, IISc Bangalore]

Q.23 The minimum frame size required for a CSMA/CD based computer network running at 1Gbps on a 200m cable with a link speed of 2×10^8 m/sec is :

(A) 125 bytes

(B) 250 bytes

(C) 500 bytes

(D) None of the above

[GATE 2008, IISc Bangalore]

Q.24 There are n stations in slotted LAN. Each station attempts to transmit with a probability p in each time slot. What is the probability that ONLY one station transmits in a given time slot?

(A) $np(1-p)^{n-1}$ (B) $(1-p)^{n-1}$

(C) $p(1-p)^{n-1}$ (D) $1-(1-p)^{n-1}$

[GATE 2007, IIT Kanpur]

Q.25 In a token ring network the transmission speed is 10^7 bps and the propagation speed is 200 meters/μs. The 1-bit delay in this network is equivalent to:

(A) 500 meters of cable.

(B) 200 meters of cable.

(C) 20 meters of cable.

(D) 50 meters of cable.

[GATE 2007, IIT Kanpur]

Q.26 The message 11001001 is to be transmitted using the CRC polynomial x^3+1 to protect it from errors. The message that should be transmitted is:

(A) 11001001000 (B) 11001001011

(C) 11001010 (D) 110010010011

[GATE 2007, IIT Kanpur]

Q.27 The distance between two stations M and N is L kilometres. All frames are K bits long. The propagation delay per kilometre is t seconds. Let R bits/second be the channel capacity. Assuming that the processing delay is negligible, the minimum number of bits for the sequence number field in a frame for maximum utilization, when the sliding window protocol is used, is :

(A) $\left\lceil \log_2 \frac{2LtR+2K}{K} \right\rceil$

(B) $\left\lceil \log_2 \frac{2LtR}{K} \right\rceil$

(C) $\left\lceil \log_2 \frac{2LtR+K}{K} \right\rceil$

(D) $\left\lceil \log_2 \frac{2LtR+2K}{2K} \right\rceil$

[GATE 2007, IIT Kanpur]

Q.28 An error correcting code has the following code words: 00000000, 00001111, 01010101, 10101010, 11110000. What is the maximum number of bit errors that can be corrected?

(A) 0 (B) 1

(C) 2 (D) 3

[GATE 2007, IIT Kanpur]

Q.29 A broadcast channel has 10 nodes and total capacity of 10 Mbps. It uses polling for medium access. Once a node finishes transmission, there is a polling delay of 80 μs to poll the next node. Whenever a node is polled, it is allowed to transmit a maximum of 1000 bytes. The maximum throughput of the broadcast channel is :

(A) 1 Mbps (B) 100/11 Mbps

(C) 10 Mbps (D) 100 Mbps

[GATE 2007, IIT Kanpur]

Q.30 Station A needs to send a message consisting of 9 packets to Station B using a sliding window (window size 3) and go-back-n error control strategy. All packets are ready and immediately available for transmission. If every 5th packet that A transmits gets lost (but no acks from B ever get lost), then what is the number of packets that A will transmit for sending the message to B?

(A) 12 (B) 14

(C) 16 (D) 18

[GATE 2006, IIT Kharagpur]

Q.31 Station A uses 32 byte packets to transmit messages to Station B using a sliding window protocol. The round trip delay between A and B is 80 milliseconds and the bottleneck bandwidth on the path between A and B is 128 kbps. What is the optimal window size that A should use?

(A) 20 (B) 40

(C) 160 (D) 320

[GATE 2006, IIT Kharagpur]

Q.32 A router has two full-duplex Ethernet interfaces each operating at 100 Mb/s. Ethernet frames are atleast 84 bytes long (including the Preamble and the Inter-Packet-Gap). The maximum packet processing time at the router for wire speed forwarding to be possible is (in micro-seconds)

(A) 0.01 (B) 3.36

(C) 6.72 (D) 8

[GATE 2006, IIT Kharagpur]

Q.33 On a wireless link, the probability of packet error is 0.2. A stop-and-wait protocol is used to transfer data across the link. The channel condition is assumed to be independent of transmission to transmission. What is the average number of transmission attempts required to transfer 100 packets?

(A) 100 (B) 125

(C) 150 (D) 200

[GATE 2006, IIT Kharagpur]

Q.34 Suppose the round trip propagation delay for a 10 Mbps Ethernet having 48-bit jamming signal is 46.4 μs. The minimum frame size is :

(A) 94 (B) 416

(C) 464 (D) 512

[GATE 2005, IIT Bombay]

Q.35 The maximum window size for data transmission using the selective reject protocol with n-bit frame sequence numbers is:

(A) 2^n (B) 2^{n-1}

(C) $2^n - 1$ (D) 2^{n-2}

[GATE 2005, IIT Bombay]

Q.36 Which of the following statements is TRUE about CSMA/CD :

(A) IEEE 802.11 wireless LAN runs CSMA/CD protocol

(B) Ethernet is not based on CSMA/CD protocol

(C) CSMA/CD is not suitable for a high propagation delay network like satellite network

(D) There is no contention in a CSMA/CD network

[GATE 2005, IIT Bombay]

Q.37 A network with CSMA/CD protocol in the MAC layer is running at 1Gbps over a 1 km cable with no repeaters. The signal speed in the cable is 2×10^{8} m/sec. The minimum frame size for this network should be:

(A) 10000 bits (B) 10000 bytes
(C) 5000 bits (D) 5000 bytes

[GATE 2005, IIT Bombay]

Q.38 A channel has a bit rate of 4 kbps and one-way propagation delay of 20 ms. The channel uses stop and wait protocol. The transmission time of the acknowledgment frame is negligible. To get a channel efficiency of at least 50%, the minimum frame size should be

(A) 80 bytes (B) 80 bits
(C) 160 bytes (D) 160 bits

[GATE 2005, IIT Bombay]

Q.39 In a TDM medium access control bus LAN, each station is assigned one time slot per cycle for transmission. Assume that the length of each time slot is the time to transmit 100 bits plus the end-to-end propagation delay. Assume a propagation speed of 2×10^{8} m/sec. The length of the LAN is 1 km with a bandwidth of 10 Mbps. The maximum number of stations that can be allowed in the LAN so that the throughput of each station can be 2/3 Mbps is

(A) 3 (B) 5
(C) 10 (D) 20

[GATE 2005, IIT Bombay]

Q.40 Consider the following message M = 1010001101. The cyclic redundancy check (CRC) for this message using the divisor polynomial $x^5+x^4+x^2+1$ is :

(A) 01110 (B) 01011
(C) 10101 (D) 10110

[GATE 2005, IIT Bombay]

Q.41 A and B are the only two stations on an Ethernet. Each has a steady queue of frames to send. Both A and B attempt to transmit a frame, collide, and A wins the first backoff race. At the end of this successful transmission by A, both A and B attempt to transmit and collide. The probability that A wins the second backoff race is :

(A) 0.5 (B) 0.625
(C) 0.75 (D) 1.0

[GATE 2004, IIT Delhi]

Q.42 In a sliding window ARQ scheme, the transmitter's window size is N and the receiver's window size is M. The minimum number of distinct sequence numbers required to ensure correct operation of the ARQ scheme is

(A) min (M,N) (B) max (M,N)
(C) M+N (D) MN

[GATE 2004, IIT Delhi]

Q.43 Consider a 10 Mbps token ring LAN with a ring latency of 400 μs. A host that needs to transmit seizes the token. Then it sends a frame of 1000 bytes, removes the frame after it has circulated all around the ring, and finally releases the token. This process is repeated for every frame. Assuming that only a single host wishes to transmit, the effective data rate is

(A) 1Mbps (B) 2Mbps
(C) 5Mbps (D) 6Mbps

[GATE 2004, IIT Delhi]

Q.44 A 20 Kbps satellite link has a propagation delay of 400 ms. The transmitter employs the “go back n ARQ” scheme with n set to 10. Assuming that each frame is 100 byte long, what is the maximum data rate possible?

(A) 5 Kbps (B) 10 Kbps
(C) 15 Kbps (D) 20 Kbps

[GATE 2004, IIT Delhi]

Q.45 Consider a simplified time slotted MAC protocol, where each host always has data to send and transmits with probability p = 0.2 in every slot. There is no backoff and one frame can be transmitted in one slot. If more than one host transmits in the same slot, then the transmissions are unsuccessful due to collision. What is the maximum number of hosts which this protocol can support if each host has to be provided a minimum throughput of 0.16 frames per time slot?

(A) 1 (B) 2
(C) 3 (D) 4

[GATE 2004, IIT Delhi]

Q.46 A 2 km long broadcast LAN has 10^7 bps bandwidth and uses CSMA/CD. The signal travels along the wire at 2×10^8 m/s. What is the minimum packet size that can be used on this network?

(A) 50 bytes

(B) 100 bytes

(C) 200 bytes

(D) None of the above

[GATE 2003, IIT Madras]

Q.47 Host A is sending data to host B over a full duplex link. A and B are using the sliding window protocol for flow control. The send and receive window sizes are 5 packets each. Data packets (sent only from A to B) are all 1000 bytes long and the transmission time for such a packet is 50 µs. Acknowledgment packets (sent only from B to A) are very small and require negligible transmission time. The propagation delay over the link is 200 µs. What is the maximum achievable throughput in this communication?

(A) 7.69×10^6 Bps (B) 11.11×10^6 Bps

(C) 12.33×10^6 Bps (D) 15.00×10^6 Bps

[GATE 2003, IIT Madras]

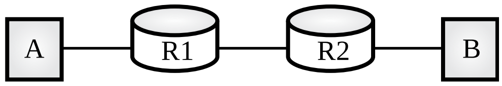
Q.48 What is the distance of the following code 000000,010101,000111,011001,111111?

(A) 2 (B) 3

(C) 4 (D) 1

[GATE 1995, IIT Kanpur]

Q.49 Consider the store and forward packet switched network given below. Assume that the bandwidth of each link is 10^6 bytes/sec. A user on host *A* sends a file of size 10^3 bytes to host *B* through routers R_1 and R_2 in three different ways. In the first case a single packet containing the complete file is transmitted from *A* to *B*. In the second case, the file is split into 10 equal parts, and these packets are transmitted from *A* to *B*. In the third case, the file is split into 20 equal parts and these packets are sent from *A* to *B*. Each packet contains 100 bytes of header information along with the user data. Consider only transmission time and ignore processing, queuing and propagation delays. Also assume that there are no errors during transmission. Let T_1, T_2 and T_3 be the times taken to transmit the file in the first, second and third case respectively. Which one of the following is CORRECT?



(A) $T_1 < T_2 < T_3$ (B) $T_1 > T_2 > T_3$

(C) $T_2 = T_3, T_3 < T_1$ (D) $T_1 = T_3, T_3 > T_2$

[GATE 2014, IIT Kharagpur]

Q.50 Consider a source computer (S) transmitting a file of size 10^6 bits to a destination computer (D) over a network of two routers (R_1 and R_2) and three links (L_1, L_2 and L_3). L_1 connects S to R_1, L_2 connects R_1 to R_2 and L_3 connects R_2 to D. Let each link be of length 100 km. Assume signals travel over each link at a speed of 10^8 meters per second. Assume that the link bandwidth on each link is 1 Mbps. Let the file be broken down into 1000 packets each of size 1000 bits. Find the total sum of transmission and propagation delays in transmitting the file from S to D?

(A) 1005 ms (B) 1010 ms

(C) 3000 ms (D) 3003 ms

[GATE 2012, IIT Delhi]

Q.51 In a packet switching network, packets are routed from source to destination along a single path having two intermediate nodes. If the message size is 24 bytes and each packet contains a header of 3 bytes, then the optimum packet size is:

(A) 4 (B) 6

(C) 7 (D) 9

[GATE 2005, IIT Bombay]

Q.52 Which one of the following statements is FALSE?

(A) Packet switching leads to better utilization of bandwidth resources than circuit switching

(B) Packet switching results in less variation in delay than circuit switching

(C) Packet switching requires more per-packet processing than circuit switching

(D) Packet switching can lead to reordering unlike in circuit switching

[GATE 2004, IIT Delhi]

Q.53 In an Ethernet local area network, which one of the following statements is TRUE?

(A) A station stops to sense the channel once it starts transmitting a frame.

(B) The purpose of the jamming signal is to pad the frames that are smaller than the minimum frame size.

(C) A station continues to transmit the packet even after the collision is detected.

(D) The exponential back off mechanism reduces the probability of collision on retransmissions.

[GATE 2016, IISc Bangalore]

Q.54 Consider a 3-bit error detection and 1-bit error correction hamming code for 4-bit data. The extra parity bits required would be ______ and the 3-bit error detection is possible because the code has a minimum distance of ______.

[GATE 1992, IIT Delhi]

Q.55 Following 7 bit single error correcting hamming coded message is received.

7	6	5	4	3	2	1	Bit No.
1	0	0	0	1	1	0	X

Determine if the message is correct (assuming that at most 1 bit could be corrupted). If the message contains an error find the bit which is erroneous and gives correct message.

[GATE 1994, IIT Kharagpur]

Q.56 In a data link protocol, the frame delimiter flag is given by 0111. Assuming that bit stuffing is employed, the transmitter sends the data sequence 01110110 as :

(A) 01101011 (B) 011010110

(C) 011101100 (D) 0110101100

[GATE 2004, IIT Delhi]

Q.57 Consider a parity check code with three data bits and four parity check bits.

Three of the Code Words are 0101011, 1001101 and 1110001.

Which of the following are also code words?

I. 0010111 II. 0110110

III. 1011010 IV. 0111010

(A) I and III (B) I, II and III

(C) II and IV (D) I, II, III and IV

[GATE 2004, IIT Delhi]

Q.58 In a communication network, a packet of length L bits takes link L_1 with a probability of p_1 or link L_2 with a probability of p_2. Link L_1 and L_2 have bit error probability of b_1 and b_2 respectively. The probability that the packet will be received without error via either L_1 or L_2 is

(A) $(1-b_1)Lp_1+(1-b_2)Lp_2$

(B) $[1-(b_1+b_2)L]\,p_1p_2$

(C) $(1-b_1)L(1-b_2)Lp_1p_2$

(D) $1-(b_1Lp_1+b_2Lp_2)$

[GATE 2005, IIT Bombay]

Q.59 In the 4B/5B encoding scheme, every 4 bits of data are encoded in a 5-bit code word. It is required that the code words have at most 1 leading and at most 1 trailing zero. How many are such code words possible?

(A) 14 (B) 16

(C) 18 (D) 20

[GATE 2006, IIT Kharagpur]

Q.60 Suppose that it takes 1 unit of time to transmit a packet (of fixed size) on a communication link. The link layer uses a window flow control protocol with a window size of N packets. Each packet causes an ack or a nak to be generated by the receiver, and ack/nak transmission times are negligible. Further, the round trip time on the link is equal to N units. Consider time $i > N$. If only acks have been received till time i (no naks), then the good put evaluated at the transmitter at time i (in packets per unit time) is

(A) $1-\frac{N}{i}$ (B) $\frac{i}{(N+i)}$

(C) 1 (D) $1-e^{\frac{i}{N}}$

[GATE 2006, IIT Kharagpur]

Q.61 Data transmitted on a link uses the following 2D parity scheme for error detection :

Each sequence of 28 bits is arranged in a 4×7 matrix (rows r_0 through r_3, and columns d_7 through d_1) and is padded with a column d_0 and row r_4 of parity bits computed using the even parity scheme. Each bit of column d_0 (respectively, row r_4) gives the parity of the corresponding row (respectively, column). These 40 bits are transmitted over the data link.

	d_7	d_6	d_5	d_4	d_3	d_2	d_1	d_0
r_0	0	1	0	1	0	0	1	1
r_1	1	1	0	0	1	1	1	0
r_2	0	0	0	1	0	1	0	0
r_3	0	1	1	0	1	0	1	0
r_4	1	1	0	0	0	1	1	0

The table shows data received by a receiver and has n corrupted bits. What is the mini-mum possible value of n?

(A) 1 (B) 2

(C) 3 (D) 4

[GATE 2008, IISc Bangalore]

Q.62 Two hosts are connected via a packet switch with 107 bits per second links. Each link has a propagation delay of 20 microseconds. The switch begins forwarding a packet 35 microseconds after it receives the same. If 10000 bits of data are to be transmitted between the two hosts using a packet size of 5000 bits, the time elapsed between the transmission of the first bit of data and the reception of the last bit of the data in microseconds is ______.

[GATE 2015, IIT Kanpur]

Practice Questions

Q.1 Which of the following options is/are TRUE?

(A) Framing protocol is used to separate frames at receiving end.

(B) Bit stuffing technique is used to identify data inside frame.

(C) Framing protocols use flag/delimiter bits pattern to separate frame.

(D) Bit stuffing is used to detect error inside frame.

Q.2 The minimum Hamming distance to correct 4-bits error is ______

Q.3 On wireless link, the probability of packet error is 0.25. Stop and wait protocol is used to transfer data. What is the average number of transmission attempts required to transfer 200 packets ______?

Q.4 Which of the following pair of window size at Sender and Receiver is NOT possible in selective repeat protocol if 4-bits sequence number field is used?

(A) (6, 6) (B) (7, 7)

(C) (8, 8) (D) (10, 10)

Q.5 A channel has bit rate of 10 kbps and one way propagation delay of 20 ms . The channel uses stop and wait protocol. The transmission time of acknowledgement is negligible to get channel efficiency of 60%, the minimum frame size should be ______ bytes?

Q.6 If a file consisting of 50,000 characters takes 40 second to send, then data rate is_____ kbps?

Q.7 In full duplex channel, if the data rate of link is 20 Mbps and signal speed is 200 m/μs. The number of bytes that can be placed on the channel of 100 km is ______?

Q.8 A network with bandwidth of 10 Mbps can pass an average of 18000 frames per minute with each frame carrying 10^4 bits. What is the through put of this network?

(A) 10 Mbps (B) 5 Mbps
(C) 3 Mbps (D) 2 Mbps

Q.9 Consider a 10 Mbps link with channel utilization of 80%. Network is using sliding window protocols. The throughput of channel is _______? (Mbps)

Q.10 Find maximum utilization of channel in polling protocol if size of frame is 10000 bits and bandwidth is 1 Mbps and propagation delay is 10 ms with 5 station in LAN is _______? (%)

Q.11 In CRC if the data unit is 100111001 and the divisor is 1011 then what is dividend at the receiver?

(A) 100111001101 (B) 100111001011
(C) 100111001 (D) 100111001110

Q.12 In Ethernet CSMA/CD, the special bit sequence transmitted by media access management to handle collision is called

(A) Preamble (B) Postamble
(C) Jam (D) None of these

Q.13 Frames of 1000 bits are sent over a 10^6 bps duplex link between two hosts. The propagation time is 25 ms. Frames are to be transmitted into this link to maximally pack them in transit (within the link).

What is the minimum number of bits (I) that will be required to represent the sequence numbers distinctly? Assume that no time gap needs to be given between transmission of two frames.

(A) $I=2$ (B) $I=3$
(C) $I=4$ (D $I=5$

Q.14 A pure ALOHA network transmits 200 bits frames using a shared channel with 200 kbps bandwidth. If the system (all stations put together) produces 500 frames per second, then the throughput of the system is _______.

(A) 0.384 (B) 0.184
(C) 0.286 (D) 0.586

Q.15 In CRC checksum method, assume that given frame for transmission is 1101011011 and the generator polynomial is $G(x)=x^4+x+1$.

After implementing CRC encoder, the encoded word sent from sender side is _______.

(A) 11010110111110 (B) 11101101011011
(C) 110101111100111 (D) 110101111001111

Q.16 A slotted ALOHA network transmits 200 bits frames using a shared channel with 200 kbps bandwidth. If the system (all stations put together) produces 1000 frames per second, then the throughput of the system is _______.

(A) 0.268 (B) 0.468'
(C) 0.368 (D) 0.568

Q.17 A link of capacity 100 Mbps is carrying traffic from a number of sources. Each source generates an on-off traffic stream; when the source is on, the rate of traffic is 10 Mbps, and when the source is off, the rate of traffic is zero. The duty cycle, which is the ratio of on-time to off-time, is 1:2. When there is no buffer at the link, the minimum number of sources that can be multiplexed on the link so that link capacity is not wasted and no data loss occurs is S_1. Assuming that all sources are synchronized and that the link is provided with a large buffer, the maximum number of sources that can be multiplexed so that no data loss occurs is S_2. The values of S_1 and S_2 are, respectively,

(A) 10 and 30 (B) 12 and 25
(C) 5 and 33 (D) 15 and 22

Q.18 A CSMA/CD network is having a bandwidth of 512 Mbps and a distance of 200 meters. Determine the minimum frame size in bits, in order to detect a collision. Assume that that the signal speed is 2×10^8m/s.

Q.19 A LAN uses CSMA/CD protocol. The end-to-end propagation delay between two stations is 50μs. Then the contention time is

(A) 150μs (B) 100μs
(C) 200μs (D) None of these

Q.20 An Ethernet channel has transmission time 2ms, contention time 4ms and idle time 2ms. Then the channel utilization is ________ (in percentage)

Q.21 Which of the following are functions of Data Link layer?

(i) Flow Control (ii) Error Control

(iii) Compression (iv) File Manager

(v) Dialog Control

(A) i and ii (B) ii, iv and v

(C) i and iii (D) iii, iv, and v

Q.22 The minimum frame size in CSMA/CD network is 24 bytes. If channel bandwidth is 1Mbps then what will be the propagation delay?

(A) 112 μs (B) 192 μs

(C) 170 μs (D) None of these

Q.23 Consider a pure ALOHA network, operating at 10 Mbps and the frame length is 100 bits. The vulnerable time in this network in μs is ________.

Q.24 A sender sends a series of packets to the same destination using 4-bit sequence numbers. If the sequence number starts with 0, what is the sequence no after bending 100 packets?

(A) 2 (B) 3

(C) 4 (D) 5

Q.25 A Go-Back-N ARQ is using 7 bits to represent the sequence number. What is the size of window (receiver's)?

(A) 127 (B) 8

(C) 14 (D) None of these

Q.26 A computer is using the following sequence numbers

0, 1, 2, 3, 4, 5, 6, 7, 0, 1, 2, 3 ....

What is the size of the window (Assume selective repeat protocol is used)

(A) 16 (B) 8

(C) 4 (D) 3

Q.27 Check sum of 10101001 11000001 (8-bit segment)

(A) 00001100 (B) 11110000

(C) 10010100 (D) 10011000

Q.28 Which of the following statements is/are FALSE?

S_1: Hamming code is used for both error detection and correction.

S_2: CRC is used for error detection.

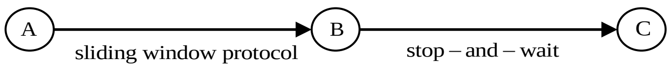
(A) S_1 only

(B) S_2 only

(C) Both S_1 and S_2

(D) Neither S_1 nor S_2

Q.29 The distance between A to B is 2000 km. The propagation delay is $2\mu s/km$ for both links. The data rate between A and B is 200 kbps. Both the links are full duplex. All data frames are 1000 bits and ACK frames are negligible. Window size is 2. If the RTT between A and C is 10ms then what is the distance between B and C? (in km)



Q.30 Consider Sliding Window Protocol for a 10 Mbps channel with one-way latency of 150μs. If the packet size is 1 KB then what is the link utilization for window size of 127?

(A) 75% (B) 100%

(C) 43% (D) None of these

Q.31 A channel connecting source and destination has data rate of 20 kbps, propagation delay of 50μs. If the transmission delay is equal to two times the propagation delay then what is the packet size for stop-and-wait protocol? Source and destination are 10 km apart.

(A) 250 bytes (B) 50 bytes

(C) 500 bytes (D) 400 bytes

Q.32 Consider the following data

$$\left.\begin{array}{l}\text{20 kbps satallite channel}\\ \text{1000 bits data frames}\\ \text{RTT} = 50\text{ms}\end{array}\right\} T_x = \frac{L}{B} = \frac{1000}{20\times10^3} = 50\text{ms}$$

What is the max throughput for window size of 4?

(A) 1 kbps (B) 4 kbps

(C) 5 kbps (D) 4 kbps

Q.33 Consider a network connecting 2 systems located 200 km apart. The bandwidth of the network is 50Mpbs. The propagation speed of the media is 2×10^8 m/s. It is needed to design a Go-Back-N Sliding Window Protocol for this network. The average packet size is 10^3 bits. The network is to be used to its full capacity. Assume that processing delays at nodes are negligible. Then the minimum size in bits of the sequence number field has to be ________.

Q.34 On a workless link, the probability of packet error is 0.4. A Stop and Wait protocol is used to transfer data across the link. The channel condition is assumed to be independent from transmission to transmission. What is the average approximate number of transmission attempts required to transfer 100 packets?

Q.35 The message 1100 1001 is to be transmitted using the CRC polynomial x^3+1 to protect it from errors. The CRC for this scenario is ________.

(A) 011 (B) 100
(C) 110 (D) 101

Q.36 Station A uses 16 bytes packets to transmit messages to station B using SWP. The latency between A and B is 40ms and the bandwidth on the path between A and B is 64 kbps. What is the optimal window size that A should use?

(A) 20 (B) 40
(C) 160 (D) 320

Q.37 Consider a 64kbps satellite communications link with one-way latency of 50ms. Selective repeat protocol is used on this link to send data with a frame size of 1KB. Neglect the transmission time of ACK. The minimum number of bits required for the sequence number field to achieve 100% utilization is ________.

Q.38 A 10kbps satellite link has propagation delay of 200ms. The sender employs the Go-Back-N ARQ scheme with n set to 10. If each frame is 50 bytes long, what is the maximum data rate possible (in kbps).

Q.39 In a CSMA/CD protocol, the signal speed is $2*10^8$ m/s and bandwidth is 10^7 bps. If the minimum size of frame is 200 bytes then find length of cable is________ (km)?

Q.40 An error detecting code has the following code words :

00000000 00001111
01010101
10101010 11110000

What is the minimum number of bits errors that can be detected?

(A) 1 (B) 2
(C) 3 (D) 4

Q.41 A and B are only two stations on an Ethernet. A and B attempt to transmit a frame and collide. What is the probability that station B wins first backoff race?

(A) 0.5 (B) 0.75
(C) 0.625 (D) 0.25

Common Data Question 42 & 43

Q.42 Station A uses 1000 bytes packet to transmit message to station B using Sliding Window Protocol. The propagation delay between A and B is 80 ms and bandwidth on the path is 4 Mbps. What is the optimal window that station A should use?

(A) 20 (B) 40
(C) 80 (D) 160

Q.43 In above questions if both the stations are using Selective Repeat Protocol, then what is the minimum number of bits required to represent Sequence Number field is

(A) 7 (B) 6
(C) 8 (D) 5

Q.44 Suppose the propagation delay for a 10 Mbps Ethernet having 64 bits jamming signal is 54.4 micro sec. What is the minimum frame size________ (bytes)?

Q.45 The sender is using Framing Protocol with delimiter flag size of 01110. If actual data is 101111110111011. Then find the number of stuff bits in transmitted data?

Q.46 In the polling media access control protocol bus LAN, each station is assigned one time slot for transmission. Each station wants to transmit a frame of size 100 bits. The propagation speed is 2* 10^8 m/s and the length of LAN is 1 km. The bandwidth is 10 Mbps. If total number of stations over this network is 5, then find the throughput of channel?

(A) 10/3 Mbps (B) 10 Mbps
(C) 20/3 Mbps (D) 5 Mbps

Q.47 Frames of 1000 bytes are sent over 2*10^6 bps full duplex link between two hosts. The propagation delay is 80 ms. Frames are to be transmitted into this link to maximally pack them in transit. If sender is using Selective repeat protocol then what is the number of bits to represent sequence number distinctly?

(A) 4 (B) 5
(C) 6 (D) 7

Q.48 A broadcast channel has total capacity of 10 Mbps and uses polling for media access. Once a node finish transmission there is polling delay of 80 micro sec to poll next node. Whenever a node is polled it is allowed to transmit a maximum 1000 bytes. The maximum number of stations that can be allowed in LAN so that throughput of each station is 10/11 Mbps.

(A) 5 (B) 10
(C) 15 (D) 20

Q.49 Consider a 10 Mbps link with propagation delay of 20 ms. The size of frame is 1000 bits and network is using selective repeat protocol. If transmission time of acknowledgement and processing time is negligible then for window size 10 what is the throughput of channel ________ (Mbps)?

Q.50 A 25 kbps satellite link has a propagation delay of 400 ms. The transmitter employs Go-Back-N sliding window protocol. If size of each frame is 100 bytes and effective bandwidth of channel is 10 kbps then find maximum window size possible?

❑❑❑

3 Network Layer

Classroom Questions

Q.1 An organization requires a range of IP address to assign one to each of its 1500 computers. The organization has approached an Internet Service Provider (ISP) for this task. The ISP uses CIDR and serves the requests from the available IP address space 202.61.0.0/17. The ISP wants to assign an address space to the organization which will minimize the number of routing entries in the ISP's router using route aggregation. Which of the following address spaces are potential candidates from which the ISP can allot any one of the organization?

I. 202.61.84.0/21 II. 202.61.104.0/21
III. 202.61.64.0/21 IV. 202.61.144.0/21

(A) I and II only (B) III and IV only
(C) II and III only (D) I and IV only

[GATE 2020, IIT Delhi]

Q.2 Consider three machines M, N, and P with IP addresses 100.10.5.2, 100.10.5.5, and 100.10.5.6 respectively. The subnet mask is set to 255.255.255.252 for all the three machines. Which one of the following is true?

(A) M, N, and P all belong to the same subnet
(B) Only M and N belong to the same subnet
(C) M, N, and P belong to three different subnets
(D) Only N and P belong to the same subnet

[GATE 2019, IIT Madras]

Q.3 Consider the following routing table at an IP router :

Network No.	Net Mask	Next Hop
128.96.170.0	255.255.254.0	Interface 0
128.96.168.0	255.255.254.0	Interface 1
128.96.166.0	255.255.254.0	R2
128.96.164.0	255.255.252.0	R3
0.0.0.0	Default	R4

For each IP address in Group I Identify the correct choice of the next hop from Group II using the entries from the routing table above.

Group I	**Group II**
(i) 128.96.171.92	(a) Interface 0
(ii) 128.96.167.151	(b) Interface 1
(iii) 128.96.164.151	(c) R2
(iv) 128.96.165.121	(d) R3
	(e) R4

(A) i-a, ii-c, iii-e, iv-d
(B) i-a, ii-d, iii-b, iv-e
(C) i-b, ii-c, iii-d, iv-e
(D) i-b, ii-c, iii-e, iv-d

[GATE 2015, IIT Kanpur]

Q.4 In the network 200.10.11.144/27, the fourth octet (in decimal) of the last IP address of the network which can be assigned to a host is ______.

(A) 158 (B) 157
(C) 156 (D) 155

[GATE 2015, IIT Kanpur]

Q.5 An IP router implementing Classless Inter-domain Routing (CIDR) receives a packet with address 131.23.151.76. The router's routing table has the following entries :

Prefix	Outer Interface Identifier
131.16.0.0/12	3
131.28.0.0/14	5
131.19.0.0/16	2
131.22.0.0/15	1

The identifier of the output interface on which this packet will be forwarded is ______.

(A) 1 (B) 2
(C) 3 (D) 4

[GATE 2014, IIT Kharagpur]

Q.6 In the IPv4 addressing format, the number of networks allowed under Class *C* addresses is:

(A) 2^{14} (B) 2^{7}

(C) 2^{21} (D) 2^{24}

[GATE 2012, IIT Delhi]

Q.7 An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it: 245.248.128.0/20. The ISP wants to give half of this chunk of addresses to Organization *A*, and a quarter to Organization *B*, while retaining the remaining with itself. Which of the following is a valid allocation of addresses to *A* and *B*?

(A) 245.248.136.0/21 and 245.248.128.0/22

(B) 245.248.128.0/21 and 245.248.128.0/22

(C) 245.248.132.0/22 and 245.248.132.0/21

(D) 245.248.136.0/24 and 245.248.132.0/21

[GATE 2012, IIT Delhi]

Q.8 Suppose computers *A* and *B* have *IP* addresses 10.105.1.113 and 10.105.1.91 respectively and they both use same subnet mask *N*. Which of the values of *N* given below should not be used if *A* and *B* should belong to the same network?

(A) 255.255.255.0

(B) 255.255.255.128

(C) 255.255.255.192

(D) 255.255.255.224

[GATE 2010, IIT Guwahati]

Q.9 If a class B network on the Internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet?

(A) 1022 (B) 1023

(C) 2046 (D) 2047

[GATE 2008, IISc Bangalore]

Q.10 Host X has IP address 192.168.1.97 and is connected through two routers R1 and R2 to another host Y with IP address 192.168.1.80. Router R1 has IP addresses 192.168.1.135 and 192.168.1.110. R2 has IP addresses 192.168.1.67 and 192.168.1.155. The subnet mask used in the network is 255.255.255.224.

Given the information above, how many distinct subnets are guaranteed to already exist in the network?

(A) 1 (B) 2

(C) 3 (D) 6

[GATE 2008, IISc Bangalore]

Q.11 Host X has IP address 192.168.1.97 and is connected through two routers R1 and R2 to another host Y with IP address 192.168.1.80. Router R1 has IP addresses 192.168.1.135 and 192.168.1.110. R2 has IP addresses 192.168.1.67 and 192.168.1.155. The subnet mask used in the network is 255.255.255.224.

Which IP address should X configure its gateway as?

(A) 192.168.1.67 (B) 192.168.1.110

(C) 192.168.1.135 (D) 192.168.1.155

[GATE 2008, IISc Bangalore]

Q.12 The address of a class B host is to be split into subnets with a 6-bit subnet number. What is the maximum number of subnets and the maximum number of hosts in each subnet?

(A) 62 subnets and 262142 hosts.

(B) 64 subnets and 262142 hosts.

(C) 62 subnets and 1022 hosts.

(D) 64 subnets and 1024 hosts.

[GATE 2007, IIT Kanpur]

Q.13 Two computers C1 and C2 are configured as follows. C1 has IP address 203.197.2.53 and netmask 255.255.128.0. C2 has IP address 203.197.75.201 and netmask 255.255.192.0. Which one of the following statements is true?

(A) C1 and C2 both assume they are on the same network

(B) C2 assumes C1 is on same network, but C1 assumes C2 is on a different network

(C) C1 assumes C2 is on same network, but C2 assumes C1 is on a different network

(D) C1 and C2 both assume they are on different networks.

[GATE 2006, IIT Kharagpur]

Q.14 A router uses the following routing table :

Destination	Mask	Interface
144.16.0.0	255.255.0.0	eth0
144.16.64.0	255.255.224.0	eth1
144.16.68.0	255.255.255.0	eth2
144.16.68.64	255.255.255.224	eth3

A packet bearing a destination address 144.16.68.117 arrives at the router. On which interface will it be forwarded?

(A) eth0 (B) eth1

(C) eth2 (D) eth3

[GATE 2006, IIT Kharagpur]

Q.15 A subnetted Class B network has the following broadcast address: 144.16.95.255 its subnet mask

(A) is necessarily 255.255.224.0

(B) is necessarily 255.255.240.0

(C) is necessarily 255.255.248.0

(D) could be any one of 255.255.224.0, 255.255.240.0,255.255.248.0

[GATE 2006, IIT Kharagpur]

Q.16 An organization has a class *B* network and wishes to form subnets for 64 departments. The subnet mask would be :

(A) 255.255.0.0 (B) 255.255.64.0

(C) 255.255.128.0 (D) 255.255.252.0

[GATE 2005, IIT Bombay]

Q.17 A company has a class *C* network address of 204.204.204.0. It wishes to have three subnets, one with 100 hosts and two with 50 hosts each. Which one of the following options represents a feasible set of subnet address/subnet mask pairs?

(A) 204.204.204.128/255.255.255.192
204.204.204.0/255.255.255.128
204.204.204.64/255.255.255.128

(B) 204.204.204.0/255.255.255.192
204.204.204.192/255.255.255.128
204.204.204.64/255.255.255.128

(C) 204.204.204.128/255.255.255.128
204.204.204.192/255.255.255.192
204.204.204.224/255.255.255.192

(D) 204.204.204.128/255.255.255.128
204.204.204.64/255.255.255.192
204.204.204.0/255.255.255.192

[GATE 2005, IIT Bombay]

Q.18 The routing table of a router is shown below :

Destination	Subnet Mask	Interface
128.75.43.0	255.255.255.0	Eth0
128.75.43.0	255.255.255.128	Eth1
192.12.17.5	255.255.255.225	Eth3
Default		Eth2

On which interface will the router forward packets addressed to destinations 128.75.43.16 and 192.12.17.10 respectively?

(A) Eth1 and Eth2 (B) Eth0 and Eth2

(C) Eth0 and Eth3 (D) Eth1 and Eth3

[GATE 2004, IIT Delhi]

Q.19 A subnet has been assigned a subnet mask of 255.255.255.192. What is the maximum number of hosts that can belong to this subnet?

(A) 14 (B) 30

(C) 62 (D) 126

[GATE 2004, IIT Delhi]

Q.20 A host is connected to a Department network which is part of a University network. The University network, in turn, is part of the Internet. The largest network in which the Ethernet address of the host is unique is

(A) the subnet to which the host belongs

(B) the department network

(C) the university network

(D) the internet

[GATE 2004, IIT Delhi]

Q.21 The subnet mask for a particular network is 255.255.31.0. Which of the following pairs of IP addresses could belong to this network?

(A) 172.57.88.62 and 172.56.87.23

(B) 10.35.28.2 and 10.35.29.4

(C) 191.203.31.87 and 191.234.31.88

(D) 128.8.129.43 and 128.8.161.55

[GATE 2003, IIT Madras]

Q.22 Match the following :

List-I

(a) 200.10.192.100

(b) 7.10.230.1

(c) 128.1.1.254

(d) 255.255.235.255

(e) 100.255.255.255

List-II

(i) Class A

(ii) Limited broadcast address

(iii) Directed broadcast address

(iv) Class C

(v) Class B

Q.23 A device has two or more IP addresses, the device is called

(A) Work station (B) Router

(C) Gateway (D) All the above

Q.24 A host with IP address 200.100.1.1 wants to send a packet to all hosts in the same network. What is SIP and DIP?

Q.25 A host with IP address 10.100.100.100 wants to use Loop testing. What is SIP and DIP?

Q.26 How many bits are allocated for NID and HID in 23.192.157.234 address

Q.27 What is not true about subnetting?

(A) It is applied for single network.

(B) It is used to improve security.

(C) Bits are borrowed from network portion.

(D) Bits are borrowed from last portion.

Q.28 What is not true about supernetting.

(A) It is used to improved security.

(B) It is applicable for two or more networks.

(C) Bits are borrowed from network portion.

(D) It is used to improve flexibility of IP address allotment.

Q.29 Which devices can use logical addressing system?

(A) Hub (B) Switch

(C) Bridge (D) Router

Q.30 Consider two hosts P and Q connected through a router R. The maximum transfer unit (MTU) value of the link between P and R is 1500 bytes, and between R and Q is 820 bytes.

A TCP segment of size 1400 bytes was transferred from P to Q through R, with IP identification value as 0x1234. Assume that the IP header size is 20 bytes. Further, the packet is allowed to be fragmented, i.e., Don't Fragment (DF) flag in the IP header is not set by P.

Which of the following statements is/are correct?

(A) Two fragments are created at R and the IP datagram size carrying the second fragment is 620 bytes.

(B) If the second fragment is lost, R will resend the fragment with the IP identification value 0x1234.

(C) If the second fragment is lost, P is required to resend the whole TCP segment.

(D) TCP destination port can be determined by analysing only the second fragment.

[GATE 2021, IIT Bombay]

Q.31 Consider the following statements about the functionality of an IP based router.

I. A router does not modify the IP packets during forwarding.

II. It is not necessary for a router to implement any routing protocol.

III. A router should reassemble IP fragments if the MTU of the outgoing link is larger than the size of the incoming IP packet.

Which of the above statements is/are TRUE?

(A) I and II only (B) II only

(C) I only (D) II and III only

[GATE 2020, IIT Delhi]

Q.32 Consider an IP packet with a length of 4,500 bytes that includes a 20-byte IPv4 header and 40-byte TCP header. The packet is forwarded to an IPv4 router that supports a Maximum Transmission Unit (MTU) of 600 bytes. Assume that the length of the IP header in all the outgoing fragments of this packet is 20 bytes. Assume that the fragmentation offset value stored in the first fragment is 0.

The fragmentation offset value stored in the third fragment is ______.

(A) 144 (B) 145

(C) 146 (D) 147

[GATE 2018, IIT Guwahati]

Q.33 The maximum number of IPv4 router addresses that can be listed in the record route (RR) option field of an IPv4 header is ______.

(A) 9 (B) 10
(C) 11 (D) 12

[GATE 2017, IIT Roorkee]

Q.34 An IP datagram of size 1000 bytes arrives at a router. The router has to forward this packet on a link whose MTU (maximum transmission unit) is 100 bytes. Assume that the size of the IP header is 20 bytes.

The number of fragments that the IP datagram will be divided into for transmission is ______.

(A) 13 (B) 14
(C) 15 (D) 16

[GATE 2016, IISc Bangalore]

Q.35 Which of the following fields of an IP header is NOT modified by a typical IP router?

(A) Check sum
(B) Source address
(C) Time to Live (TTL)
(D) Length

[GATE 2015, IIT Kanpur]

Q.36 Host A (on TCP/IP v4 network A) sends an IP datagram D to host B (also on TCP/IP v4 network B). Assume that no error occurred during the transmission of D. When D reaches B, which of the following IP header field(s) may be different from that of the original datagram D?

i. TTL ii. Checksum
iii. Fragment offset

(A) i only (B) i and ii only
(C) ii and iii only (D) i, ii and iii

[GATE 2014, IIT Kharagpur]

Q.37 Every host in an IPv4 network has a 1 second resolution real-time clock with battery backup. Each host needs to generate up to 1000 unique identifiers per second. Assume that each host has a globally unique IPv4 address. Design a 50–bit globally unique ID for this purpose. After what period (in seconds) will the identifiers generated by a host wrap around?

(A) 256 (B) 257
(C) 258 (D) 259

[GATE 2014, IIT Kharagpur]

Q.38 An IP router with a Maximum Transmission Unit (MTU) of 1500 bytes has received an IP packet of size 4404 bytes with an IP header of length 20 bytes. The values of the relevant fields in the header of the third IP fragment generated by the router for this packet are :

(A) MF bit : 0, Datagram Length : 1444; Offset : 370
(B) MF bit : 1, Datagram Length : 1424; Offset : 185
(C) MF bit : 1, Datagram Length : 1500; Offset : 370
(D) MF bit : 0, Datagram Length : 1424; Offset : 2960

[GATE 2014, IIT Kharagpur]

Q.39 In an IPv4 datagram, the M bit is 0, the value of HLEN is 10, the value of total length is 400 and the fragment offset value is 300. The position of the datagram, the sequence numbers of the first and the last bytes of the payload, respectively are :

(A) Last fragment, 2400 and 2789
(B) First fragment, 2400 and 2759
(C) Last fragment, 2400 and 2759
(D) Middle fragment, 300 and 689

[GATE 2013, IIT Bombay]

Q.40 One of the header fields in an IP datagram is the Time-to-Live (TTL) field. Which of the following statements best explains the need for this field?

(A) It can be used to prioritize packets.
(B) It can be used to reduce delays.
(C) It can be used to optimize throughput.
(D) It can be used to prevent packet looping.

[GATE 2010, IIT Guwahati]

Q.41 For which one of the following reasons does internet protocol (IP) use the time-to-live (TTL) field in IP datagram header?

(A) Ensure packets reach destination within that time
(B) Discard packets that reach later than that time
(C) Prevent packets from looping indefinitely
(D) Limit the time for which a packet gets queued in intermediate routers

[GATE 2006, IIT Kharagpur]

Q.42 Which of the following statements is TRUE?
(A) Both Ethernet frame and IP packet include checksum fields
(B) Ethernet frame includes a checksum field and IP packet includes a CRC field
(C) Ethernet frame includes a CRC field and IP packet includes a checksum field
(D) Both Ethernet frame and IP packet include CRC fields

Common Data Question 43 & 44

Consider three IP networks A, B and C. Host H_A in network A sends messages each containing 180 bytes of application data to a host H_C in network C. The TCP layer prefixes a 20 byte header to the message. This passes through an intermediate network B. The maximum packet size, including 20 byte IP header, in each network is :

A: 1000 bytes

B: 100 bytes

C: 1000 bytes

The networks A and B are connected through a 1 Mbps link, while B and C are connected by a 512 Kbps link (bps = bits per second).

Q.43 Assuming that the packets are correctly delivered, how many bytes, including headers, are delivered to the IP layer at the destination for one application message, in the best case? Consider only data packets.
(A) 200 (B) 220
(C) 240 (D) 260

[GATE 2004, IIT Delhi]

Q.44 What is the rate at which application data is transferred to host H_C? Ignore errors, acknowledgements and other overheads?
(A) 325.5 Kbps (B) 354.5 Kbps
(C) 409.6 Kbps (D) 512.0 Kbps

[GATE 2004, IIT Delhi]

Q.45 In the TCP/IP protocol suite, which one of the following is NOT part of the IPv4 header?
(A) Fragment Offset
(B) Source IP address
(C) Destination IP address
(D) Destination port number

[GATE 2004, IIT Delhi]

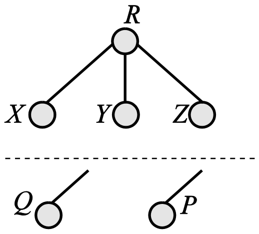
Q.46 A TCP message consisting of 2100 bytes is passed to IP for delivery across two networks. The first network can carry a maximum payload of 1200 bytes per frame and the second network can carry a maximum payload of 400 bytes per frame, excluding network overhead. Assume that IP overhead per packet is 20 bytes. What is the total IP overhead in the second network for this transmission?
(A) 40 bytes (B) 80 bytes
(C) 120 bytes (D) 160 bytes

[GATE 2004, IIT Delhi]

Q.47 Which of the following assertions is FALSE about the Internet Protocol (IP)?
(A) It is possible for a computer to have multiple IP addresses
(B) IP packets from the same source to the same destination can take different routes in the network
(C) IP ensures that a packet is discarded if it is unable to reach its destination within a given number of hops
(D) The packet source cannot set the route of an outgoing packets; the route is determined only by the routing tables in the routers on the way.

[GATE 2003, IIT Madras]

Q.48 Consider a computer network using the distance vector routing algorithm in its network layer. The partial topology of the network is as shown below.



The objective is to find the shortest cost path from the router R to router P and Q. Assume that R does not initially know the shortest routes to P and Q. Assume that R has three neighbouring routers denoted as X, Y and Z. During one iteration, R measures its distance to its neighbours X, Y and Z as 3, 2 and 5, respectively. Router R gets routing vectors from its neighbors that indicate that the distance to router P from routers X, Y and Z are 7, 6 and 5 respectively. The routing vector also indicates that the distance to router Q from routers X, Y and Z are 4, 6 and 8 respectively. Which of the following statement (s) is/are correct with respect to the new routing table of R, after updation during this iteration?

(A) The distance from R to P will be stored as 10.

(B) The distance form R to Q will be stored as 7.

(C) The next lop router for a packet from R to P is Y.

(D) The next lop router for a packet from R to Q is Z.

Q.49 Consider the following statements about the routing protocols. Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) in an IPv4 network.

I. RIP uses distance vector routing

II. RIP packets are sent using UDP

III. OSPF packets are sent using TCP

IV. OSPF operation is based on link-state routing

Which of the above statements are CORRECT?

(A) I and IV only

(B) I, II and III only

(C) I, II and IV only

(D) II, III and IV only

[GATE 2017, IIT Roorkee]

Q.50 Host A sends a UDP datagram containing 8880 bytes of user data to host B over an Ethernet LAN. Ethernet frames may carry data up to 1500 bytes (i.e. MTU = 1500 bytes). Size of UDP header is 8 bytes and size of IP header is 20 bytes. There is no option field in IP header. How many total number of IP fragments will be transmitted and what will be the contents of offset field in the last fragment?

(A) 6 and 925 (B) 6 and 7400

(C) 7 and 1110 (D) 7 and 8880

[GATE 2015, IIT Kanpur]

Q.51 Consider the following three statements about link state and distance vector routing protocols, for a large network with 500 network nodes and 4000 links.

[S_1] : The computational overhead in link state protocols is higher than in distance vector protocols.

[S_2] : A distance vector protocol (with split horizon) avoids persistent routing loops, but not a link state protocol.

[S_3] : After a topology change, a link state protocol will converge faster than a distance vector protocol.

Which one of the following is correct about S_1, S_2, and S_3?

(A) S_1, S_2, and S_3 are all true.

(B) S_1, S_2, and S_3 are all false.

(C) S_1 and S_2 are true, but S_3 is false.

(D) S_1 and S_3 are true, but S_2 is false.

[GATE 2014, IIT Kharagpur]

Q.52 Which of the following is TRUE about the interior gateway routing protocols – Routing Information Protocol (RIP) and Open Shortest Path First (OSPF)

(A) RIP uses distance vector routing and OSPF uses link state routing

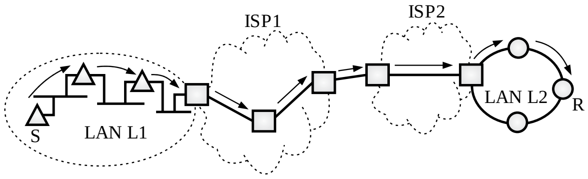
(B) OSPF uses distance vector routing and RIP uses link state routing

(C) Both RIP and OSPF use link state routing

(D) Both RIP and OSPF use distance vector routing

[GATE 2014, IIT Kharagpur]

Q.53 In the diagram shown below, L1 is an Ethernet LAN and L2 is a Token-Ring LAN. An IP packet originates from sender S and traverses to R, as shown. The links within each ISP and across the two ISPs, are all point-to-point optical links. The initial value of the TTL field is 32. The maximum possible value of the TTL field when R receives the datagram is ______.

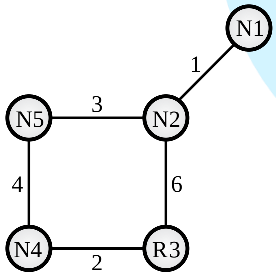


(A) 26 (B) 27
(C) 28 (D) 29

[GATE 2015, IIT Kanpur]

Common Data Question 54 & 55

Consider a network with five nodes, N1 to N5, as shown as below.



The network uses a Distance Vector Routing protocol. Once the routes have been stabilized, the distance vectors at different nodes are as follows.

N1 : (0,1,7,8,4)
N2 : (1,0,6,7,3)
N3 : (7,6,0,2,6)
N4 : (8,7,2,0,4)
N5 : (4,3,6,4,0)

Each distance vector is the distance of the best known path at that instance to nodes, N1 to N5, where the distance to itself is 0. Also, all links are symmetric and the cost is identical in both directions. In each round, all nodes exchange their distance vectors with their respective neighbors. Then all nodes update their distance vectors. In between two rounds, any change in cost of a link will cause the two incident nodes to change only that entry in their distance vectors.

Q.54 The cost of link N2-N3 reduces to 2 (in both directions). After the next round of updates, what will be the new distance vector at node, N3?

(A) 3, 2, 0, 2, 5 (B) 3, 2, 0, 2, 6
(C) 7, 2, 0, 2, 5 (D) 7, 2, 0, 2, 6

[GATE 2011, IIT Madras]

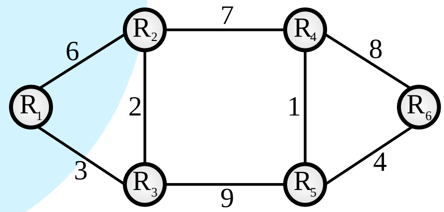
Q.55 The cost of link N2-N3 reduces to 2 (in both directions). After the next round of updates, the link N1-N2 goes down. N2 will reflect this change immediately in its distance vector as cost, ∞. After the NEXT ROUND of update, what will be the cost to N1 in the distance vector of N3?

(A) 3 (B) 9
(C) 10 (D) ∞

[GATE 2011, IIT Madras]

Common Data Question 56 & 57

Consider a network with 6 routers R1 to R6 connected with links having weights as shown in the following diagram.



Q.56 All the routers use the distance vector based routing algorithm to update their routing tables. Each router starts with its routing table initialized to contain an entry for each neighbour with the weight of the respective connecting link. After all the routing tables stabilize, how many links in the network will never be used for carrying any data?

(A) 4 (B) 3
(C) 2 (D) 1

[GATE 2010, IIT Guwahati]

Q.57 Suppose the weights of all unused links are changed to 2 and the distance vector algorithm is used again until all routing tables stabilize. How many links will now remain unused?

(A) 0 (B) 1
(C) 2 (D) 3

[GATE 2010, IIT Guwahati]

Q.58 Two popular routing algorithms are Distance Vector (DV) and Link State (LS) routing. Which of the following are true?

[S_1] : Count to infinity is a problem only with DV and not LS routing

[S_2] : In LS, the shortest path algorithm is run only at one node

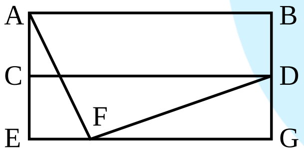
[S_3] : In DV, the shortest path algorithm is run only at one node

[S_4] : DV requires lesser number of network messages than LS

(A) S_1, S_2 and S_4 only
(B) S_1, S_3 and S_4 only
(C) S_2 and S_3 only
(D) S_1 and S_4 only

[GATE 2008, IISc Bangalore]

Q.59 For the network given in the figure below, the routing tables of the four nodes *A*, *E*, *D* and *G* are shown. Suppose that *F* has estimated its delay to its neighbours, *A*, *E*, *D* and *G* as 8, 10, 12 and 6 msecs respectively and updates its routing table using distance vector routing technique.



Routing table of A		Routing table of D		Routing table of E		Routing table of G	
A	0	A	20	A	24	A	21
B	40	B	8	B	27	B	24
C	14	C	30	C	7	C	22
D	17	D	0	D	20	D	19
E	21	E	14	E	0	E	22
F	9	F	7	F	11	F	10
G	24	G	22	G	22	G	0

(A)

A	8
B	20
C	17
D	12
E	10
F	0
G	6

(B)

A	21
B	8
C	7
D	19
E	14
F	0
G	22

(C)

A	8
B	20
C	17
D	12
E	10
F	16
G	6

(D)

A	8
B	8
C	7
D	12
E	10
F	0
G	6

[GATE 2007, IIT Kanpur]

Q.60 Count to infinity is a problem associated with

(A) link state routing protocol.
(B) distance vector routing protocol
(C) DNS while resolving host name
(D) TCP for congestion control

[GATE 2005, IIT Bombay]

Q.61 Consider the following two statements.

S_1 : Destination MAC address of an ARP reply is a broadcast address.

S_2 : Destination MAC address of an ARP request is a broadcast address.

Which one of the following choices is correct?

(A) S_1 is true and S_2 is false.
(B) Both S_1 and S_2 are true.
(C) S_1 is false and S_2 is true.
(D) Both S_1 and S_2 are false.

[GATE 2021, IIT Bombay]

Q.62 Suppose that in an IP-over-Ethernet network, a machine X wishes to find the MAC address of another machine Y in its subnet. Which one of the following techniques can be used for this?

(A) X sends an ARP request packet to the local gateway's IP address which then finds the MAC address of Y and sends to X
(B) X sends an ARP request packet to the local gateway's MAC address which then finds the MAC address of Y and sends to X
(C) X sends an ARP request packet with broadcast MAC address in its local subnet
(D) X sends an ARP request packet with broadcast IP address in its local subnet

[GATE 2019, IIT Madras]

Q.63 Which one of the following protocols is NOT used to resolve one form of address to another one?

(A) DNS (B) ARP
(C) DHCP (D) RARP

[GATE 2016, IISc Bangalore]

Q.64 The ARP is used for

(A) Finding the IP address from the DNS.

(B) Finding the IP address of the default gateway.

(C) Finding the IP address that corresponds to a MAC address.

(D) Finding the MAC address that corresponds to an IP address.

[GATE 2005, IIT Bombay]

Q.65 Trace route reports a possible route that is taken by packets moving from some host *A* to some other host *B*. Which of the following options represents the technique used by trace route to identify these hosts :

(A) By progressively querying routers about the next router on the path to *B* using ICMP packets, starting with the first router

(B) By requiring each router to append the address to the ICMP packet as it is forwarded to *B*. The list of all routers en-route to *B* is returned by *B* in an ICMP reply packet

(C) By ensuring that an ICMP reply packet is returned to *A* by each router en-route to *B*, in the ascending order of their hop distance from *A*

(D) By locally computing the shortest path from *A* to *B*

[GATE 2005, IIT Bombay]

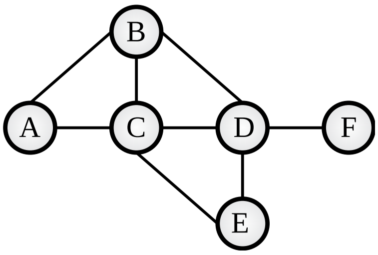
Common Data Question 66 & 67

Consider a simple graph with unit edge costs. Each node in the graph represents a router. Each node maintains a routing table indicating the next hop router to be used to relay a packet to its destination and the cost of the path to the destination through that router. Initially, the routing table is empty. The routing table is synchronously updated as follows. In each updated interval, three tasks are performed.

i. A node determines whether its neighbours in the graph are accessible. If so, it sets the tentative cost to each accessible neighbour as 1. Otherwise, the cost is set to ∞.

ii. From each accessible neighbour, it gets the costs to relay to other nodes via that neighbour (as the next hop).

iii. Each node updates its routing table based on the information received in the previous two steps by choosing the minimum cost.



Q.66 For the graph given above, possible routing tables for various nodes after they have stabilized, are shown in the following options. Identify the correct table.

(A)

Table for node A		
A	–	–
B	B	1
C	C	1
D	B	3
E	C	3
F	C	4

(B)

Table for node C		
A	A	1
B	B	1
C	–	–
D	D	1
E	E	1
F	E	3

(C)

Table for node B		
A	A	1
B	–	–
C	C	1
D	D	1
E	C	2
F	D	2

(D)

Table for node D		
A	B	3
B	B	1
C	C	1
D	–	–
E	E	1
F	F	1

[GATE 2005, IIT Bombay]

Q.67 Continuing from the earlier problem, suppose at some time *t*, when the costs have stabilized, node *A* goes down. The cost from node *F* to node *A* at time $(t+100)$ is :

(A) > 100 but finite

(B) ∞

(C) 3

(D) > 3 and ≤ 100

[GATE 2005, IIT Bombay]

Practice Questions

Q.1 Range of IP Address from 224.0.0.0 to 239.255.255.255 are

(A) Reserved for loopback

(B) Reserved for broadcast

(C) Used for multicast packets

(D) Reserved for future addressing

Q.2 On a LAN, where are IP datagrams transported?

(A) In the LAN header

(B) In the application field

(C) In the information field of the LAN frame

(D) After the TCP header

Q.3 Which of the following transmission media is not readily suitable to CSMA operation?

(A) Radio (B) Optical fibers

(C) Coaxial cable (D) Twisted pair

Q.4 The broadcast address for IP network 172.16.0.0 with subnet mask 255.255.0.0 is

(A) 172.16.0.255

(B) 172.16.255.255

(C) 255.255.255.255

(D) 172.255.255.255

Q.5 What is IP class and number of sub-networks if the subnet mask is 255.224.0.0?

(A) Class A, 3 (B) Class A, 8

(C) Class B, 3 (D) Class B, 32

Q.6 The process of modifying IP address information in IP packet headers while in transit across a traffic routing device is called

(A) Port address translation (PAT)

(B) Network address translation (NAT)

(C) Address mapping

(D) Port mapping

Q.7 An IP packet has arrived with the first 8 bits as 0100 0010. Which of the following is correct?

(A) The number of hops this packet can travel is 2

(B) The total number of bytes in header is 16 bytes

(C) The upper layer protocol is ICMP

(D) The receiver rejects the packet

Q.8 A supernet has a first address of 205.16.32.0 and a supernet mask of 255.255.248.0. A router receives 4 packets with the following destination addresses. Which packet belongs to this supernet?

(A) 205.16.42.56 (B) 205.17.32.76

(C) 205.16.31.10 (D) 205.16.39.44

Q.9 An organization is granted the block 130.34.12.64/26. It needs to have 4 subnets. Which of the following is not an address of this organization?

(A) 130.34.12.124 (B) 130.34.12.89

(C) 130.34.12.70 (D) 130.34.12.132

Q.10 Which network protocol allows hosts to dynamically get a unique IP number on each bootup

(A) DHCP (B) BOOTP

(C) RARP (D) ARP

Q.11 The address of a class B host is to be split into subnets with a 6 bit subnet number. What is the maximum number of subnets and the maximum number of hosts in each subnet?

(A) 62 subnets and 262142 hosts

(B) 64 subnets and 262142 hosts

(C) 62 subnets and 1022 hosts

(D) 64 subnets and 1024 hosts

Q.12 Which of the following statement(s) is/are false?

[S1] : ARP request is broadcast and ARP reply is unicast

[S2] : DHCP is used as resolver to map address from one form to another

(A) S1 only (B) S2 only

(C) Both (D) None of these

Q.13 Which of the following protocol uses Path Vector Routing?

(A) OSPF (B) RIP

(C) BGP (D) None of these

Q.14 A mask has created 30 subnets. What is the particular byte of the mask where the extra 1's are added?

(A) 11110000 (B) 11111000

(C) 11000000 (D) 11111100

Q.15 If we have an IP host address of 201.222.5.120 and a subnet mask of 255.255.255.240. What is the broadcast address?

(A) 201.222.5.121 (B) 201.222.5.122

(C) 201.222.5.127 (D) 201.222.5.122

Q.16 Given the address 172.16.2.120 and the subnet mask of 255.255.248.0. How many hosts are available?

Q.17 Find the mask that creates 21 subnets in class-A

(A) 255.128.0.0 (B) 255.192.0.0

(C) 255.248.0.0 (D) 255.240.0.0

Q.18 Given the mask 255.255.252.0, how many hosts per subnet does this create?

(A) 252 (B) 1024

(C) 1022 (D) None of these

Q.19 Which of the following IP addresses are considered 'network' addresses with a /28 prefix?

(A) 165.203.5.192 (B) 165.203.6.255

(C) 165.203.6.240 (D) 165.203.6.255

Q.20 A class C network address has been subnetted with a /28 mask. Which of the following addresses is a broadcast address for one of the resulting subnets?

(A) 198.57.78.33 (B) 198.57.78.64

(C) 198.57.78.97 (D) 198.57.78.15

Q.21 You are a network administrator and have been assigned the IP address of 201.222.5.0. You need to have 24 subnets with 6 hosts per subnet. The subnet mark is 255.255.255.248. What is the address of 4^{th} subnet?

(A) 201.222.5.10 (B) 201.222.5.1

(C) 201.222.5.16 (D) 201.222.5.24

Q.22 A datagram of 400 bytes (3980 bytes of IP Payload plus 20 bytes of IP Header) arrives at a router and must be forwarded to a link with MTU of 1500 bytes. What will be the offset of the second fragment?

Q.23 BGP is based on

(A) Link state routing

(B) Distance vector routing

(C) Dijkstra's Algorithm

(D) Path vector routing

Q.24 Which of the following protocol(s) is/are used for routing within an autonomous system?

(A) OSPF (B) RIP

(C) Both A & B (D) BGP

Q.25 Which one of the following hosts in any subnet of 192.168.32.0 is not valid. Assume the subnet mask used is 255.255.255.240

(A) 192.168.32.33 (B) 192.168.32.112

(C) 192.168.32.119 (D) 192.168.32.126

Q.26 You have a network ID 131.107.0.0 with 8 subnets. You need to allow the largest possible number of hosts/subnet. Which subnet mask should you use?

(A) 255.255.240.0 (B) 255.255.248.0

(C) 255.255.252.0 (D) 255.255.224.0

Q.27 A router has the following CIDR entries in the routing table

Address/mask	Next hop
135.46.56.0/22	Interface 0
135.46.58.0/22	Interface 1
135.53.40.0/23	Router 1
Default	Router 2

What is the next hop the router routes to, if a packet with the IP address 135.46.63.10 arrives at it?

(A) Interface 0 (B) Interface 1

(C) Router 1 (D) Router 2

Q.28 IP Address : 124.133.112.66

Mask : 255.255.224.0

Which of the following is not a broadcast address of the subnet?

(A) 124.133.255.255

(B) 124.133.63.255

(C) Both (A) and (B)

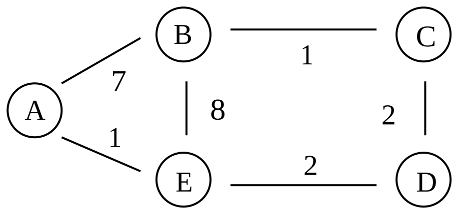
(D) None of these

Q.29 A datagram of 4000 bytes (20 bytes of IP Header + 3980 bytes of IP Payload) arrives at a router and must be forwarded the link with MTU of 1500 bytes then at what byte the 2^{nd} fragment is ended?

(A) 2960 (B) 1480

(C) 2959 (D) 1479

Q.30 Consider the following graph:



The initial state of routing table is

	A	B	C	D	E
A	0	7	∞	∞	1
B	7	0	1	∞	8
C	∞	1	0	2	∞
D	∞	∞	2	0	2
E	1	8	∞	2	0

If Node B sends vector table to C, then what is updated table result at node C (Distance vector rating protocol)?

(A) (8, 1, 0, 2, 1) (B) (8, 1, 0, 2, 9)

(C) (8, 1, 0, 4, 2) (D) None of these

Q.31 An IPv4 packet has arrived with offset value (60) and value of HLEN field $(10)_{10}$. The value of total length field is $(250)_{10}$. What are the first and last byte numbers respectively?

(A) 60 and 310 (B) 480 and 689

(C) 60 and 289 (D) 480 and 309

Q.32 An IPv4 packet has the first few hexadecimal digits as shown below.

0X450000 5C 000 30000 9606.....

How many hops can this packet travel before being dropped?

(A) 105 (B) 150

(C) 95 (D) None of these

Q.33 The IP Address of a host is 201.15.130.137 and its subnet mask is 255.255.255.224. The IP address of the last host of the same subnet is

(A) 201.15.130.143 (B) 201.15.130.158

(C) 201.15.131.192 (D) 201.12.130 .16 3

Q.34 What will be the valid broadcast address's fourth byte for subnet 172.16.176.0/20?

(A) 255 (B) 254

(C) 252 (D) 250

Q.35 Suppose a subnet x has a subnet mask 255.255.192.0 and a system A has IP address 157.106.46.234. Which of the following belongs to same network as A?

(A) 157.106.63.3

(B) 157.106.132.71

(C) Both (A) and (B)

(D) None of these

Q.36 Find total number of networks in class B IP-Addressing?

(A) 2^{16} (B) 2^{14}

(C) 2^{24} (D) 2^{12}

Q.37 In a network with broadcast ID 200.200.200.159. What is the possible subnet mask in the same network?

(A) /24 (B) /25

(C) /26 (D) /27

Q.38 Which of the following subnet mask can't be used if two host H(A): x.y.z.155 and H(B): x.y.z.162 belongs to the same network?

(A) /24 (B) /25

(C) /26 (D) /27

Q.39 Which one of the following field does not change while movement of IP-Packet?

(A) Total Length

(B) Identification No.

(C) TTL

(D) Checksum

Q.40 You need 500 subnets, each with 100 usable host address per subnet.

What network mask will you assign using a class B network address?

(A) 255.255.255.252

(B) 255.255.255.128

(C) 255.255.255.0

(D) 255.255.254.0

Q.41 The four byte IP-address consist of
(A) Network Address only
(B) Host address only
(C) Neither Network nor Host address
(D) Both Network and Host Address

Q.42 What is the slash notation (/n) of subnet mask 255.255.224.0?
(A) /16 (B) /18
(C) /20 (D) /19

Q.43 In which class the IP-address 239.175.56.0 belongs?
(A) Class A (B) Class B
(C) Class C (D) Class D

Q.44 Which of the following is the last IP Address of class B which can be assigned to a Host?
(A) x.y.z.254 (B) x.y.z.255
(C) x.y.255.254 (C) x.y.255.255

Q.45 Which of the following IP-Address are NOT used for Private Network?
(A) 10.0.0.0 (B) 172.16.0.0
(C) 128.0.0.0 (D) 192.168.0.0

Q.46 Which of the following is NOT address resolution technique?
(A) DHCP (B) DNS
(C) ARP (D) RARP

Q.47 Which of the following statement is FALSE regarding Distance Vector (DV) and Link State (LS) routing protocols?
(A) In DV, every nodes share it's routing table with it's neighbor periodically.
(B) In LS, every node build it's own minimum spanning tree (MST).
(C) In DV, every node broadcast it's routing table to get distance of other.
(D) In LS, every node broadcast it's query message to get distance of other.

Q.48 Suppose a Host(A) with IP-address 200.200.200.170 belongs to a network with subnet mask 255.255.255.224.
What is the fourth acted of network ID in which Host(A) belongs ?

Q.49 Which one of the following field of IP Header may not change on visiting each router?
(A) Checksum (B) TTL
(C) Offset (D) None

Q.50 Which of the following assertion is FALSE about Internet Protocol (IP)?
(A) IP Packet from source to destination can take different route in the path.
(B) The checksum filed in IP Header help to detect error in header only not in data.
(C) The length of IP Packet remains same throughout it's journey.
(D) TTL inside IP Header prevents it to goes into infinite loop.

Q.51 A IP Packet of size 1000 bytes is visiting to a router having maximum transmission unit (MTU) is 200 byes without header. What is the maximum overhead inside IP Packet if size of network header is 20 bytes (Bytes)?

Q.52 An ISP has the following CIDR based IP-address available with it: 200.200.128.0/20.The ISP wants to give half of this IP- address to Org-A and quarter to Org-B. If first IP-Address will be assigned to a network which consumes more number of IP-address, then what is possible value of 3rd octed of Org-B ________?

Q.53 Suppose Host A wants to send a packet of size 5020 bytes at network layer to Host B. The maximum transmission unit (MTU) of underlying layer of Host A is 1400 bytes without header. If size of header at network layer is 20 bytes, then what is the offset of 3rd fragment that Host B will receive?

Q.54 Suppose an organization wants to create sub-network containing 35,25,10 hosts in each sub-networks. What is the maximum length of subnet mask that organization should use?
(A) /22 (B) /24
(C) /25 (D) /26

Q.55 An IP datagram of size 2000 bytes arrives at a router. The router has to forward this packet on a link whose MTU is 300 bytes. Assume size of IP-Header is 20 bytes. The number of fragments that IP datagram will be divided into transmission is?

Q.56 Which of the following statement is NOT TRUE about Distance Vector Routing (DVR) Protocol?

(A) Every node share it's routing table periodically with it's immediate neighbor if there is a change.

(B) It uses RIP

(C) Count to infinity problems always occur in DVR

(D) Count to infinity problems occurs if there a link failure.

Q.57 Which of the following assertion is NOT TRUE about Internet Control Message Protocol (ICMP)?

(A) It uses error reporting message to find whether router or host is alive or not.

(B) It uses query message to get specific information from router or host.

(C) The debugging tools like Ping or tracert uses ICMP packet.

(D) ICMP packet is used to get distance of another node.

Q.58 Which one of the following protocols is NOT implemented in Transport Layer?

(A) Slow Start (B) UDP

(C) ARP (D) Fast Retransmit

❑❑❑

4 Transport Layer

Classroom Questions

Q.1 A TCP server application is programmed to listen on port number P on host S. A TCP client is connected to the TCP server over the network.

Consider that while the TCP connection was active, the server machine S crashed and rebooted. Assume that the client does not use the TCP keepalive timer.

Which of the following behaviors is/are possible?

(A) The TCP server application on S can listen on P after reboot.

(B) If the client was waiting to receive a packet, it may wait indefinitely.

(C) If the client sends a packet after the server reboot, it will receive a RST segment.

(D) If the client sends a packet after the server reboot, it will receive a FIN segment.

[GATE 2021, IIT Bombay]

Q.2 Consider the three-way handshake mechanism followed during TCP connection establishment between hosts P and Q. Let X and Y be two random 32-bit starting sequence numbers chosen by P and Q respectively. Suppose P sends a TCP connection request message to Q with a TCP segment having SYN bit = 1, SEQ number = X, and ACK bit = 0. Suppose Q accepts the connection request. Which one of the following choices represents the information present in the TCP segment header that is sent by Q to P?

(A) SYN bit = 1 , SEQ number = X+1 , ACK bit =0 , ACK number =Y, FIN bit =0

(B) SYN bit = 1 , SEQ number = Y, ACK bit = 1 , ACK number = X, FIN bit = 0

(C) SYN bit = 1 , SEQ number = Y, ACK bit = 1 , ACK number = X+1, FIN bit = 0

(D) SYN bit = 0 , SEQ number = X+ 1, ACK bit = 0 , ACK number = Y, FIN bit = 1

[GATE 2021, IIT Bombay]

Q.3 Consider a TCP connection between a client and a server with the following specifications; the round trip time is 6 ms, the size of the receiver advertised window is 50 KB, slow-start threshold at the client is 32 KB, and the maximum segment size is 2 KB. The connection is established at time t = 0. Assume that there are no timeouts and errors during transmission. Then the size of the congestion window (in KB) at time t + 60 ms after all acknowledgements are processed is ________.

[GATE 2020, IIT Delhi]

Q.4 Consider a long-lived TCP session with an end-to-end bandwidth of 1 Gbps (= 10^9 bits-per-second). The session starts with a sequence number of 1234. The minimum time (in seconds, rounded to the closet integer) before this sequence number can be used again is ______.

(A) 33 (B) 34

(C) 35 (D) 36

[GATE 2018, IIT Guwahati]

Q.5 Match the following :

Field		Length in bits	
P.	UDP Header's Port Number	I.	48
Q.	Ethernet MAC Address	II.	8
R.	IPv6 Next Header	III.	32
S.	TCP Header's Sequence Number	IV.	16

(A) P-III, Q-IV, R-II, S-I
(B) P-II, Q-I, R-IV, S-III
(C) P-IV, Q-I, R-II, S-III
(D) P-IV, Q-I, R-III, S-II

[GATE 2018, IIT Guwahati]

Q.6 Consider the following statements regarding the slow start phase of the TCP congestion control algorithm. Note that CWND stands for the TCP congestion window and MSS window denotes the Maximum Segments Size:

(i) The CWND increases by 2 MSS on every successful acknowledgment
(ii) The CWND approximately doubles on every successful acknowledgement
(iii) The CWND increases by 1 MSS every round trip time
(iv) The CWND approximately doubles every round trip time

Which one of the following is correct?

(A) Only (ii) and (iii) are true
(B) Only (i) and (iii) are true
(C) Only (iv) is true
(D) Only (i) and (iv) are true

[GATE 2018, IIT Guwahati]

Q.7 Consider a TCP client and a TCP server running on two different machines. After completing data transfer, the TCP client calls close to terminate the connection and a FIN segment is sent to the TCP server. Server-side TCP responds by sending an ACK, which is received by the client-side TCP. As per the TCP connection state diagram (RFC 793), in which state does the client-side TCP connection wait for the FIN from the server-side TCP?

(A) LAST-ACK (B) TIME-WAIT
(C) FIN-WAIT-1 (D) FIN-WAIT-2

[GATE 2017, IIT Roorkee]

Q.8 Consider socket API on a Linux machine that supports connected UDP sockets. A connected UDP socket is a UDP socket on which connect function has already been called. Which of the following statements is/are CORRECT?

I. A connected UDP socket can be used to communicate with multiple peers simultaneously.
II. A process can successfully call connect function again for an already connected UDP socket.

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

[GATE 2017, IIT Roorkee]

Q.9 For a host machine that uses the token bucket algorithm for congestion control, the token bucket has a capacity of 1 megabyte and the maximum output rate is 20 megabytes per second. Tokens arrive at a rate to sustain output at a rate of 10 megabytes per second. The token bucket is currently full and the machine needs to send 12 megabytes of data. The minimum time required to transmit the data is _______ seconds.

(A) 1.1 sec (B) 1.2 sec
(C) 1.3 sec (D) 1.4 sec

[GATE 2016, IISc Bangalore]

Q.10 Suppose two hosts use a TCP connection to transfer a large file. Which of the following statements is/are FALSE with respect to the TCP connection?

I. If the sequence number of a segment is m, then the sequence number of the subsequent segment is always m+1.
II. If the estimated round trip time at any given point of time is t sec, the value of the retransmission timeout is always set to greater than or equal to t sec.
III. The size of the advertised window never changes during the course of the TCP connection.
IV. The number of unacknowledged bytes at the sender is always less than or equal to the advertised window.

(A) III only (B) I and III only
(C) I and IV only (D) II and IV only

[GATE 2015, IIT Kanpur]

Q.11 Identify the correct order in which a server process must invoke the function calls accept, bind, listen, and recv according to UNIX socket API.

(A) listen, accept, bind, recv
(B) bind, listen, accept, recv
(C) bind, accept, listen, recv

(D) accept, listen, bind, recv

[GATE 2015, IIT Kanpur]

Q.12 Consider the following statements.

I. TCP connections are full duplex

II. TCP has no option for selective acknowledgement

III. TCP connections are message streams

(A) Only I is correct

(B) Only I and III are correct

(C) Only II and III are correct

(D) All of I, II and III are correct

[GATE 2015, IIT Kanpur]

Q.13 Let the size of congestion window of a TCP connection be 32 KB when a timeout occurs. The round trip time of the connection is 100 msec and the maximum segment size used is 2 KB. The time taken (in msec) by the TCP connection to get back to 32 KB congestion window is ________.

(A) 1100 to 1300 (B) 1101 to 1301

(C) 1102 to 1302 (D) 1103 to 1303

[GATE 2014, IIT Kharagpur]

Q.14 Which of the following socket API functions converts an unconnected active TCP socket into a passive socket?

(A) connect (B) bind

(C) listen (D) accept

[GATE 2014, IIT Kharagpur]

Q.15 The transport layer protocols used for real time multimedia, file transfer, DNS and email, respectively are

(A) TCP, UDP, UDP and TCP

(B) UDP, TCP, TCP and UDP

(C) UDP, TCP, UDP and TCP

(D) TCP, UDP, TCP and UDP

[GATE 2013, IIT Bombay]

Q.16 Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a timeout occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.

(A) 8 MSS (B) 14 MSS

(C) 7 MSS (D) 12 MSS

[GATE 2012, IIT Delhi]

Q.17 While opening a TCP connection, the initial sequence number is to be derived using a time-of-day (ToD) clock that keeps running even when the host is down. The low order 32 bits of the counter of the ToD clock is to be used for the initial sequence numbers. The clock counter increments once per milliseconds. The maximum packet lifetime is given to be 64s.

Which one of the choices given below is closest to the minimum permissible rate at which sequence numbers used for packets of a connection can increase?

(A) 0.015/s (B) 0.064/s

(C) 0.135/s (D) 0.327/s

[GATE 2009, IIT Roorkee]

Q.18 What is the maximum size of data that the application layer can pass on to the TCP layer below?

(A) Any size

(B) 2^{16} bytes - size of TCP header

(C) 2^{16} bytes

(D) 1500 bytes

[GATE 2008, IISc Bangalore]

Q.19 Which of the following system calls results in the sending of SYN packets?

(A) socket (B) bind

(C) listen (D) connect

[GATE 2008, IISc Bangalore]

Q.20 In the slow start phase of the TCP congestion control algorithm, the size of the congestion window :

(A) does not increase

(B) increase linearly

(C) increases quadratically

(D) increases exponentially

[GATE 2008, IISc Bangalore]

Q.21 A computer on a 10 Mbps network is regulated by a token bucket. The token bucket is filled at a rate of 2 Mbps. It is initially filled to capacity with 16 Megabits. What is the maximum duration for which the computer can transmit at the full 10 Mbps?

(A) 1.6 seconds (B) 2 seconds

(C) 5 seconds (D) 8 seconds

[GATE 2008, IISc Bangalore]

Q.22 A client process P needs to make a TCP connection to a server process S. Consider the following situation: the server process S executes a socket(), a bind() and a listen() system call in that order, following which it is preempted. Subsequently, the client process P executes a socket() system call followed by connect() system call to connect to the server process S. The server process has not executed any accept() system call. Which one of the following events could take place?

(A) connect() system call returns successfully

(B) connect() system call blocks

(C) connect() system call returns an error

(D) connect() system call results in a core dump

[GATE 2008, IISc Bangalore]

Q.23 Which of the following statements are TRUE?

S1 : TCP handles both congestion and flow control

S2 : UDP handles congestion but not flow control

S3 : Fast retransmit deals with congestion but not flow control

S4 : Slow start mechanism deals with both congestion and flow control

(A) S1, S2 and S3 only

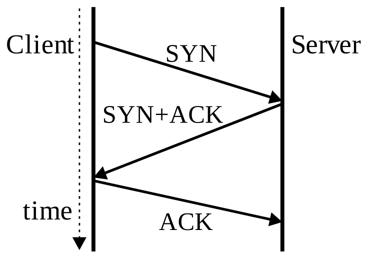
(B) S1 and S3 only

(C) S3 and S4 only

(D) S1, S3 and S4 only

[GATE 2008, IISc Bangalore]

Q.24 The three way handshake for TCP connection establishment is shown below.



Which of the following statements are TRUE?

S1 : Loss of SYN+ACK from the server will not establish a connection

S2 : Loss of ACK from the client cannot establish the connection

S3 : The server moves LISTEN→ SYN_RCVD→SYN_SENT→ESTABLISHED in the state machine on no packet loss

S4 : The server moves LISTEN→ SYN_RCVD→ESTABLISHED in the state machine on no packet loss

(A) S2 and S3 only (B) S1 and S4 only

(C) S1 and S3 only (D) S2 and S4 only

[GATE 2008, IISc Bangalore]

Q.25 Consider the following statements about the timeout value used in TCP.

(i) The timeout value is set to the RTT (Round Trip Time) measured during TCP connection establishment for the entire duration of the connection.

(ii) Appropriate RTT estimation algorithm is used to set the timeout value of a TCP connection.

(iii) Timeout value is set to twice the propagation delay from the sender to the receiver.

Which of the following choices hold?

(A) (i) is false, but (ii) and (iii) are true

(B) (i) and (iii) are false, but (ii) is true

(C) (i) and (ii) are false, but (iii) is true

(D) (i), (ii) and (iii) are false

[GATE 2007, IIT Kanpur]

Q.26 Consider a TCP connection in a state where there are no outstanding ACKs. The sender sends two segments back to back. The sequence numbers of the first and second segments are 230 and 290 respectively. The first segment was lost, but the second segment was received correctly by the receiver. Let X be the amount of data carried in the first segment (in bytes), and Y be the ACK number sent by the receiver.

The values of X and Y (in that order) are

(A) 60 and 290 (B) 230 and 291

(C) 60 and 231 (D) 60 and 230

[GATE 2007, IIT Kanpur]

Q.27 A program on machine X attempts to open a UDP connection to port 5376 on a machine Y, and a TCP connection to port 8632 on machine Z. However, there are no applications listening at the corresponding ports on Y and Z. An ICMP Port Unreachable error will be generated by

(A) Y but not Z (B) Z but not Y
(C) Neither Y nor Z (D) Both Y and Z

[GATE 2006, IIT Kharagpur]

Q.28 Packets of the same session may be routed through different paths in:

(A) TCP, but not UDP
(B) TCP and UDP
(C) UDP, but not TCP
(D) Neither TCP nor UDP

[GATE 2005, IIT Bombay]

Q.29 On a TCP connection, current congestion window size is Congestion Window = 4 KB. The window size advertised by the receiver is Advertise Window = 6 KB. The last byte sent by the sender is LastByteSent = 10240 and the last byte acknowledged by the receiver is LastByteAcked = 8192. The current window size at the sender is:

(A) 2048 bytes (B) 4096 bytes
(C) 6144 bytes (D) 8192 bytes

[GATE 2005, IIT Bombay]

Q.30 Which one of the following statements is FALSE?

(A) TCP guarantees a minimum communication rate
(B) TCP ensures in-order delivery
(C) TCP reacts to congestion by reducing sender window size
(D) TCP employs retransmission to compensate for packet loss

[GATE 2004 : IIT Delhi]

Q.31 In TCP, a unique sequence number is assigned to each

(A) byte (B) word
(C) segment (D) message

[GATE 2004, IIT Delhi]

Q.32 Suppose that the maximum transmit window size for a TCP connection is 12000 bytes. Each packet consists of 2000 bytes. At some point in time, the connection is in slow-start phase with a current transmit window of 4000 bytes. Subsequently, the transmitter receives two acknowledgments. Assume that no packets are lost and there are no time-outs. What is the maximum possible value of the current transmit window?

(A) 4000 bytes (B) 8000 bytes
(C) 10000 bytes (D) 12000 bytes

[GATE 2004, IIT Delhi]

Q.33 Which one of the following statements is FALSE?

(A) TCP guarantees a minimum communication rate
(B) TCP ensures in-order delivery
(C) TCP reacts to congestion by reducing sender window size
(D) TCP employs retransmission to compensate for packet loss

[GATE 2004, IIT Delhi]

Q.34 Assume that the bandwidth for a TCP connection is 1048560 bits/sec. Let α be the value of RTT in milliseconds (rounded off to the nearest integer) after which the TCP window scale option is needed. Let β be the maximum possible window size with window scale option. Then the values of α and β are

(A) 63 milliseconds, 65535×2^{14}
(B) 63 milliseconds, 65535×2^{16}
(C) 500 milliseconds, 65535×2^{14}
(D) 500 milliseconds, 65535×2^{16}

[GATE 2015, IIT Kanpur]

Practice Questions

Q.1 Suppose you are browsing the world wide web using a web browser and trying to access the web servers. What is the underlying protocol and port number that are being used?

(A) UDP, 80 (B) TCP, 80
(C) TCP, 25 (D) UDP, 25

Q.2 An ACK number of 1000 in TCP always means that

(A) 999 bytes have been successfully received
(B) 1000 bytes have been successfully received
(C) 1001 bytes have been successfully received
(D) None of the above

Q.3 Assuming that for a given network layer implementation, connection establishment overhead is 100 bytes and disconnection overhead is 28 bytes. What would be the minimum size of the packet the transport layer needs to keep up, if it wishes to implement a datagram service above the network needs to keep its overhead to a minimum of 12.5%. (Ignore transport layer overhead)

(A) 512 bytes (B) 768 bytes
(C) 1152 bytes (D) 1024 bytes

Q.4 A simple and reliable data transfer can be accomplished by using the 'handshake protocol'. It accomplishes reliable data transfer because for every data item sent by the transmitter _____.

Q.5 Consider "Additive Increase Multiplicative Decrease" approach of TCP congestion control. One MSS (Maximum segment size) is 512 bytes. If the TCP sender doesn't perceive the path as not congested, then what will be congestion window after 2 RTTS provided the congestion window is assigned 1MSS initially?

(A) 512 bytes (B) 2048 bytes
(C) 2560 bytes (D) 1536 bytes

Q.6 A TCP machine is sending window size of 65,535 bytes over 1Gbps channel that has a 20ms one-way latency. What is the maximum throughput?

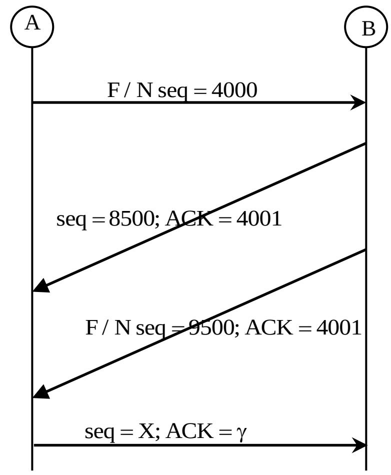
Q.7 A computer on a 6Mbps network regulated by a token bucket. The token bucket is filled at a rate of 2 Mbps. It is initially filled to capacity with 4 megabits. How long can the computer transfer at the full 4Mbps? (in seconds)

Q.8 Suppose that the TCP congestion window is set to 18KB and a timeout occurs. How big will the window be in the 6^{th} transmission if the next 4 transmission bursts are all successful? (Assume 1MSS=1KB)

Q.9 In a 4Gbps line, it takes 2 seconds to wrap around the sequence number. How many bits were used for sequence Number?

(A) 37 bits (B) 34 bits
(C) 30 bits (D) 32 bits

Q.10 TCP Connection using 3-way handshaking is shown below



Consider the following statement about this termination:

S_1: X's Value is 4000
S_2: Y's Value is 9500

Which of the above statements is/are TRUE?

(A) S_1 Only
(B) S_2 Only
(C) Both
(D) Neither S_1 and S_2

Q.11 A TCP host has receiver window size of 1 KB and a congestion window size of 2 KB. What is the window size of that host?

(A) 2 KB (B) 1 KB
(C) 3 KB (D) None of these

Q.12 Which of the following statements is/are FALSE?

S_1: TCP Provides end to end connection.
S_2: TCP ensures that data is delivered reliably in sequence and without error.

(A) S_1 Only
(B) S_2 Only
(C) Both S_1 & S_2
(D) Neither S_1 and S_2

Q.13 In TCP, sequence number is used for

(A) Rearranging the data of receiver end before passing to the application.
(B) Detecting duplicate data
(C) Both
(D) None of these.

Q.14 Which of the following statement/s is/are TRUE?

S_1: TCP is a message-oriented protocol
S_2: UDP is an unreliable connectionless protocol

(A) S_1 only (B) S_2 only
(C) Both (D) None of these

Q.15 Which of the following is a feature of UDP?

(A) Flow control (B) Error control
(C) Both A & B (D) None

Q.16 For a system using TCP, the sender window size is determined by ________ window size.

(A) Receiver (B) Congestion
(C) Both (D) None of these

Q.17 In connection establishment phase of TCP, for confirmation, the TCP header fields SYN(s), ACK(A) contain values

(A) S=0, A=0 (B) S=0, A=1
(C) S=1, A=0 (D) S=1, A=1

Q.18 The number of packets required in TCP connection establishment phase is

(A) 2 (B) 3
(C) 4 (D) 1

Q.19 In a TCP connection, current congestion window size is 2 KB. The window size advertised by the receiver is 3 KB. The last byte sent by the sender is 5120 and the last byte acknowledged by the receiver is 4096. The current window size at the sender is _____ (in KB).

Q.20 Which of the following is CORRECT?
S_1: Transport layer is responsible for process to process delivery
S_2: TCP is a connectionless practice.
S_3: UDP ensures in-order delivery

(A) S_1 only (B) S_2 only
(C) S_2 and S_3 (D) None of these

Q.21 TCP sender received an ACK with sequence number as 1024 from receiver what would be the segment number of the next segment that will be sent by the sender to receiver?

(A) 1025 (B) 1024
(C) 1023 (D) None of these

Q.22 Which of the following statement/s is/are TRUE?
S_1: In UDP, checksum is used to detect errors over both header & data
S_2: In UDP, inclusion of checksum is optional

(A) S_1 only (B) S_2 only
(C) Both (D) Neither S_1 nor S_2

Q.23 Which of the following statements is FALSE about TCP?

(A) It uses sequence number field for inorder delivery of data segments
(B) It is connection oriented protocols so all packets follow same path.
(C) It is responsible for end to end delivery of data segments
(D) It is used in an application in which reliability is must.

Q.24 Which of the following is NOT part of UDP header?

(A) Checksum
(B) Source port
(C) Sequence number
(D) Total length

Q.25 Which of the following statement is NOT True?

(A) TCP assigns sequence number to each byte
(B) TCP assign sequence number only to the first byte of segment
(C) TCP uses 32- bits for sequence number field.
(D) Byte numbering is used for flow control

Q.26 Which of the following field is NOT included in TCP Header?

(A) Sequence number
(B) Acknowledgement number
(C) Checksum
(D) Protocol

Q.27 Consider the following assertions :
S1: TCP uses checksum field for error control
S2: TCP uses sequence number field for flow control
S3: TCP window size is varying depending upon congestion of networks.
Which of the following is/are TRUE?

(A) S1, S2 (B) S2, S3
(C) S1, S3 (D) All

Q.28 Suppose size of receiver window is 20 KB and size of congestion window is 10 KB. What should be the size of sender window to maintain flow control?

(A) 30 KB

(B) 20 KB

(C) 10 KB

(D) Could be 10 KB or 20 KB

Q.29 Suppose size of current window of sender is 2MSS (Maximum segment size) and sender is using Slow Start congestion control protocol. If threshold value of networks is 8MSS then find the sender window size after 5 RTT?

Q.30 Suppose size of each segment is 2000 bytes and current sender window size is 6000 bytes. Sender received three ACK subsequently then what is the sender window size if Slow Start protocol is used?

(A) 6000 byes (B) 8000 bytes

(C) 10000 bytes (D) 12000 bytes

Q.31 Initially sender detected that size of receiver window is 6 MSS and congestion window size is 4 MSS. After one RTT, what will be the sender window size if Slow Start protocol is used (in MSS)?

Q.32 Which of the following statements is/are TRUE about UDP?

S1 : It uses three way handshaking process to established connection.

S2 : Header size of UDP packet is fixed and of 8 bytes.

S3 : It is used for application layer where reliability is not required.

(A) S1, S2 (B) S3

(C) S2, S3 (D) All

Q.33 In which of the following, the UDP protocol is NOT suitable to use?

(A) FTP (B) TFTP

(C) SNMP (C) RIP

Q.34 Suppose TCP connection is transferring file size of 5000 bytes. The first byte number is 2001. What is the sequence number of third segment if each segment carry 1000 byte?

Q.35 Consider the following assertion :

S1 : Slow start phase follows exponential growth.

S2 : Congestion avoidance phase follows additive increase.

S3 : In fast retransmit if there is loss of segment at any point of time then it moves to slow start phase again.

(A) S1, S2 (B) S2, S3

(C) S1, S3 (D) All

Q.46 Suppose sender has sent bytes upto 202. Assume Congestion window size is 20 bytes. The receiver has sent an ACK No of 200 with receiver window size of 9 bytes. What is the sequence number of last byte inside sender window?

Q.37 Suppose initial window size of sender is 1 MSS and it uses Slow Start phase with threshold value of 16 MSS. Assume that the lost is detected at 8^{th} RTT by timeout. What is the sender window size after 12^{th} RTT ________ (MSS)?

Q.38 Consider the following assertion :

S1 : In congestion control protocol the loss can be detected by timeout or by three duplicate acknowledgement.

S2 : If loss is detected by timeout then new slow start phase is started with sender window size 1MSS.

S3 : If loss is detected by three duplicate ACK then new congestion avoidance phase with sender window size half of current window size.

(A) S1, S2 (B) S1, S3

(C) S2, S3 (D) All

❑❑❑

5 Application Layer

Classroom Questions

Q.1 Which of the following protocol pairs can be used to send and retrieve e-mails (in that order)?

(A) SMTP MIME (B) SMTP, POP3

(C) IMAP POP3 (D) IMAP, SMTP

[GATE 2019, IIT Madras]

Q.2 Assume that you have made a request for a web page through your web browser to a web server. Initially the browser cache is empty. Further, the browser is configured to send HTTP requests in non-persistent mode. The web page contains text and five very small images. The minimum number of TCP connections required to display the web page completely in your browser is ________.

[GATE 2020, IIT Delhi]

Q.3 Which of the following is/are example(s) of stateful application layer protocol?

(i) HTTP (ii) FTP

(iii) TCP (iv) POP3

(A) (i) and (ii) only

(B) (ii) and (iii) only

(C) (ii) and (iv) only

(D) (iv) only

[GATE 2016, IISc Bangalore]

Q.4 Identify the correct sequence in which the following packets are transmitted on the network by a host when a browser requests a webpage from a remote server, assuming that the host has just been restarted.

(A) HTTP GET request, DNS query, TCP SYN

(B) DNS query, HTTP GET request, TCP SYN

(C) DNS query, TCP SYN, HTTP GET request

(D) TCP SYN, DNS query, HTTP GET request

[GATE 2016, IISc Bangalore]

Q.5 Which one of the following statements is NOT correct about HTTP cookies?

(A) A cookies is a piece of code that has the potential to compromise the security of an Internet user

(B) A cookie gains entry to the user's work area through an HTTP header

(C) A cookie has an expiry date and time

(D) Cookies can be used to track the browsing pattern of a user at a particular site

[GATE 2015, IIT Kanpur]

Q.6 Identify the correct order in which the following actions take place in an interaction between a web browser and a web server.

1. The web browser requests a webpage using HTTP.
2. The web browser establishes a TCP connection with the web server.
3. The web server sends the requested webpage using HTTP.
4. The web browser resolves the domain name using DNS.

(A) 4,2,1,3 (B) 1,2,3,4

(C) 4,1,2,3 (D) 2,4,1,3

[GATE 2014, IIT Kharagpur]

Q.7 The protocol data unit (PDU) for the application layer in the Internet stack is:

(A) Segment (B) Datagram

(C) Message (D) Frame

[GATE 2012, IIT Delhi]

Q.8 Which of the following transport layer protocols is used to support electronic mail?

(A) SMTP (B) IP

(C) TCP (D) UDP

[GATE 2012, IIT Delhi]

Q.9 Which one of the following is not a client server application?

(A) Internet chat (B) Web browsing
(C) E-mail (D) Ping

[GATE 2010, IIT Guwahati]

Q.10 Which one of the following uses UDP as the transport protocol?
(A) HTTP (B) Telnet
(C) DNS (D) SMTP

[GATE 2007, IIT Kanpur]

Q.11 Match the following :

P.	SMTP	1.	Application layer
Q.	BGP	2.	Transport layer
R.	TCP	3.	Data link layer
S.	PPP	4.	Network layer
		5.	Physical layer

(A) P-2, Q-1, R-3, S-5
(B) P-1, Q-4, R-2, S-3
(C) P-1, Q-4, R-2, S-5
(D) P-2, Q-4, R-1, S-3

[GATE 2007, IIT Kanpur]

Q.12 Consider the following clauses :
i. Not inherently suitable for client authentication.
ii. Not a state sensitive protocol.
iii. Must be operated with more than one server.
iv. Suitable for structured message organization.
v. May need two ports on the serve side for proper operation.

The option that has the maximum number of correct matches is
(A) IMAP-i; FTP-ii; HTTP-iii; DNS-iv; POP3-v
(B) FTP-i; POP3-ii; SMTP-iii; HTTP-iv; IMAP-v
(C) POP3-i; SMTP-ii; DNS-iii; IMAP-iv; HTTP-v
(D) SMTP-i; HTTP-ii; IMAP-iii; DNS-iv; FTP-v

[GATE 2007, IIT Kanpur]

Q.13 Which one of the following statements is FALSE?
(A) HTTP runs over TCP
(B) HTTP describes the structure of web pages
(C) HTTP allows information to be stored in a URL
(D) HTTP can be used to test the validity of a hypertext link

[GATE 2004, IIT Delhi]

Q.14 Consider the three commands : PROMPT, HEAD and RCPT. Which of the following options indicate a correct association of these commands with protocols where these are used?
(A) HTTP, SMTP, FTP
(B) FTP, HTTP, SMTP
(C) HTTP, FTP, SMTP
(D) SMTP, HTTP, FTP

[GATE 2005, IIT Bombay]

Q.15 Assume that “host1.mydomain.dom” has an IP address of 145.128.16.8. Which of the following options would be most appropriate as a subsequence of steps in performing the reverse lookup of 145.128.16.8? In the following options “NS” is an abbreviation of “nameserver”.
(A) Query a NS for the root domain and then NS for the “dom” domains
(B) Directly query a NS for “dom” and then a NS for “mydomain.dom” domains
(C) Query a NS for in-addr.arpa and then a NS for 128.145.in-addr.arpa domains
(D) Directly query a NS for 145.in-addr.arpa and then a NS for 128.145.in-addr.arpa domains

[GATE 2005, IIT Bombay]

Q.16 HELO and PORT, respectively, are commands from the protocols:
(A) FTP and HTTP
(B) TELNET and POP3
(C) HTTP and TELNET
(D) SMTP and FTP

[GATE 2006, IIT Kharagpur]

Q.17 Consider the different activities related to email.
m1 : Send an email from mail client to mail server
m2 : Download an email from mailbox server to a mail client
m3 : Checking email in a web browser
Which is the application level protocol used in each activity?
(A) m1: HTTP m2 : SMTP m3 : POP
(B) m1: SMTP m2 : FTP m3 : HTTP
(C) m1: SMTP m2 : POP m3 : HTTP
(D) m1: POP m2 : SMTP m3 : IMAP

[GATE 2011, IIT Madras]

Q.18 In one of the pairs of protocols given below, both the protocols can use multiple TCP connections between the same client and the server. Which one is that?

(A) HTTP, FTP (B) HTTP, TELNET
(C) FTP, SMTP (D) HTTP, SMTP

[GATE 2015, IIT Kanpur]

Practice Questions

Q.1 A packet filtering firewall can
(A) Deny certain users from accessing a service
(B) Block worms and viruses from entering the network
(C) Disallow some files from being accessed through FTP
(D) Block some hosts from accessing the network

Q.2 Which of the following protocol is used for transferring electronic mail messages from one machine to another?
(A) TELNET (B) FTP
(C) SNMP (D) SMTP

Q.3 Generally TCP is reliable and UDP is not reliable. DNS which has to be reliable uses UDP because
(A) UDP is slower
(B) DNS servers has to keep connections
(C) DNS requests are generally very small and fit well within UDP segments
(D) None of these

Q.4 Consider the set of activities related to e-mail

A : Send an e-mail from a mail client to mail server.

B : Download e-mail headers from mailbox and retrieve mails from server to cache.

C : Checking e-mail through a web browser.

The application level protocol used for each activity in the same sequence is
(A) SMTP, HTTPS, IMAP
(B) SMTP, POP, IMAP
(C) SMTP, IMAP, HTTPS
(D) SMTP, IMAP, POP

Q.5 A packet whose destination is outside the local TCP/IP network segment is sent to _______.
(A) File server
(B) DNS server
(C) DHCP server
(D) Default gateway

Q.6 Distance vector routing algorithm is a dynamic routing algorithm. The routing tables in distance vector routing algorithm are updated _______.
(A) automatically
(B) by server
(C) by exchanging information with neighbour nodes
(D) with back up database

Q.7 Which of the following statement(s) is/are false?

S1: DNS maps the IP address to URL address

S2: FTP uses UDP as a transport protocol

(A) S1 only (B) S2 only
(C) Both (D) None of these

Q.8 Which of the following is TRUE about DNS?
(A) DNS uses only UDP protocol.
(B) DNS uses only TCP protocol.
(C) DNS can use either TCP or UDP depending upon response of message size.
(D) None

Q.9 Which of the following are FALSE?
(A) DNS can use either TCP or UDP protocol.
(B) DNS uses well known port 53.
(C) DNS is used to find IP-address with MAC address.
(D) DNS uses TCP if response message size is less than 512 byes.

Q.10 Which of the following statement is FALSE about DNS operation?
(A) DNS is client-server architecture.
(B) DNS organizes the name space in decentralize manner.

(C) DNS matches name space from root to leaf or from leaf to root of tree.

(D) Each node in tree has domain name.

Q.11 Which of the following program is not used in internet to provide electronic mail services?

(A) SMTP (B) POP3

(C) IMAP4 (D) ICMP

Q.12 Which of the following statements is incorrect?

(A) In electronics mail, MIME allows the transfer of multimedia messages.

(B) In electronic mail, MTA transfers the mail across internet.

(C) POP3 and IMAP4 protocols are used for polling messages from mail server.

(D) None

Q.13 Consider the following statements :

S1 : WWW is repository of information.

S2 : Hypertexts are documents.

S3 : Browser interpret and display a web documents.

S4 : Common Gateway Interface (CGI) is a protocol.

(A) S1, S2, S4 (B) S1, S2, S3

(C) S1, S3, S4 (D) All

Q.14 Which of the following statements is incorrect about HTTP?

(A) HTTP uses TCP connection transfer files.

(B) HTTP is a protocol used to access world wide web (WWW)

(C) TCP uses any port number for HTTP connection.

(D) HTTP message is similar to SMTP message.

❑❑❑

Operating System

1 Process Management - I

Classroom Questions

Q.1 Which combination of the following features will suffice to characterize an OS as a multi programmed OS?

i. More than one program may be loaded into main memory at the same time for execution.

ii. If a program waits for certain events such as I/O another program is immediately scheduled for execution.

iii. If the execution of a program terminates, another program is immediately scheduled for execution.

(A) (i) (B) (i) and (ii)

(C) (i) and (iii) (D) (i), (ii) and (iii).

[GATE 2002, IISc Bangalore]

Q.2 What is the objective of multiprogramming?

(A) Have a process running at all time

(B) Have multiple programs waiting in a queue ready to run

(C) To increase CPU utilization

(D) None of the mentioned

Q.3 In operating system, each process has its own _______

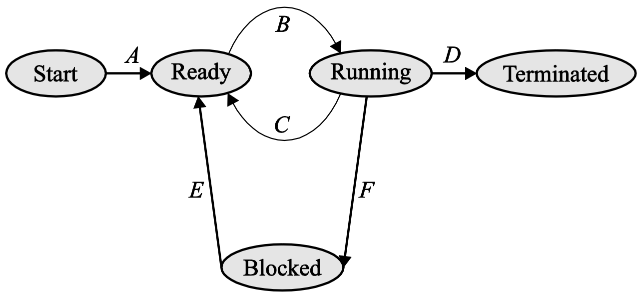
(A) Address space and global variables

(B) List of open files

(C) Process Control Block

(D) all of the mentioned

Q.4 In the following process state transition diagram for a uniprocessor system, assume that there are always some processes in the ready state:



Now consider the following statements:

I. If a process makes a transition D, it would result in another process making transition A immediately.

II. A process P2 in blocked state can make transition E while another process P1 is in running state.

III. The OS uses preemptive scheduling.

IV. The OS uses non-preemptive scheduling.

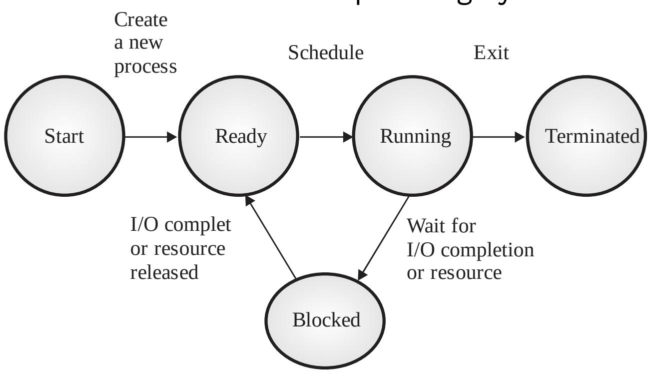
Which of the above statements are true?

(A) I. and II. (B) I. and III.

(C) II and III. (D) II. and IV.

[GATE 2009 : IIT Roorkee]

Q.5 The process state transition diagram of an operating system is as given below. Which of the following must be FALSE about the above operating system?



(A) It is a multiprogrammed operating system

(B) It uses preemptive scheduling

(C) It uses non-preemptive scheduling

(D) It is a multi-user operating system

[GATE 2006 : IIT Kharagpur]

Q.6 Which of the following does not interrupt a running process?

(A) A device

(B) Timer

(C) Scheduler process

(D) Power failure

[GATE 2001 : IIT Kanpur]

Q.7 System calls are usually invoked by using :
(A) A software interrupt
(B) Polling
(C) An indirect jump
(D) A privileged instruction

[GATE 1999 : IIT Bombay]

Q.8 A processor needs software interrupt to
(A) Test the interrupt system of the processor.
(B) Implement co- routines.
(C) Obtain system services which need execution of privileged instructions.
(D) Return from subroutine.

[GATE 2001 : IIT Kanpur]

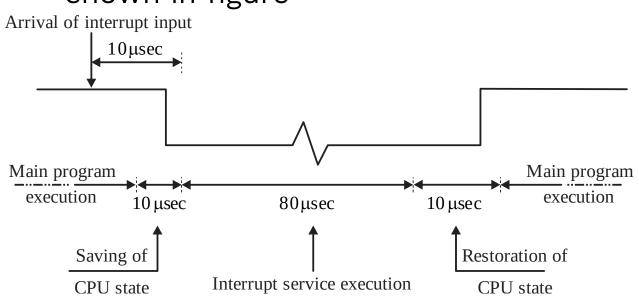
Q.9 A Computer handles several interrupt sources of which the following are relevant for this question:

- Interrupt from CPU temperature sensor (raises interrupt if CPU temperature is too high)
- Interrupt from Mouse (raises interrupt if the mouse is moved or a button is pressed)
- Interrupt from keyboard (raises interrupt when a key is pressed or release)
- Interrupt from Hard Disk (raise interrupt when a disk read is completed)

Which one these will be handled at the HIGHEST priority?
(A) Interrupt from Hard Disk
(B) Interrupt from Mouse
(B) Interrupt from Keyboard
(D) Interrupt from CPU temperature sensor.

[GATE 2011 : IIT Madras]

Q.10 The details of an interrupt cycle are shown in figure



Given that an interrupt input arrives every 1msec, what is the percentage of the total time that the CPU devotes for the main program execution.

[GATE 1993 : IIT Bombay]

Q.11 Let the time taken to switch between user and kernel modes of execution be t_1 while the time taken to switch between two processes be t_2. Which of the following is TRUE?
(A) $t_1 > t_2$
(B) $t_1 = t_2$
(C) $t_1 < t_2$
(D) Nothing can be said about the relation between t_1 and t_2.

[GATE 2011 : IIT Madras]

Q.12 A process executes the following segment of code :

```
for(i = 1; i < = n; i++)
fork ();
```

The number of new processes created is
(A) n (B) ((n(n + 1))/2)
(C) $2^n - 1$ (D) $3^n - 1$

[GATE 2004 : IIT Delhi]

Q.13 The following C program is executed on a Unix/Linux system:

```
#include <unistd.h>
int main ()
{
    int i;
    for (i=0; i<10; i++)
    if (i% 2 = = 0) fork();
    return 0;
}
```

The total number of child processes created is_______.

[GATE 2019 : IIT Madras]

Q.14 Consider the following code fragment:

```
if (fork() == 0)
{
    a = a + 5;
    printf(“%d,%d\n”, a, &A);
}
else { a = a –5; printf(“%d, %d\n”, a, &A); }
```

Let u, v be the values printed by the parent process, and x, y be the values printed by the child process. Which one of the following is TRUE?

(A) u = x + 10 and v = y

(B) u = x + 10 and v != y

(C) u + 10 = x and v = y

(D) u + 10 = x and v != y

[GATE 2005 : IIT Bombay]

Q.15 Consider the following statements about user level threads and kernel threads. Which one of the following statements is FALSE?

(A) Context switch time is longer for kernel level threads than for user level threads.

(B) User level threads do not need any hardware support.

(C) Related kernel level threads can be scheduled on different processors in a multi- processor system.

(D) Blocking one kernel level thread blocks all related threads.

[GATE 2007 : IIT Kanpur]

Q.16 A thread is usually defined as a “light weight process “because an operating system (OS) maintains smaller data structures for a thread than for a process. In relation to this, which of the following is TRUE?

(A) On per-thread basis, the OS maintains only CPU register state.

(B) The OS does not maintain a separate stack for each thread

(C) On per-thread basis, the OS does not maintain virtual memory state.

(D) On per-thread basis, the OS maintains only scheduling and accounting information.

[GATE 2011 : IIT Madras]

Q.17 Which one of the following is FALSE?

(A) User level threads are not scheduled by the kernel.

(B) When a user level thread is blocked, all other threads of its process are blocked.

(C) Context switching between user level threads is faster than context switching between kernel level threads.

(D) Kernel level threads cannot share the code segment.

[GATE 2014 : IIT Kharagpur]

Q.18 Threads of a process share

(A) Glob al variables but not heap

(B) Head but not global variables

(C) Neither global variables nor heap

(D) Both heap and global variables

[GATE 2017 : IIT Roorkee]

Q.19 Which of the following is/are shared by all the threads in a process?

I. Program counter

II. Stack

III. Address space

IV. Registers

(A) I and II only (B) III only

(C) IV only (D) III and IV only

[GATE 2017 : IIT Roorkee]

Q.20 Consider the following table of arrival time and burst time for three processes P0, P1 and P2.

Processes	Arrival time	Burst time
P0	0 ms	9 ms
P1	1 ms	4 ms
P2	2 ms	9 ms

The pre-emptive shortest job first scheduling algorithm is used. Scheduling is carried out only at arrival or completion of processes.

What is the average waiting time for the three processes?

(A) 5.0 ms (B) 4.33 ms

(C) 6.33 ms (D) 7.33 ms

[GATE 2011 : IIT Madras]

Q.21 An operating system uses shortest remaining time first scheduling algorithm for pre-emptive following of processes. Consider the following set of processes with their arrival time and CPU burst time (in millisecond) :

Process	Arrival time	Burst time
P1	0	12
P2	2	4
P3	3	6
P4	8	5

The average waiting time (in milliseconds) of the processes is _______.

[GATE 2014 : IIT Kharagpur]

Q.22 Consider the following set of processes that need to be scheduled on a single CPU. All the times are given in milliseconds.

Process Name	Arrival time	Execution Time
A	0	6
B	3	2
C	5	4
D	7	6
E	10	3

Using the shortest remaining time first scheduling algorithm, the average process turnaround time (in mseC) is_________.

[GATE 2014 : IIT Kharagpur]

Q.23 An operating system uses Shortest Remaining Time First (SRT) process scheduling algorithm. Consider the arrival times and execution times for the following processes:

Process	Execution time	Arrival Time
P1	20	0
P2	25	15
P3	10	30
P4	15	45

What is the total waiting time for process P2?

(A) 5 (B) 15

(C) 40 (D) 55

[GATE 2007 : IIT Kanpur]

Q.24 Consider three CPU-intensive processes, which require 10, 20 and 30 time units and arrive at times 0, 2 and 6 respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm?(Do not count the context switches at time zero and at the end)

(A) 1 (B) 2

(C) 3 (D) 4

[GATE 2006 : IIT Kharagpur]

Q.25 Consider the following four processes with arrival times (in milliseconds) and their length of CPU bursts (in milliseconds) as shown below :

Process	P1	P2	P3	P4
Arrival time	0	1	3	4
CPU burst time	3	1	3	Z

These processes are run on a single processor using preemptive Shortest Remaining Time First scheduling algorithm. If the average waiting time of the processes is 1 milliseconds, then the value of Z is _________.

[GATE 2019 : IIT Madras]

Q.26 Consider three processes, all arriving at time zero, with total execution time of 10, 20 and 30 units, respectively. Each process spends the first 20% of execution time doing I/O, the next 70% of time doing computation and the last 10% of time doing I/O again. The operating system uses a shortest remaining compute time first scheduling algorithm and schedules a new process either when the running processes gets blocked on I/O or when the running process finishes its compute burst. Assume that all I/O operations can be overlapped as much as possible. For what percentage of time does the CPU remain idle?

(A) 0% (B) 10.6%

(C) 30.0% (D) 89.4%

[GATE 2006 : IIT Kharagpur]

Q.27 Consider the set of processes with arrival time (in milliseconds). CPU burst time (in milliseconds). And priority (0 is the highest priority) shown below. None of the processes have I/O burst time.

Process	Arrival Time	Burst Time	Priority
P_1	0	11	2
P_2	5	28	0
P_3	12	2	3
P_4	2	10	1
P_5	9	16	4

The average waiting time (in millisecond) of all the processes using preemptive priority scheduling algorithm is ______________.

[GATE 2017 : IIT Roorkee]

Q.28 The arrival time, priority, and duration of the CPU and I/O bursts for each of three processes P1, P2 and P3 are given in the table below. Each process has a CPU burst followed by an I/O burst followed by another CPU burst. Assume that each process has its own I/O resource.

Process	Arrival Time	Priority	Burst Duration CPU, I/O, CPU
P_1	0	2	1, 5, 3
P_2	2	3 (lowest)	3, 3, 1
P_3	3	1 (highest)	2, 3, 1

The multi-programmed operating system uses preemptive priority scheduling. What are the finish times of the processesP1, P2 andP3 ?

(A) 11, 15, 9 (B) 10, 15, 9

(C) 11, 16, 10 (D) 12, 17, 11

[GATE 2006 : IIT Kharagpur]

Q.29 Consider a uniprocessor system executing three task T_1, T_2 and T_3 each of which is composed of an infinite sequence of jobs (or instances) which arrive periodically at intervals of 3, 7 and 20 milliseconds, respectively. The priority of each task is the inverse of its period, and the available tasks are scheduled in order of priority, with the highest priority task scheduled first. Each instance of T_1, T_2 & T_3 requires an execution time of 1, 2 and 4 milliseconds, respectively. Given that all task initially arrive at the beginning of the 1st millisecond and task pre-emption are allowed. The first instance of T_3 completes its execution at the end of ______ millisecond.

[GATE 2015 : IIT Kanpur]

Q.30 Four jobs to be executed on a single processor system arrive at time 0^+ in the order A, B, C, D. Their burst CPU time requirements are 4,1,8,1 time units respectively. The completion time of A under round robin scheduling with time slice of one time units is

(A) 10 (B) 4

(C) 8 (D) 9

[GATE 1996 : IISc Bangalore]

Q.31 Assume that the following jobs are to be executed on a single processor system.

Job Id	CPU Burst Time
p	4
q	1
r	8
s	1
t	2

The jobs are assumed to have arrived at time 0^+ and in the order p, q, r, s, t. Calculate the departure time (completion time) for job p if scheduling is round robin with time slice 1.

(A) 4 (B) 10

(C) 11 (D) 12

[GATE 1993 : IIT Bombay]

Q.32 Consider the 3 processes, P1, P2 and P3 shown in the table.

Process	Arrival Time	Time Units Required
P1	0	5
P2	1	7
P3	3	4

The completion order of the 3 processes under the policies FCFS and RR2 (round robin scheduling with CPU quantum of 2 time units) are

(A) FCFS: P1, P2, P3 RR2: P1, P2, P3

(B) FCFS: P1, P3, P2 RR2: P1, P3, P2

(C) FCFS: P1, P2, P3 RR2: P1, P3, P2

(D) FCFS: P1, P3, P2 RR2: P1, P2, P3

[GATE 2012 : IIT Delhi]

Q.33 Which scheduling policy is most suitable for time-shared operating systems?

(A) Shortest Job First

(B) Round Robin

(C) First come First serve

(D) Elevator

[GATE 1995 : IIT Kanpur]

Q.34 For the processes listed in the following scheduling schemes will give the lowest turnaround time?

Process	Arrival Time	Processing Time
A	0	3
B	1	6
C	4	4
D	6	2

(A) First Come First Serve

(B) Non- preemptive Shortest Job First

(C) Shortest Remaining Time

(D) Round Robin with Quantum value two

[GATE 2015 : IIT Kanpur]

Q.35 If the time-slice used in the round-robin scheduling policy is more than the maximum time required to execute any process, then the policy will

(A) degenerate to shortest job first

(B) degenerate to priority scheduling

(C) degenerate to first come first serve

(D) None of the above

[GATE 2008 : IISc Bangalore]

Q.36 Which of the following statements are true?

I. Shortest remaining time first scheduling may cause starvation.

II. Preemptive scheduling may cause starvation

III. Round robin is better than FCFS in terms of response time.

(A) I only (B) I and III only

(C) II and III only (D) I, II and III

[GATE 2010 : IIT Guwahati]

Q.37 Three processes A, B and C each execute a loop of 100 iteration. In each iteration of the loop, a process performs a single computation that requires t_c CPU millisecond and then initiates a single I/O operation that lasts for t_{io} millisecond. It is assumed that the computer where the processes execute has sufficient number of I/O devices and the OS of the computer assigns different I/O devices to each process. Also the scheduling overhead of the OS is negligible. The process have the following characteristics:

Process	t_c	t_{i0}
A	100 ms	500 ms
B	350 ms	500 ms
C	200 ms	500 ms

The processes A, B and C are started at times 0,5 and 10 millisecond respectively in a pure time sharing system (round robin scheduling that uses a time slice of 50 milliseconds. The time in millisecond at which process C would complete its first I/O operation is ________?

[GATE 2014 : IIT Kharagpur]

Q.38 The sequence __________ is an optimal non- preemptive scheduling sequence for the following jobs which leaves the CPU idle for ____________ unit (s) of time

Job	Arrival Time	Burst Time
1	0.0	9
2	0.6	5
3	1.0	1

(A) {3, 2, 1}, 1 (B){2, 1, 3}, 0

(C) {3, 2, 1}, 0 (D){1, 2, 3}, 5

[GATE 1995 : IIT Kanpur]

Q.39 Consider three processes (process id 0, 1, 2 respectively) with compute time bursts 2, 4 and 8 time units. All processes arrive at time zero. Consider the longest remaining time first (LRTF) scheduling algorithm. In LRTF ties are broken by giving priority to the process with the lowest process id. The average turnaround time is:

(A) 13 units. (B) 14 units.
(C) 15 units. (D) 16 units.

[GATE 2006 : IIT Kharagpur]

Q.40 Which of the following scheduling algorithms is non- preemptive?

(A) Round Robin
(B) First-In First-out
(C) Multilevel Queue Scheduling
(D) Multilevel Queue Scheduling with Feedback.

[GATE 2002 : IISc Bangalore]

Q.41 Which one of the following statements is/are TRUE?

(A) Threads of a process share global variables but not heap
(B) The OS does not maintain a separate stack for each thread
(C) ISR is Interrupt Service Routine
(D) Preemptive scheduling may cause starvation

Q.42 Which of th following is/are the correct definition of a valid process transition in an operating system?

(A) Wake up: ready → running
(B) Dispatch: ready → running
(C) Block: ready → running
(D) Timer runout: running → ready

Q.43 Three processes arrive at time zero with CPU bursts time of 16, 20 & 10 milliseconds. If the scheduler has prior knowledge about the length of CPU bursts, Minimum achievable average waiting time for this three processes in a non-preemptive scheduler (rounded to nearest integer) is ______ milliseconds.

[GATE 2021 : IIT Bombay]

Q.44 Statements correct in context of CPU scheduling?

(A) Round Robin policy can be used even when the CPU time required by each of the processes is not known in apriori.
(B) Implementing preemptive scheduling needs hardware support.
(C) The goal is to only maximize CPU utilization and minimize throughput.
(D) Turnaround time includes waiting time.

[GATE 2021 : IIT Bombay]

Q.45 Which of the following standard C library functions will always invoke a system call when executed from a single-threaded process in a UNIX/LINUX Operating system?

(A) sleep (B) strlen
(C) malloc (D) exit

[GATE 2021 : IIT Bombay]

Q.46 Which of the following need not necessarily be saved on a context switch between processes?

(A) General purpose registers
(B) Translation look-aside buffer
(C) Program counter
(D) All of the above

[GATE 2000 : IIT Kharagpur]

Q.47 Which of the following actions is/are typically not performed by the operating system when switching context from process A to process B?

(A) Saving current register values and restoring saved register values for process B.
(B) Changing address translation tables.
(C) Swapping out the memory image of process A to the disk.
(D) Invalidating the translation look-aside buffer

Q.48 Consider n processes sharing the CPU in a round- robin fashion. Assuming that each process switch takes s second, what must be the quantum size such that the overhead resulting from process switching is minimized but, at the same time each process is guaranteed to get its turn at the CPU at least every t second?

(A) $q \le \frac{t-ns}{n-1}$ (B) $q \ge \frac{t-ns}{n-1}$

(C) $q \le \frac{t-ns}{n+1}$ (D) $q \ge \frac{t-ns}{n+1}$

[GATE 1998 : IIT Delhi]

Practice Questions

Q.1 What is the degree of multiprogramming?

(A) the number of processes executed per unit time

(B) the number of processes in the ready queue

(C) the number of processes in the I/O queue

(D) the number of processes in memory

Q.2 Which one of the following statements is/are TRUE?

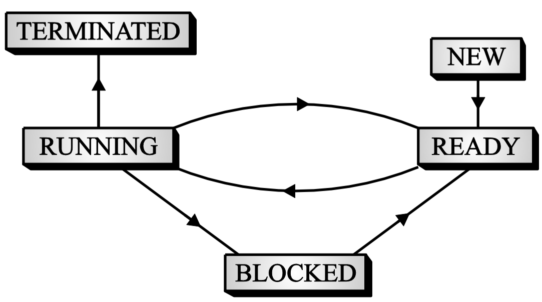
(A) If all processes I/O bound, the ready queue will almost always be empty and the Short term Scheduler will have a little to do.

(B) If all processes I/O bound, the ready queue will almost always be full and the Short term Scheduler will have a lot to do.

(C) If all processes CPU bound, the I/O queue will almost always be empty.

(D) A CPU bound process is one that spends more of its time doing computations.

Q.3 The process state transition diagram in Figure is representative of



(A) A batch operating system.

(B) An operating system with a preemptive scheduler.

(C) An operating system with a non-preemptive scheduler.

(D) A uni-programmed operating system.

[GATE 1996 : IISc Bangalore]

Q.4 A process stack does not contain ________

(A) Function parameters

(B) Local variables

(C) Return addresses

(D) PID of child process

Q.5 The address of the next instruction to be executed by the current process is provided by the ________

(A) CPU registers

(B) Program counter

(C) Process stack

(D) Pipe

Q.6 Suppose that a process is in "Blocked" state waiting for some I/O service. When the service is completed, it goes to the ________

(A) Running state (B) Ready state

(C) Suspended state (D) Terminated state

Q.7 How does the software trigger an interrupt?

(A) Sending signals to CPU through bus

(B) Executing a special operation called system call

(C) Executing a special program called system program

(D) Executing a special program called interrupt trigger program

Q.8 Which one of the following statements is/are TRUE?

(A) Blocking one kernel level thread blocks all related threads.

(B) The OS does not maintain a separate stack for each thread

(C) ISR is Interrupt Service Routine

(D) Preemptive scheduling may cause starvation

Q.9 When a process creates a new process two possibilities exits-

(A) The parent continue to executes concurrently with its children

(B) The parent stop to executes concurrently with its children

(C) The parent waits until some or all of its children have terminated

(D) The parent do not wait until some or all of its children have terminated

Q.10 A process executes the code

fork ();

fork ();

fork ();

The total number of child processes created is

(A) 3 (B) 4

(C) 7 (D) 8

[GATE 2012 : IIT Delhi]

Q.11 A process executes the following code

for (i=0; i<n; i++)

fork ();

The total number of child processes created is

(A) n (B) $2^n - 1$

(C) 2^n (D) $2^{n+1} - 1$

Q.12 Which of the following is an example of a spooled device?

(A) The terminal used to enter the input data for the C program being executed.

(B) An output device used to print the output of a number of jobs.

(C) The secondary memory device in a virtual storage system.

(D) The swapping area on a disk used by the swapper.

[GATE 1998 : IIT Delhi]

Q.13 Which of the following is an example of spooled device?

(A) A line printer used to print the output of a number of jobs.

(B) A terminal used to enter input data to a running program.

(C) A secondary storage device in a virtual memory system.

(D) A graphic display device.

[GATE 1996 : IISc Bangalore]

Q.14 Which of the following is an example of a spooled device?

(A) The terminal used to the input data for a program being executed.

(B) The secondary memory device in a virtual memory system

(C) A line printer used to print the output of a number of jobs.

(D) None of the above

[GATE 1992 : IIT Delhi]

Q.15 A CPU has two modes-privileged and non-privileged. In order to change the mode from privileged to non-privileged

(A) a hardware interrupt is needed.

(B) a software interrupt is needed.

(C) a privileged instruction (which does not generate an interrupt) is needed.

(D) a non-privileged instruction (which does not generate an interrupt) is needed

[GATE 2001 : IIT Kanpur]

Q.16 Consider the following processes, with the arrival time and the length of the CPU burst given in millisecond. The scheduling algorithm used is pre-emptive shortest remaining-time first.

Process	Arrival time	Burst time
P1	0	10
P2	3	6
P3	7	1
P4	8	2

The average turnaround time of these processes is ________ milliseconds.

[2016]

Q.17 Consider the following CPU processes with arrival time (in milliseconds)and length of CPU bursts (in millisecond) As given below :

Process	Arrival time	Burst time
P1	0	7
P2	3	3
P3	5	5
P4	6	2

If the pre-emptive shorted remaining time first scheduling algorithm is used to schedule the process, than the average waiting time across all process is _____ milliseconds.

[GATE 2017 : IIT Roorkee]

Q.18 Consider the following set of processes, with the arrival times and the CPU- burst times given in milliseconds.

Process	Arrival time	Burst time
P1	0	5
P2	1	3
P3	2	3
P4	4	1

What is average turnaround time for these processes with the pre-emptive shortest remaining processing time first (SRPT) algorithm?

(A) 5.50 (B) 5.75

(C) 6.00 (D) 6.25

[GATE 2014 : IIT Kharagpur]

Q.19 Consider the following set of processes, assumed to have arrived at time 0. Consider the CPU scheduling algorithms Shortest Job First (SJF) and Round Robin (RR). For RR, assume that the processes are scheduled in the order P1, P2, P3 P4

Processes	P_1	P_2	P_3	P_4
Burst time (in ms)	8	7	2	4

If the time quantum for RR is 4 ms, then the absolute value of the difference between the average turnaround times (in ms) of SJF and RR (round off to 2 decimal places) is ____.

[GATE 2020 : IIT Delhi]

Q.20 Four jobs are waiting to be run. Their expected run times are 6, 3, 5 and x. In what order should they be run to minimize the average response time?

Q.21 Consider a set of n tasks with knows runtimes $r_1, r_2, \ldots\ldots\ldots\ldots, r_n$ to be on a uniprocessor machine. Which of the following processor scheduling algorithms will result in the maximum throughput?

(A) Round-Robin

(B) Shortest-job-First

(C) Highest-Response-Ratio-Next

(D) First-Come-First-Served

[GATE 2001 : IIT Kanpur]

Q.22 A uni-processor computer system only has two processes, both of which alternate 10 msms CPU bursts with 90 msms I/O bursts. Both the processes were created at nearly the same time. The I/O of both processes can proceed in parallel. Which of the following scheduling strategies will result in the *least* CPU utilization (over a long period of time) for this system?

(A) First come first served scheduling

(B) Shortest remaining time first scheduling

(C) Static priority scheduling with different priorities for the two processes

(D) Round robin scheduling with a time quantum of 5 ms.

[GATE 2003 : IIT Madras]

Q.23 A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (the lowest priority). The scheduler re-evaluates the process priorities every T time units and decides the next process to schedule. Which one of the following is TRUE if the processes have no I/O operations and all arrive at time zero?

(A) This algorithm is equivalent to the first-come-first-serve algorithm.

(B) This algorithm is equivalent to the round-robin algorithm.

(C) This algorithm is equivalent to the shortest-job-first algorithm.

(D) This algorithm is equivalent to the shortest-remaining-time-first algorithm

[GATE 2013 : IIT Bombay]

Q.24 The highest-response ratio next scheduling policy favours...........jobs, but it also limits the waiting time of ....jobs

[GATE 1999 : IIT Bombay]

Q.25 Group 1 contains some CPU scheduling algorithms and Group 2 contains some applications. Match entries in Group 1 to entries in Group 2.

Group I	**Group II**
(P) Gang Scheduling	(1) Guaranteed Scheduling
(Q) Rate Monotonic Scheduling	(2) Real-time Scheduling
(R) Fair Share Scheduling	(3) Thread Scheduling

(A) P – 3 Q – 2 R – 1

(B) P – 1 Q – 2 R – 3

(C) P – 2 Q – 3 R – 1

(D) P – 1 Q – 3 R – 2

2 Process Management - II

Classroom Questions

Q.1 The following two functions P1 and P2 that share a variable B with an initial value of 2 execute concurrently.

P1 ()	P2 ()
{ C = B – 1; B = 2 * C; }	{ D = 2 * B; B = D – 1; }

The number of distinct values that B can possibly take after the execution is _________.

[GATE 2000 : IIT Kharagpur]

Q.2 Consider three concurrent processes P1, P2 and P3 as shown below, which access a shared variable D that has been initialized to 100

P1	P2	P3
: : D=D+20 : :	: : D = D – 50 : :	: : D = D + 10 : :

The processes are executed on a uniprocessor system running a time – shared operating system .If the minimum and maximum possible values of D after the three process have completed are X of Y respectively, then the value of Y-X is _________.

[GATE 2019 : IIT Madras]

Q.3 A critical region is:

(A) One which is enclosed by a pair of P and V operations on semaphores.

(B) A program segment that has not been proved bug-free.

(C) A program segment that often causes unexpected system crashes.

(D) A program segment where shared resources are accessed.

[GATE 1987 : IIT Bombay]

Q.4 Processes P_1 and P_2 use critical flag in the following routine to achieve mutual exclusion. Assume that critical flag is initialized to FALSE in the main program.

```
get_exclusive_access ( )
{
if (critical _flag == FALSE) {
critical_flag = TRUE ;
critical_region () ;
critical_flag = FALSE;  }
}
```

Consider the following statements.
i. It is possible for both P_1 and P_2 to access critical region concurrently.
ii. This may lead to a deadlock
Which of the following holds?
(A) (i) is false and (ii) is true
(B) Both (i) and (ii) are false
(C) (i) is true and (ii) is false
(D) Both (i) and (ii) are true

[GATE 2007 : IIT Kanpur]

Q.5 Consider the following two-process synchronization solution.

Process 0	Process 1
Entry: loop while (turn == 1); (Critical section) Exit: turn = 1;	Entry: loop while (turn ==0); (Critical section) Exit: turn = 0;

The shared variable turn is initialized to zero. Which one of the following is TRUE?

(A) This is a correct two- process synchronization solution.

(B) This solution violates mutual exclusion requirement.

(C) This solution violates progress requirement.

(D) This solution violates bounded wait requirement.

[GATE 2016 : IISc Bangalore]

Q.6 Consider the method used by processes P1 and P2 for accessing their critical sections whenever needed, as given below. The initial values of shared Boolean variables S1 and S2 are randomly assigned.

Method used by P1	Method used by P2
While (S1==S2); Critical section S1 = S2;	While (S1! = S2); Critical section S2 = not (S1);

Which one of the following statements describes the properties achieved?

(A) Mutual exclusion but not progress.

(B) Progress but not mutual exclusion.

(C) Neither mutual exclusion nor progress

(D) Both mutual exclusion and progress.

[GATE 2010 : IIT Guwahati]

Q.7 Two processes X and Y need to access a critical section. Consider the following synchronization construct used by both the processes.

Process X /* other code for process X*/	Process Y/* /*other code for process Y

while (true){
varP = true;
while (varQ == true)
{
/* Critical Section */
varP = false;
}}
	while (true){
	varQ = true;
	while (varP == true)
	{
	/* Critical Section */
	varQ = false;
	}}

Here, varP and varQ are shared variables and both are initialized to false. Which one of the following statements is true?

(A) The proposed solution prevents deadlock but fails to guarantee mutual exclusion

(B) The proposed solution guarantees mutual exclusion but fails to prevent deadlock

(C) The proposed solution guarantees mutual exclusion and prevents deadlock

(D) The proposed solution fails to prevent deadlock and fails to guarantee mutual exclusion

[GATE 2010 : IIT Guwahati]

Q.8 The enter _CS () and leave _CS () function to implement critical section of a process are realized using test-and-set instruction as follows:

```
    void enter _ CS(x)
{
    while (test-and-set(x));
}
    void leave _ CS(x)
{
    x=0;
}
```

In the above solution, X is a memory location associated with the CS and is initialized to 0. Now consider the following statements:

(I) The above solution to CS problem is deadlock- free.

(II) The solution is starvation free

(III) The processes enter CS in FIFO order.

(IV) More than one process can enter CS at the same time.

Which of the above statements are TRUE?

(A) I only (B) I and II

(C) II and III (D) IV only

[GATE 2009 : IIT Roorkee]

Q.9 At a particular time of computation the value of a counting semaphore is 7. Then 20P operations and 15V operations were completed on this semaphore. The resulting value of the semaphore is;

(A) 42 (B) 2

(C) 7 (D) 12

[GATE 1992 : IIT Delhi]

Q.10 Consider a non-negative counting semaphore S. The operation P(S) decrements S, and V(S) increments S. During an execution, 20 P(S) operations and 12 V(S) operation are issued in some order. The largest initial value of S for which at least one P(S) operation will remain blocked is __________.]

[GATE 2014 : IIT Kharagpur]

Q.11 Given below is a program which when executed spawns two concurrent processes :

semaphore X : = 0 ;

/* Process now forks into concurrent processes P1 & P2 */

P1: repeat forever	**P2: repeat forever**
V (X) ; Compute ; P(X) ;	P(X) ; Compute ; V(X) ;

Consider the following statements about processes P1 and P2:

I. It is possible for process P1 to starve.

II. It is possible for process P2 to starve.

Which of the following holds?

(A) Both I and II are true

(B) I is true but II is false

(C) II is true but I is false

(D) Both I and II are false

[GATE 2005 : IIT Bombay]

Q.12 A shared variable x, initialized to zero, is operated on by four concurrent processes W, X, Y, Z as follows. Each of the processes W and X reads x from memory, increments by one, stores it to the memory, and then terminates. Each of the processes Y and Z reads X from memory, decrements by two, stores it to memory, and terminates. Each process before reading x invokes the P operation (i.e., wait) on a counting semaphore S and invokes the V operation (i.e., signal) on the semaphore S after storing x to memory. Semaphore S is initialized to two. What is the maximum possible value of x after all processes complete execution?

(A) – 2 (B) – 1

(C) 1 (D) 2

[GATE 2013 : IIT Bombay]

Q.13 Each of a set of *n* processes executes the following code using two semaphores *a* and *b* initialized to 1 and 0, respectively. Assume that count is a shared variable initialized to 0 and not used in CODE SECTION P.

CODE SECTION P

wait (*a*); count = count + 1;

if (count = = n) signal (*b*);

signal (*a*);

wait (*b*); signal (*b*);

CODE SECTION Q

What does the code achieve?

(A) It ensures that no process executes CODE SECTION Q before every process has finished CODE SECTION P.

(B) It ensures that at most two processes are in CODE SECTION Q at any time.

(C) It ensures that all processes execute CODE SECTION P mutually exclusively.

(D) It ensures that at most $n-1$ processes are in CODE SECION P at any time.

[GATE 2020 : IIT Delhi]

Q.14 The following program consists of 3 concurrent processes and 3 binary semaphores. The semaphores are initialized as S0 = 1, S1 = 0, S2 = 0.

Process P0	**Process P1**	**Process P2**

while (true) {
wait (S0);
print ‘0’;
release (S1);
release(S2);
}
	wait (S1);
	release (S0);
		wait (S2);
		release (S0);

How many times will process P0 print ‘0’?

(A) Atleast twice (B) Exactly twice

(C) Exactly thrice (D) Exactly once

[GATE 2008 : IISc Bangalore]

Common Data for Questions 15 & 16

Suppose we want to synchronize two concurrent processes P and Q using binary semaphores S and T. The code for the processes P and Q is shown below.

Process P:	**Process Q:**

while (1) {
W:
print ‘0’;
print ‘0’;
X:
}
	while (1) {
	Y:
	print ‘1’
	print ‘1’
	Z:
	}

Synchronization statements can be inserted only at point W, X, Y and Z.

Q.15 Which of the following will always lead to an output staring with 001100110011?

(A) P(S) at W, V(S) at X, P(T) at Y, V(T) at Z, S and T initially 1

(B) P(S) at W, V(T) at X, P(T) at Y, V(S) at Z, S initially 1, and T initially 0

(C) P(S) at W, V(T) at X, P(T) at Y, V(S) at Z, S and T initially 1

(D) P(S) at W, V(S) at X, P(T) at Y, V(T) at Z, S initially 1, and T initially 0

[GATE 2003 : IIT Madras]

Q.16 Which of the following will ensure that the output string never contains a substring of the form 01^nor 10^n where n is odd?

(A) P(S) at W, V(S) at X, P(T) at Y, V(T) at Z, S and T initially 1

(B) P(S) at W, V(S) at X, P(T) at Y, V(T) at Z, S and T initially 1

(C) P(S) at W, V(S) at X, P(S) at Y, V(S) at Z, S initially 1

(D) V(S) at W, V(T) at X, P(S) at Y, P(T) at Z, S and T initially 1.

[GATE 2003 : IIT Madras]

Q.17 Two concurrent processes P1 and P2 use four shared resources R1, R2, R3 and R4, as shown below.

P1	P2
Compute: Use R1; Use R2; Use R3; Use R4;	Compute; Use R1; Use R2; Use R3;. Use R4;

Both processes are started at the same time, and each resource can be accessed by only one process at a time The following scheduling constraints exist between the access of resources by the processes:

P2 must complete use of R1 before P1 gets access to R1

P1 must complete use of R2 before P2 gets access to R2.

P2 must complete use of R3 before P1 gets access to R3.

P1 must complete use of R4 before P2 gets access to R4.

There are no other scheduling constraints between the processes. If only binary semaphores are used to enforce the above scheduling constraints, what is the minimum number of binary semaphores needed?

(A) 1 (B) 2

(C) 3 (D) 4

[GATE 2005 : IIT Bombay]

Q.18 Three concurrent processes X, Y and Z execute three different code segments that access X executes the P operation (i.e., wait) on semaphores a, b and c; process Y executes the P operation on semaphores b, c and d; process Z executes the P operation on semaphores c, d and a before entering the respective code segments. After completing the execution of its code segment, each process invokes the V operation (i.e., signal) on its three semaphores. All semaphores are binary semaphores initialized to one. Which one of the following represents a deadlock- free order of invoking the P operations by the processes?

(A) X: P(a)P(b)P(c), Y:P(b)P(c)P(d), Z: P(c)P(d)P(a)

(B) X: P(b)P(a)P(c), Y:P(b)P(c)P(d), Z:P(a)P(c)P(d)

(C) X: P(b)P(a)P(c), Y: P(c)P(b)P(d), Z:P(a)P(c)P(d)

(D) X: P(a)P(b)P(c), Y:P(c)P(b)P(d) Z:P(c)P(d)P(a)

[GATE 2013 : IIT Bombay]

Q.19 The atomic fetch-and-set x, y instruction unconditionally sets the memory location x to 1 and fetches the old value of x in y without allowing any intervening access to the memory location x. Consider the following implementation of P and V function on a binary semaphore S.

```
    void P (binary_ semaphore * S)
{
    unsigned y;
    unsigned * x = &(S →value);
    do
{
    fetch-and-set x, y;
}   while (y);
}
    void V (binary_ semaphore * S)
{
    S→ value = 0;
}
```

Which one of the following is true?

(A) The implementation may not work if context switching is disabled in P
(B) Instead of using fetch- and- set, a pair of normal load/ store can be used
(C) The implementation of V is wrong.
(D) The code does not implement a binary semaphore.

[GATE 2006 : IIT Kharagpur]

Q.20 Consider the following solution to the producer-consumer synchronization problem. The shared buffer size is *N*. Three semaphores *empty*, *full* and mutex are defined with respective initial values of 0, *N* and 1. Semaphore empty denotes the number of available slots in the buffer, for the consumer to read from. Semaphore full denotes the number of available slots in the buffer, for the producer to write to. The placeholder variables, denoted by *P*, *Q*, *R*, and *S*, in the code below can be assigned either *empty* or *full*. The valid semaphore operations are: wait ( ) and signal ( ).

Producer :	**Consumer :**

do {
wait (P);
wait (mutex);
/ / Add item to buffer
signal (mutex);
signal(Q);
} while (1);
	do {
	wait (R);
	wait (mutex);
	/ / Consume item from buffer
	signal (mutex);
	signal(S);
	} while (1);

Which one of the following assignments to *P*, *Q*, *R* and *S* will yield the correct solution?
(A) P : full, Q : full, R : empty, S : empty
(B) P : empty, Q : empty, R : full, S : full
(C) P : full, Q : empty, R : empty, S : full
(D) P : empty, Q : full, R : full, S : empty

[GATE 2018 : IIT Guwahati]

Q.21 The semaphore variables full, empty and mutex are initialized to 0, n and 1, respectively. Process P1 repeatedly adds one item at a time to a buffer of size n, and process P2 repeatedly removes one item at a time from the same buffer using the programs given below. In the programs, K, L, M and N are unspecified statements.

P1	**P2**

while (1) {
K;
P(mutex);
Add an item to the buffer;
V(mutex);
L;
}
	while (1) {
	M;
	P(mutex);
	Remove an item from the buffer;
	V(mutex);
	N;
	}

The statements K, L, M and N are respectively
(A) P(full), V(empty), P(full), V(empty)
(B) P(full), V(empty), P(empty), V(full)
(C) P(empty), V(full), P(empty), V(full)
(D) P(empty), V(full), P(full), V(empty)

[GATE 2004 : IIT Delhi]

Q.22 Consider the procedure below for the producer-Consumer problem which uses semaphores:
Semaphore n = 0;
Semaphore s = 1;

void producer ()	void consumer ()

{
while (true){
produce ();
semWait (s);
addToBuffer ();
semSignal (s);
semSignal (n)
}
}
	{
	while (true){
	semWait (s);
	semWait (n)’
	remove FromBuffer ();
	semSignal (s);
	consume ();
	}
	}

Which one of the following is **TRUE?**
(A) The producer will be able to add an item to the buffer, but the consumer can never consume it.
(B) The consumer will remove no more than one item from the buffer.
(C) Deadlock occurs if the consumer succeeds in acquiring semaphore s when the buffer is empty.
(D) The starting value for the semaphore n must be 1 and not 0 for deadlock- free operation.

[GATE 2014 : IIT Kharagpur]

Q.23 A solution to the Dining Philosophers Problem which avoids deadlock is
(A) Ensure that all philosophers pick up the left fork before the right fork.
(B) Ensure that all philosophers pick up the right fork before the left fork.

(C) Ensure that one particular philosophers picks up the left fork before the right fork, and that all other philosophers pick up the right fork before the left fork.

(D) None of the above.

[GATE 2014 : IIT Kharagpur]

Q.24 Let m[0]...........m[4] be mutexes (binary semaphores) and P[0]..........P[4] be processes. Suppose each process P[i] executes the following:

Wait (m[i]); wait (m[(i+1) mod 4];...............................

Release (m[i]); release (m[(i+1)mod 4]);

This could cause

(A) Thrashing

(B) Deadlock

(C) Starvation, but not deadlock

(D) None of the above

[GATE 2014 : IIT Kharagpur]

Q.25 Synchronization in the classical readers and writers problem can be achieved through use of semaphores. In the following incomplete code for readers-writers problem, two binary semaphores mutex and wrt are used to obtain synchronization

```
wait (wrt)
writing is performed
signal (wrt)
wait (mutex)
readcount = readcount + 1
if readcount = 1 then S1
S2
reading is performed
S3
readcount = readcount - 1
if readcount = 0 then S4
signal (mutex)
```

The values of S1, S2, S3, S4, (in that order) are

(A) signal (mutex), wait (wrt), signal (wrt), wait (mutex)

(B) signal (wrt), signal (mutex), wait (mutex), wait (wrt)

(C) wait (wrt), signal (mutex), wait (mutex), signal (wrt)

(D) signal (mutex), wait (mutex), signal (mutex), wait (mutex)

[GATE 2006 : IIT Kharagpur]

Q.26 Consider the following proposed solution for the critical section problem. There are n processes : P_0....................P_{n-1}. In the code, function pmax returns an integer not smaller than any of if arguments. For all i, t[i] is initialized to zero.

```
    Code for P1
    do {
    c[i]= 1;
    t[i]= pmax(t[0],.......,t[n-1])+1; c[i]=0
    for every j≠i in {0,.....,n-1}
{
    while (c[j]);
    while (t[j]) != 0 && t[j]<=t[i]);
}
    Critical Section;
    t[i] = 0;
    Remainder Section
} while (true);
```

While one of the following is TRUE about the above solution?

(A) At most one process can be in the critical section at any time

(B) The bounded wait condition is satisfied

(C) The progress condition is satisfied

(D) It cannot cause a deadlock

[GATE 2016 : IISc Bangalore]

Q.27 Each process P_i, i = 1.........9 is coded as follows

```
    Repeat
    P(mutex)
{
    critical section
}
    v(mutex)
    forever
```

The code for P_{10} is identical except that it uses V(mutex) in place of P(mutex). What is the largest number of processes that can be inside the critical section at any moment?

(A) 1 (B) 2

(C) 3 (D) None

[GATE 1997 : IIT Madras]

Q.28 Consider the following pseudocode, where S is a semaphore initialized to 5 in line#2 and counter is a shared variable initialized to 0 in line#1. Assume that the increment operation in line#7 is not atomic.

```
int counter = 0;
Semaphore S = init(5);
Void parop(void){
wait(S);
wait(S);
```

```
    counter++;
    signal(S);
    signal(S);
}
```

If live threads execute the function parop concurrently, which of the following program behavior(s) is/are possible?

(A) The value of counter is 0 after all the threads successfully complete the execution of parop.

(B) The value of counter is 1 after all the threads successfully complete the execution of parop.

(C) There is a deadlock involving all the threads.

(D) The value of counter is 5 after all the threads successfully complete the execution of parop.

[GATE 2021 : IIT Bombay]

Q.29 Consider a computer system with multiple shared resource types, with one instance per resource type. Each instance can be owned by only one process at a time. Owning and freeing of resources are done by holding a global lock (L). The following scheme is used to own a resource instance:

```
function OWNRESOURCE (Resource R)
    Acquire lock L // a global lock
    if R is available then
    Acquire R
    Release lock L
    else
    if R is owned by another process P
then
Terminate P, after releasing all
resources owned by P
    Acquire R
    Restart P
    Release lock L
    end if
    end if
    end function
```

Which of the following choice(s) about the above scheme is/are correct?

(A) The scheme ensures that deadlocks will not occur.

(B) The scheme violates the mutual exclusion property.

(C) The scheme may lead to live-lock.

(D) The scheme may lead to starvation.

[GATE 2021 : IIT Bombay]

Common Data for Questions 30 & 31

Barrier is a synchronization construct where a set of processes synchronizes globally i.e., each process in the set arrive at the barrier and waits for all other to arrive and then all processes leave the barrier. Let the number of processes in the set be three and S be a binary semaphore with the usual P and V function. Consider the following C implementation of a barrier with line numbers shown on the left.

```
void barrier (void){
1.      P(S);
2.      process _arrived++;
3.      V(S);
4.      while (process _arrived! = 3)
5.      P(S);
6.      process _left++:
7.      if(process_left==3) {
8.      process _arrived =0;
9.      process_left=0;
10.     }
11.     V(S);
        }
```

The variables process _arrived and process _left are shared among all processes and are initialized to zero. In a concurrent program all the three processes call the barrier function when they need to synchronize globally.

Q.30 The above implementation of barrier is incorrect. Which one of the following is true?

(A) The barrier implementation is wrong due to the use of binary semaphore S

(B) The barrier implementation may lead to a deadlock if two barrier invocation are used in immediate succession

(C) Lines 6 to 10 need not be inside a critical section.

(D) The barrier implementation is correct if there are only two processes instead of three.

[GATE 2006 : IIT Kharagpur]

Q.31 Which one of the following rectifies the problem in the implementation?

(A) Lines 6 to 10 are simply replaces by process _arrived

(B) At the beginning of the barrier the first process to enter the barrier waits until process arrived becomes zero before proceeding to execute P(S)

(C) Context switch is disabled at the beginning of the barrier and re-enabled at the end.

(D) The variable process _left is mad private instead of shared.

[GATE 2006 : IIT Kharagpur]

Q.32 The P and V operations on counting semaphores, where s is a counting semaphore are defined as follows:

P(s): s = s – 1;
If s < 0 then wait;
V(s): s = s + 1;
If s<=0 then wake up a process waiting on s;

Assume that P_b and V_b the wait and signal operations on binary semaphores X_b and Y_b are used to implement the semaphore operations P(s) and V(s) as follows :

P(s):$P_b(x_b)$;	**V(s):** $P_b(x_b)$;

s = s – 1;
if (s < 0)
{
$V_b(x_b)$;
$P_b(y_b)$;
}
else $V_b(x_b)$;
	s = s + 1;
	if (s <= 0) $V_b(y_b)$;
	$V_b(x_b)$;

The initial values of x_b and y_b are respectively

(A) 0 and 0 (B) 0 and 1
(C) 1 and 0 (D) 1 and 1

[GATE 2008 : IISc Bangalore]

Q.33 Consider the following solution to the producer-consumer problem suing a buffer of size 1. Assume that the initial value of account is 0. Also assume that the testing of count and assignment to count are atomic operations.

[GATE 1999 : IIT Bombay]

Show that in this solution it is possible that both the processes are sleeping at the same time.

Producer:	**Consumer:**

Repeat
Produce an item;
if count = 1 then sleep;
place item in buffer.
count = 1;
Wakeup(Consumer);
Forever
	Repeat
	if count = 0 then sleep;
	Remove item from buffer;
	count = 0;
	Wakeup(Producer);
	Consume item;
	Forever;

Practice Questions

Q.1 A critical section is a program segment.

(A) Which should run in certain specified amount of time
(B) Which avoids resources are accessed
(C) Where shared resources are accessed
(D) Which must be enclosed by a pair of semaphore operations, P and V

[GATE 1996 : IISc Bangalore]

Q.2 When several processes access the same data concurrently and the outcome of the execution depends on the particular order in which the access takes place is called ________

(A) dynamic condition
(B) race condition
(C) essential condition
(D) critical condition

[GATE 1996 : IISc Bangalore]

Q.3 Mutual exclusion can be provided by the ___________

(A) mutex locks
(B) binary semaphores
(C) Semaphore only
(D) Spin lock

[GATE 1996 : IISc Bangalore]

Q.4 Process synchronization can be done on ___________

(A) hardware level
(B) software level
(C) both hardware and software level
(D) none of the mentioned

GATE 1996 : IISc Bangalore]

Q.5 Consider Peterson's algorithm for mutual exclusion between two concurrent processes i and j. The program executed by process i is shown below.

```
    Repeat
    flag [i] = true;
    turn = j;
    while (P) do no-op;
Enter critical section, perform actions,
Then exit critical section
    flag [i] = false;
Perform other non- critical section action.
    until false;
```

For the program to guarantee mutual exclusion, the predicate P in the while loop should be

(A) flag [j] = true and turn = i
(B) flag [j] = true and turn = j
(C) flag [i] = true and turn = j
(D) flag [i] = true and turn = i

[GATE 2001 : IIT Kanpur]

Q.6 Two processes, P1 and P2, need to access a critical section of code. Consider the following synchronization construct used by the processes: Here, wants1 and wants2 are shared variables, which are initialized to false. Which one of the following statements is TRUE about the above construct?

/* P1 */	/* P2 */

while (true) {
wants1 = true;
while (wants2 == true);
/* Critical Section */
wants1=false;
}
/* Remainder section */
	while (true) {
	wants2 = true;
	while (wants1==true);
	/* Critical Section */
	wants2 = false;
	}
	/* Remainder section */

(A) It does not ensure mutual exclusion.
(B) It does not ensure bounded waiting.
(C) It requires that processes enter the critical section in strict alternation.
(D) It does not prevent deadlocks, but ensures mutual exclusion.

[GATE 2007 : IIT Kanpur]

Q.7 A Counting semaphore was initialized to 10. Then 6P (Wait) operations and 4V (Signal) operations were completed on this semaphore. The resulting value of the semaphore is _____.

(A) 0 (B) 8
(C) 10 (D) 12

[GATE 1998 : IIT Delhi]

Q.8 Consider two processes P_1 and P_2 accessing the shared variables X and Y protected by two binary semaphores S_X and S_Y and respectively, both initialized to 1. P and V denote the usual semaphore operators, where P decrements the semaphore value, and V increments the semaphore value. The pseudo-code of P_1 and P_2 is as follows :

```
    P1:
    where true do{
    L1: ........................
    L2: ........................
    X = X + 1;
    Y = Y – 1:
    V(SX);
    V(SY);
}
    P2:
    where true do
{
    L3: ........................
    L4: ........................
    Y = Y + 1;
    X = X – 1:
    V(SY);
    V(SX);
    }
```

In order to avoid deadlock, the operators at L_1, L_2, L_3 and L_4 are respectively .

(A) $P(S_Y), P(S_X); P(S_X), P(S_Y)$
(B) $P(S_X), P(S_Y); P(S_Y), P(S_X)$
(C) $P(S_X), P(S_X); P(S_Y), P(S_Y)$
(D) $P(S_X), P(S_Y); P(S_X), P(S_Y)$

[GATE 2004 : IIT Delhi]

Q.9 Select the correct statements regarding mutex lock to prevent race condition:

(A) A process must acquire the lock before entering a critical section
(B) A process need not acquire the lock before entering a critical section;
(C) It releases the lock when it exits the critical section;
(D) A process must acquire the lock when it exits the critical section

[GATE 2004 : IIT Delhi]

Q.10 A certain computation generates two arrays a and b such that $a[i] = f(i)$ for $0 \le i < n$ and $b[i] = g(a[i])$ for $0 \le i < n$. Suppose this computation is decomposed into two concurrent processes X and Y such that X computes the array a and Y computes the array b. The processes employ two binary semaphores R and S, both initialized to zero. The array a is shared by the processes are shown below.

Process X:	**Process Y:**

private i;
for(i=0;i<n; i++)
{
a[i] = f(i);
exit X (R,S);
}
	private i;
	for(i = 0; i<n; i++)
	{
	entry Y (R,S);
	b[i] = g(a[i]);
	}

Which one of the following represents the CORRECT implementation of Exit X and Entry Y?

(A)	ExitX(R, S) {
	P(R);
	V(S);
	}
	
	EntryY (R, S) {
	P(S);
	V(R);
	}
		(B)	ExitX(R, S) {
			V(R);
			V(S);
			}
			
			EntryY(R, S) {
			P(R);
			P(S);
			}

(C)	ExitX(R, S)
	{
	P(S);
	V(R);
	}
	EntryY(R, S) {
	V(S);
	P(R);
	}
		(D)	ExitX(R, S) {
			V(R);
			P(S);
			}
			EntryY(R, S) {
			V(S);
			P(R);
			}

[GATE 2013 : IIT Bombay]

Q.11 Fetch _And _Add (X, i) is an atomic Read- Modify- write instruction that reads the value of memory location X, increments it by the value 1 and returns the old value of X. It is used in the pseudo code shown below to implement a busy- wait lock. L is an unsigned integer shared variable initialized to 0. The value of 0 corresponds to lock being available, while any non-zero value corresponds to the lock being not available.

```
AcquireLock(L)
{
    while (Fetch _ And _Add (L,1))
    L = 1;
}
ReleaseLock(L)
{
    L = 0;
}
```

This implementation

(A) fails as L can overflow

(B) fails as L can take on a non-zero value when the lock is actually available

(C) work correctly but may starve some processes

(D) works correctly without starvation

[GATE 2012 : IIT Delhi]

Q.12 Consider the following two-process synchronization solution

Process 0 Entry: loop while (turn == 1);
(critical section) Exit: turn = 1;

Process 1 Entry: loop while (turn == 0);
(critical section) Exit: turn = 0;

The shared variable turn is initialized to zero. Which one of the following is TRUE?

(A) This is a correct two-process synchronization solution.

(B) This solution violates mutual exclusion requirement.

(C) This solution violates progress requirement.

(D) This solution violates bounded wait requirement.

[GATE 2016 : IISc Bangalore]

Q.13 Fill in the boxes below to get a solution for the readers-writers problem, using a single binary semaphore, mutex (initialized to 1) and busy waiting. Write the box numbers (1, 2 and 3), and their contents in your answer book.

L1	L2

int R = 0, W = 0;
Reader () {
wait (mutex);
if (W == 0) {
R = R + 1;
□
______________(1)
	Writer () {
	wait (mutex);
	if (□) { ________ (3)
	signal (mutex);
	goto L2;
	}
	W=1;

```
    }
    else {
     ▯▭
________________(2)
        goto L1;
    }
    ..../* do the
read*/
    wait (mutex);
    R = R - 1;
    signal (mutex);
```

```
    signal (mutex);
    ...../*do the write*/
    wait( mutex);
    W=0;
    signal (mutex);
}
```

3 Deadlock

Classroom Questions

Q.1 An operating system contains 3 user processes each requiring 2 units of resource R. The minimum number of units of R such that no deadlocks will ever arise is:

(A) 3 (B) 5

(C) 4 (D) 6

[GATE 1997 : IIT Madras]

Q.2 A Computer system has 6 tape drives, with n processes competing for them. Each process may need 3 tape drives. The maximum value of n for which the system is guaranteed to be deadlock free is

(A) 2 (B) 3

(C) 4 (D) 1

[GATE 1992 : IIT Delhi]

Q.3 Consider a system having m resources of the same type. These resources are shared by 3 processes A, B and C, which have peak demands of 3, 4 and 6 respectively. For what value of m deadlock will not occur?

(A) 7 (B) 9

(C) 10 (D) 13

[GATE 1993 : IIT Bombay]

Q.4 A system has 6 identical resources and N processes competing for them. Each process can request atmost 2 resources. Which one of the following values of N could lead to a deadlock?

(A) 1 (B) 2

(C) 3 (D) 4

[GATE 2015 : IIT Kanpur]

Q.5 Consider a system with 3 processes that share 4 instances of the same resource type. Each process can request a maximum of K instances. Resource instances can be requested and released only one at a time. The largest value of K that will always avoid deadlock is ___________.

[GATE 2018 : IIT Guwahati]

Q.6 Consider a system with 4 types of resources R1 (3 units), R2 (2 units), R3 (3 units), R4 (2 units). A non-preemptive resource allocation policy is used. At any given instance, a request is not entertained if it cannot be completely satisfied. Three process P1, P2, P3 request the resources as follows if executed independently.

Process P1:	**Process P2:**	**Process P3:**
t = 0 : requests 2 units of R2 t = 1 : requests 1 unit of R3 t = 3 : request 2 units of R1 t = 5 : releases 1 unit of R2 and 1 unit of R1 t = 7 : releases 1 unit of R3 t = 8 : requests 2 units of r4 t = 10 : finishes	t = 0 : requests 2 units of R3 t = 2 : requests 1 unit of R4 t = 4 : request 1 unit of R1 t = 6 : releases 1 unit of R3 t = 8 : finishes	t = 0 : requests 1 unit of R4 t = 2 : requests 2 unit of R1 t = 5 : releases 2 units of R1 t = 7 : requests 1 unit of R2 t = 8 : requests 1 units of R3 t = 9 : Finishes

Which one of the following statements is **TRUE** if all three processes run concurrently starting at time t = 0?

(A) All processes will finish without any deadlock

(B) Only P1 and P2 will be in deadlock

(C) Only p1 and P3 will be in deadlock

(D) All three processes will be in deadlock

[GATE 2009 : IIT Roorkee]

Q.7 An operating system implements a policy that requires a process to release all resources before making a request for another resource. Select the TRUE statement from the following:

(A) Both starvation and deadlock can occur

(B) Starvation can occur but deadlock cannot occur

(C) Starvation cannot occur but deadlock can occur

(D) Neither starvation nor deadlock can occur

[GATE 2008 : IISc Bangalore]

Q.8 Consider the following policies for preventing deadlock in a system with mutually exclusive resources.

I. Processes should acquire all their resources at the beginning of execution. If any resource acquired so far are released.

II. The resources are numbered uniquely, and processes are allowed to request for resources only in increasing resource numbers.

III. The resources are numbered uniquely, and processes are allowed to request for resources only in decreasing resource numbers.

IV. The resources are numbered uniquely, a processes is allowed to request only for a resources with resource number larger than its currently held resources.

Which of the above policies can be used for preventing deadlock?

(A) Any one of I and III but not II or IV

(B) Any one of I, III, and IV but not II

(C) Any one of II and III but not I or IV

(D) Any one of I, II, III, and IV

[GATE 2015 : IIT Kanpur]

Q.9 Which of the following is NOT a valid deadlock prevention scheme?

(A) Release all resources before requesting a new resource

(B) Number the resources uniquely and never request a lower numbered resource than the last one requested.

(C) Never request a resource after releasing any resource.

(D) Request and all required resources allocated before execution.

[GATE 2000 : IIT Kharagpur]

Q.10 Suppose n processes, $p_1.......p_n$ share m identical resource units, which can be reserved and released on at time. The maximum resource requirement of process p_i is s_i where $s_i > 0$. Which one of the following is a sufficient condition for ensuring that deadlock does not occur?

(A) $\forall i, s_i < m$

(B) $\forall i, s_i < n$

(C) $\sum_{i=1}^{n} s_i < (m+n)$

(D) $\sum_{i=1}^{n} s_i < (m \times n)$

[GATE 2005 : IIT Bombay]

Q.11 Consider the following snapshot of a system running n processes. Process i is holding x_i instances of a resource R, for $1 \le i \le n$. Currently, all instances of R are occupied. Further, for all I, process I has placed a request for an additional y_i instances while holding the x_i instances it already has. There are exactly two processes P and q such that $y_p = y_q = 0$.

Which one of the following conditions guarantees that no other process apart from p and q can complete execution?

(A) $X_p + X_q <$ Min $\{Y_k: 1 \le k \le n, k \ne p, k \ne q\}$

(B) Min $(X_p, X_q) \ge$ Min $\{Y_k : 1 \le k \le n, k \ne p, k \ne q\}$

(C) Min $(X_p, X_q) \le$ Max $\{Y_k : 1 \le k \le n, k \ne p, k \ne q\}$

(D) $X_p + X_q <$ Max $\{Y_k : 1 \le k \le n, k \ne p, k \ne q$

[GATE 2019 : IIT Madras]

Q.12 A system has n resources $R_{0.........}R_{n-1,}$ and K processes $P_{0.........}P_{k-1}$. The implementation of the resource request logic of each process $P_{i,}$ is as follows:

```
if (i%2==0)
{
    if (i<n) request Ri ;
    if(i+2<n) request Ri+2;
}
else
{
```

if (i<n) request R_{n-1};
if (i+2<n) request R_{n-i-2};
}

In which one of the following situations is a deadlock possible?

(A) n =40, k = 26 (B) n = 21, k = 12
(C) n = 20, k = 10 (D) n = 41, k = 9

[GATE 2010 : IIT Guwahati]

Q.13 A system shares 9 tape drives. The current allocation and maximum requirement of allocation and maximum requirement of tape drives for three processes are shown below:

Process	Current Allocation	Maximum Requirement
P1	3	7
P2	1	6
P3	3	5

Which of the following best describes current state of the system?

(A) Sate, Deadlocked
(B) Sate, Not Deadlocked
(C) Not Sate, Deadlocked
(D) Not Sate, Not Deadlocked

[GATE 2017 : IIT Roorkee]

Q.14 An operating system uses the Banker' algorithm for deadlock avoidance when managing the allocation of three resource types X, Y, and Z to three processes PO, P1, and P2. The table given below presents the current system state. Here, the allocation matrix shows the current number of resources of each type allocated to each process and the max matrix shows the maximum number of resources of each type required by each process during its execution.

	Allocation			Max		
	X	Y	Z	X	Y	Z
P0	0	0	1	8	4	3
P1	3	2	1	6	2	0
P2	2	1	1	3	3	3

There are 3 units of X, 2 units of type Y and 2 units of type Z still available. The system is currently in a safe state. Consider the following independent requests for additional resources in the current state:

REQ1: P0 requests 0 units of X, 0 units of Y and 2 units of Z

REQ2: P1 requests 2 units of X, 0 units of Y and 0 units of Z

Which one of the following is **TRUE**?

(A) Only REQ1 can be permitted.
(B) Only REQ2 can be permitted.
(C) Both REQ1 and REQ2 can be permitted.
(D) Neither REQ1 nor REQ2 can be permitted.

[GATE 2014 : IIT Kharagpur]

Q.15 In a system, there are three types of resources : *E*, *F* and *G*. Four processes P_0, P_1, P_2 and P_3 execute concurrently. At the outset, the processes have declared their maximum resource requirements using a matrix named Max as given below. For example, Max $[P_2, F]$ is the maximum number of instances of *F* that P_2 would require. The number of instances of the resources allocated to the various processes at any given state is given by a matrix named Allocation.

Consider a state of the system with the Allocation matrix as shown below, and in which 3 instances of *E* and 3 instances of *F* are the only resources available.

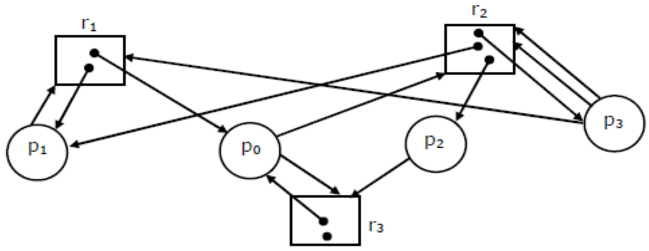
Allocation				Max			
	E	F	G		E	F	G
P_0	1	0	1	P_0	4	3	1
P_1	1	1	2	P_1	2	1	4
P_2	1	0	3	P_2	1	3	3
P_3	2	0	0	P_3	5	4	1

From the perspective of deadlock avoidance, which one of the following is true?

(A) The system is in *safe* state.
(B) The system is not in *safe* state, but would be *safe* if one more instance of *E* were available
(C) The system is not in *safe* state, but would be *safe* if one more instance of *F* were available
(D) The system is not in *safe* state, but would be *safe* if one more instance of *G* were available

[GATE 2018 : IIT Guwahati]

Q.16 Consider the resource allocation graph given in the figure. Which of these processes will finish LAST?



(A) P0
(B) P3
(C) P2
(D) None of the above, the system is in deadlock state.

[GATE 1994 : IIT Kharagpur]

Practice Questions

Q.1 A Computer has six tape drives, with n processes competing for them. Each process may need two drives. What is the maximum value of n for the system to be deadlock free?

(A) 6 (B) 5
(C) 4 (D) 3

[GATE 1998 : IIT Delhi]

Q.2 A system contains three programs and each requires three tape units for its operation. The minimum number of tape units which the system must have such that deadlocks never arise is ________.

[GATE 2014 : IIT Kharagpur]

Q.3 A system is having 10 user processes P1, P2, P3 . . . P10 where P1 requires 2 units of resource R, P2 requires 3 units of resource R, P3 requires 4 units of resource R and so on. The minimum number of units of R that ensures no deadlock is ________

Q.4 Which of the following is **NOT** true of deadlock prevention and deadlock avoidance schemes?

(A) In deadlock prevention, the request for resources is always granted if the resulting state is safe
(B) In deadlock avoidance, the request for resources is always granted if the resulting state is safe
(C) Deadlock avoidance is less restrictive than deadlock prevention.
(D) Deadlock avoidance requires knowledge of resource requirements a priori.

[GATE 2008 : IISc Bangalore]

Q.5 The circular wait condition can be prevented by __________

(A) defining a linear ordering of resource types
(B) using thread
(C) using pipes
(D) all of the mentioned

[GATE 2008 : IISc Bangalore]

Q.6 A single processor system has three resource types X, Y, and Z which are shared by three processes. There are 5 units of each resource type. Consider the following scenario, where the column alloc denotes the number of units of each resource type allocated to each process, and the column request denotes the number of units of each resource type requested by a process in order to complete execution, Which of these processes will finish last?

	Alloc			request		
	X	Y	Z	X	Y	Z
P0	1	2	1	1	0	3
P1	2	0	1	0	1	2
P2	2	2	1	1	2	0

(A) P0
(B) P1
(C) P2
(D) None of the above, since system is in a deadlock

[GATE 2005 : IIT Bombay]

Q.7 Which of the following phenomena are responsible for deadlock?

(A) A process holding at least one resource is waiting to acquire additional resourced held by other processes.
(B) Only one process at a time can use a resource.
(C) A resource can be released only voluntarily by the process holding it, after that process has completed its task.
(D) All of the above

[GATE 2005 : IIT Bombay]

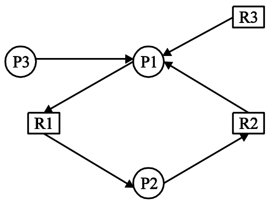
Q.8 Match the following columns while considering the case of deadlock prevention.

I.	Mutual Exclusion	**P.**	Low resource utilization and possibility of starvation
II.	Hold and wait	**Q.**	Not practical for most systems
III.	No preemption	**S.**	Must hold for non-sharable resources, thus can't be used for deadlock prevention

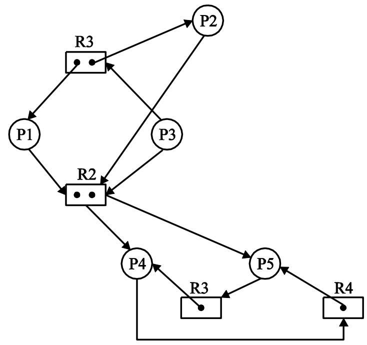
(A) I-S, II-Q, III-P
(B) I-Q, II-S, III-P
(C) I-S, II-P, III-Q
(D) None of the above

[GATE 2005 : IIT Bombay]

Q.9 Consider the resource allocation graphs G1 and G2. What can be said about deadlock condition?



Graph G1



Graph G2

(A) Both are deadlocked
(B) G1 deadlocked, G2 not
(C) G1 not, G2 deadlocked
(D) None is deadlocked

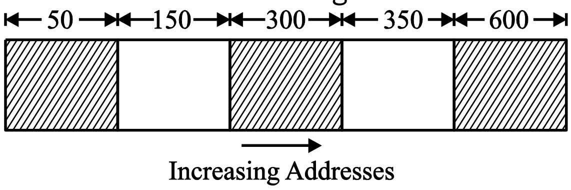
[GATE 2005 : IIT Bombay]

❑❑❑

4 Memory Management and Virtual Memory

Classroom Questions

Q.1 Consider the following heap (in given figure) in which blank regions are not in use and hatched regions are in use.



The sequence of requests for blocks of sizes 300, 25, 125, 50 can be satisfied if we use:

(A) Either first fit or best fit policy (any one)
(B) First fit but not best fit policy
(C) Best fit but not first fit policy
(D) None of the above

[*GATE 1994 : IIT* Kharagpur]

Q.2 Consider six memory partitions of sizes 200 KB, 400 KB, 600 KB 500 KB, 300 KB and 250 KB, where KB refers to kilobyte. These partitions need to be allotted to four processes of size 357 KB, 210 KB, 468 KB and 491 KB in that order. If the best fit algorithm is used, which partitions are NOT allotted to any process?

(A) 200 KB and 300 KB
(B) 200 KB and 250 KB
(C) 250 KB and 300 KB
(D) 300 KB and 400 KB

[GATE 2015 : IIT Kanpur]

Q.3 Let a memory have four free blocks of sizes 4k, 8k, 20k, 2k. These blocks are allocated following the best-fit strategy. The allocation requests are stored in a queue as shown below.

Request No	Request sizes	Usage Time
J_1	2k	4
J_2	14k	10
J_3	3k	2
J_4	6k	8
J_5	6k	4
J_6	10k	1
J_7	7k	8
J_8	20k	6

The time at which the request for J7 will be completed will be

(A) 16 (B) 19
(C) 20 (D) 37

[GATE 2007 : IIT Kanpur]

Q.4 Consider allocation of memory to a new process. Assume that none of the existing holes in the memory will exactly fit the process's memory requirement. Hence, a new hole of smaller size will be created if allocation is made in any of the existing holes. Which one of the following statement is TRUE?

(A) The hole created by first fit is always larger than the hole created by next fit.
(B) The hole created by worst fit is always larger than the hole created by first fit.
(C) The hole created by best fit is never larger than the hole created by first fit.
(D) The hole created by next fit is never larger than the hole created by best fit.

[GATE 2020 : IIT Delhi]

Q.5 Consider a system with byte-addressable memory, 32-bit logical addresses, 4 kilobyte page size and page table entries of 4 bytes each. The size of the page table in the system in megabytes is ________.

[GATE 2015 : IIT Kanpur]

Q.6 Consider a machine with 64 MB physical memory and a 32- bit virtual address space. If the page size is 4KB, what is the approximate size of the page table?

(A) 16 M (B) 8 MB
(C) 2MB (D) 24MB

[GATE 2001 : IIT Kanpur]

Q.7 In a virtual memory system, size of virtual address is 32-bit, size of physical address is 30-bit, page size is 4 Kbyte and size of each page table entry is 32-bit. The main memory is byte addressable. Which one of the following is the maximum number of bits that can be used for storing protection and other information in each page table entry?

(A) 2 (B) 10
(C) 12 (D) 14

[GATE 2004 : IIT Delhi]

Q.8 Consider a computer system with 40-bit virtual addressing and page size of sixteen kilobytes. If the computer system has a one- level page table per process and each page table entry requires 48 bits, then the size of the per- process page table is ________ megabytes.

[GATE 2016 : IISc Bangalore]

Q.9 Let the page fault service time be 10ms in a computer with average memory access time being 20ns. If one page fault is generated for every 10^6 memory accesses, what is the effective access time for the memory?

(A) 21ns (B) 30ns
(C) 23ns (D) 35ns

[GATE 2011 : IIT Madras]

Q.10 If an instruction takes i microseconds and a page fault takes an additional j microseconds, the effective instruction time if on the average a page fault occurs every k instruction is ________.

(A) i + (j/k) (B) i + j*k
(C) (i + j) / k (D) (i + j) *k

[GATE 1998 : IIT Delhi]

Q.11 Consider a system with a two- level paging scheme in which a regular memory access takes 150 nanoseconds and servicing a page fault takes 8 millisecond. An average instruction takes 100 nanoseconds of CPU time and two memory accesses. The TLB hit ratio is 90 % and the page fault rate is one in every 10,000 instruction. What is the effective average instruction execution time?

(A) 645 nanoseconds
(B) 1050 nanoseconds
(C) 1251 nanoseconds
(D) 1230 nanoseconds

[GATE 2004 : IIT Delhi]

Q.12 A CPU generates 32-bit virtual addresses. The page size is 4 KB. The processor has a translation look-aside buffer (TLB) which can hold a total of 128 page table entries and is 4-way set associative. The minimum size of the TLB tag is:

(A) 11 bits (B) 13 bits
(C) 15 bits (D) 20 bits

[GATE 2006 : IIT Kharagpur]

Q.13 A computer system implements a 40 bit virtual address, page size of 8 kilobytes, and a 128-entry translation look-aside buffer (TLB) organized into 32 sets each having four ways. Assume that the TLB tag does not store any process id. The minimum length of the TLB tag in bits is ________

(A) 20 (B) 10
(C) 11 (D) 22

[GATE 2015 : IIT Kanpur]

Q.14 Consider a process executing on an operating system that uses demand paging. The average time for a memory access in the system is M units if the corresponding memory page is available in memory, and D units if the memory access causes a page fault. It has been experimental measured that the average time taken for a memory access in the process is X units.

Which one of the following is the correct expression for the page fault rate experienced by the process?

(A) (D – M) / (X – M)
(B) (X – M) / (D – M)
(C) (D – X) / (D – M)
(D) (X – M) / (D – X)

[GATE 2018 : IIT Guwahati]

Q.15 A demand paging system takes 100 time units to service a page fault and

300 time units to replace a dirty page. Memory access time is 1 time unit. The probability of a page fault is p. In case of a page fault, the probability of page being dirty is also p. It is observed that the average access time is 3 time units. Then the value of p is

(A) 0.194 (B) 0.233

(C) 0.514 (D) 0.981

[GATE 2007 : IIT Kanpur]

Q.16 Assume that in a certain computer, the virtual addresses are 64 bits long and the physical addresses are 48 bits long. The memory is word addressable. The page size is 8k Band the word size is 4 bytes. The Translation Look-aside Buffer (TLB) in the address translation path has 128 valid entries. At most how many distinct virtual addresses can be translated without any TLB miss?

(A) 16×2^{10} (B) 8×2^{20}

(C) 4×2^{20} (D) 256×2^{10}

[GATE 2019 : IIT Madras]

Q.17 Consider a paging system that uses 1-level page table residing in main memory and a TLB for address translation. Each main memory access takes 100 ns and TLB lookup takes 20 ns. Each page transfer to/from the disk takes 5000 ns. Assume that the TLB hit ratio is 95%, page fault rate is 10%. Assume that for 20% of the total page faults, a dirty page has to be written back to disk before the required page is read from disk. TLB update time is negligible.

The average memory access time in ns (round off to 1 decimal places) is ________

[GATE 2020 : IIT Delhi]

Q.18 A processor uses 2- level page table for virtual to physical address translation. Page tables for both levels are stored in the main memory. Virtual and physical address are both 32 bits wide. The memory is byte addressable. For virtual to physical address translation, the 10 most significant bits for the virtual address are used as index into the first level page table while the next 10 bits are used as index into the second level page table.

The 12 least significant bits of the virtual address are used as offset within the page. Assume that the page table entries in both levels of page tables are 4 bytes wide. Further, the processor has a translation look- aside a buffer (TLB), with a hit a rate of 96%. The TLB caches recently used virtual page number and the corresponding physical page numbers. The processor also has a physically address cache with a hit rate of 90%. Main memory access time is 10 ns, cache access time 1 ns, and TLB access time also 10 ns.

Assuming that no page faults occur, the average time taken to access a virtual address is approximately (to the nearest 0.5 ns)

(A) 1.5 ns (B) 2 ns

(C) 3 ns (D) 4 ns

[GATE 2003 : IIT Madras]

Q.19 Consider the virtual page reference string

1, 2, 3, 2, 4, 1, 3, 2, 4, 1

On a demand paged virtual memory system running on a computer system that has main memory size of 3 page frames which are initially empty. Let LRU, FIFO and under the corresponding page replacement policy. Then

(A) OPTIMAL < LRU < FIFO

(B) OPTIMAL < FIFO < LRU

(C) OPTIMAL = LRU

(D) OPTIMAL = FIFO

[GATE 2012 : IIT Delhi]

Q.20 Consider a main memory with five page frames and the following sequence of page references: 3, 8, 2, 3, 9, 1, 6, 3, 8, 9, 3, 6, 2, 1, 3. which one of the following is true with respect to page replacement policies First in First out (FIFO) and Least Recently Used (LRU)?

(A) Both incur the same number of page fault

(B) FIFO incurs 2 more page fault than LRU

(C) LRU incurs 2 more page faults than FIFO

(D) FIFO incurs 1 more page faults than LRU

[GATE 2015 : IIT Kanpur]

Q.21 A system user FIFO policy for page replacement. It has 4 pages frames with no pages loaded to begin with. The system first accesses 100 distinct pages in some order and then reverse order. How many page faults will occur?

(A) 196 (B) 192

(C) 197 (D) 195

[GATE 2010 : IIT Guwahati]

Q.22 The Address sequence generated by tracing a particular program executing in a pure demand paging system with 100 records per page with 1 free main memory frame is recorded as following. What is the number of page faults?

0100,0200,0430,0499,0510,0530,0560,0120,0220,0240,0260,0320,0370

(A) 13 (B) 8

(C) 7 (D) 10

[GATE 1995 : IIT Kanpur]

Q.23 The address sequence generated by tracing a particular program executing in a pure demand paging system with 100 bytes per page is 0100, 0200, 0430, 0499, 0510, 0530, 0560, 0120, 0220, 0240, 0260, 0320, 0410.

Suppose that the memory can store only one page and if x is the address which causes a page fault then the bytes from addresses x to x + 99 are loaded on to the memory. How many page faults will occur ?

(A) 0 (B) 4

(C) 7 (D) 8

[GATE 2007 : IIT Kanpur]

Q.24 Assume that a main memory with only 4 pages, each of 16 bytes, is initially empty. The CPU generates the following sequence of virtual addresses and uses the Least Recently Used (LRU) page replacement policy.

0, 4, 8, 20, 24, 36, 44, 12, 68, 72, 80, 84, 28, 32, 88, 92

How many page faults does this sequence cause? What are the page numbers of the pages present in the main memory at the end of the sequence?

(A) 6 and 1, 2, 3, 4

(B) 7 and 1, 2, 4, 5

(C) 8 and 1, 2, 4, 5

(D) 9 and 1, 2, 3, 5

[GATE 2008 : IISc Bangalore]

Q.25 Which page replacement policy sometimes leads to more page faults when size of memory is increased?

(A) Optimal (B) LRU

(C) FIFO (D) None of these

[GATE 1992 : IIT Delhi]

Q.26 The optimal page replacement algorithm will select the page that.

(A) Has not been used for the longest time in the past

(B) Will not be used for the longest time in the future

(C) Has been used least number of times

(D) Has been used most number of times

[GATE 2002 : IISc Bangalore]

Q.27 In which one of the following page replacement policies, Belady's anomaly may occur?

(A) FIFO (B) Optimal

(C) LRU (D) MRU

[GATE 2009 : IIT Roorkee]

Q.28 Recall that Belady's anomaly is that the page-fault rate may increase as the number of allocated frames increases. Now, consider the following statements:

S1: Random page replacement algorithm (where a page chosen at random is replaced)
Suffers from Belady's anomaly

S2: LRU page replacement algorithm suffers from Belady's anomaly

Which of the following is CORRECT?

(A) S1 is true, S2 is true

(B) S1 is true, S2 is false

(C) S1 is false, S2 is true

(D) S1 is false, S2 is false

[GATE 2017 : IIT Roorkee]

Q.29 Consider a computer system with ten physical page frames. The system is provided with an access sequence ($a_1, a_2, ..., a_{20}, a_1, a_2, ...a_{20}$), where each a_i is a distinct virtual page number. The difference in the number of page faults between the last – in- first- out page replacement policy and the optimal page replacement policy is ________.

[GATE 2016 : IISc Bangalore]

Q.30 A processor uses 2- level page table for virtual to physical address translation. Page tables for both levels are stored in the main memory. Virtual and physical address are both 32 bits wide. The memory is byte addressable. For virtual to physical address translation, the 10 most significant bits for the virtual address are used as index into the first level page table while the next 10 bits are used as index into the second level page table. The 12 least significant bits of the virtual address are used as offset within the page. Assume that the page table entries in both levels of page tables are 4 bytes wide. Further, the processor has a translation look- aside a buffer (TLB), with a hit a rate of 96%. The TLB caches recently used virtual page number and the corresponding physical page numbers. The processor also has a physically address cache with a hit rate of 90%. Main memory access time is 10 ns, cache access time 1 ns, and TLB access time also 10 ns.

Suppose a process has only the following pages in its virtual address space: two contiguous code pages starting at virtual address 0x00000000, two contiguous data pages starting at virtual address 0x00400000, and a stack page starting at virtual address 0xFFFFF000. The amount of memory required for storing the page tables of this process is

(A) 8 KB (B) 12 KB

(C) 16 KB (D) 20 KB

[GATE 2003 : IIT Madras]

Q.31 A computer uses 64- bit virtual address, 32- bit physical address, and a three- level paged page table organization. The page table base register stores the base address of the first- level table(T_1), which occupies exactly one page. Each entry of T_1 store the base address of a page of the second- level table(T_2). Each entry of T_2 store the base address of a page of the third-level table(T_3). Each entry of T_3 store a page table entry (PTE). The PTE is 32 bits in size. The processor used in the computer has a 1 MB 16- way set associative virtually indexed physically tagged cache. The cache block size is 64 bytes.

What is the size of a page in KB in this computer?

(A) 2 (B) 4

(C) 8 (D) 16

[GATE 2013 : IIT Bombay]

Q.32 In the context of operating systems, which of the following statements is/are correct with respect to paging?

(A) Page size has no impact on internal fragmentation.

(B) Paging incurs memory overheads.

(C) Multi-level paging is necessary to support pages of different sizes.

(D) Paging helps solve the issue of external fragmentation.

[GATE 2021 : IIT Bombay]

Q.33 Consider a 3 level page table to translate a 39 bits virtual address to a physical address As shown below:

Level 1 offset	Level 2 offset	Level 3 offset	Page offset
9bit	9bit	9 bit	12 bit

The page size is 4KB (1 KB= 2^10 bytes) and page table entry size at every level is 8 bytes. A

Process P is currently using 2GB (1 GB=2^30 bytes) virtual memory which is mapped to 2GB

Of physical memory. The minimum amount of memory required for the page table of P across

All level is_____ KB.

[GATE 2021 : IIT Bombay]

Q.34 For each of the four processes P1, P2, P3 and P4. The total size in kilobytes

(KB) and the number of segments are given below.

Process	Total size (in KB)	Number of segments
P1	195	4
P2	254	5
P3	45	3
P4	364	8

The page size is 1 KB. The size of an entry in the page table is 4 bytes. The size of an entry in the segment table is 8 bytes. The maximum size of a segment is 256 KB. The paging method for memory management uses two-level paging, and its storage overhead is P. The storage overhead for the segmentation method is S. The storage overhead for the segmentation and paging method is T. What is the relation among the overheads for the different methods of memory management in the concurrent execution of the above four processes ?

(A) P < S < T (B) S < P < T

(C) S < T < P (D) T < S < P

[GATE 2006 : IIT Kharagpur]

Q.35 Which of the following statements is false?

(A) Virtual memory implements the translation of a program's address space into physical memory address space.

(B) Virtual memory allows each program to exceed the size of the primary memory.

(C) Virtual memory increases the degree of multiprogramming.

(D) Virtual memory reduces the context switching overhead.

[GATE 2001 : IIT Kangpur]

Q.36 Which of the following is/are advantages of virtual memory?

(A) Faster access to memory on an average.

(B) Processes can be given protected address spaces.

(C) Linker can assign addresses independent of where the program will be loaded in physical memory.

(D) Programs larger than the physical memory size can be run.

[GATE 1999 : IIT Bombay]

Q.37 The minimum number of page frames that must be allocated to a running process in a virtual memory environment is determined by.

(A) The instruction set architecture

(B) Page size

(C) Physical memory size

(D) Number of processes in memory

[GATE 2004 : IIT Delhi]

Q.38 A Computer system supports 32- bit virtual addresses as well as 32- bit physical addresses. Since the virtual address space is of the same size as the physical address space, the operating system designers decide to get rid of the virtual memory entirely. Which one of the following is true?

(A) Efficient implementation of multi-user support is no longer possible

(B) The processor cache organization can be made more efficient now

(C) Hardware support for memory management is no longer needed

(D) CPU scheduling can be made more efficient now

[GATE 2006 : IIT Kharagpur]

Q.39 Locality of reference implies that the page reference being made by a process

(A) Will always be to the page used in the previous page reference

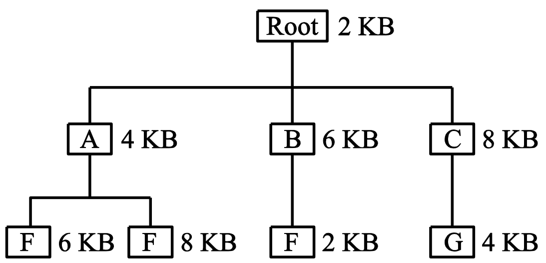
(B) Is likely to be one of the pages used in the last few page references

(C) Will always be to one of the pages existing in memory

(D) Will always lead to page fault

[GATE 1997 : IIT Madras]

Q.40 The overlay tree for a program is as shown below:



What will be the size of the partition (in physical memory) required to load (and run) this program?

(A) 12 KB (B) 14 KB

(C) 10KB (D) 8KB

[GATE 1998 : IIT Delhi]

Q.41 The capacity of memory units is defined by the number of words multiplied by the number of bits/work. How many separate address and data lines are needed for a memory of 4K × 16?

(A) 10 address, 16 data lines

(B) 11 address, 8 data lines

(C) 12 address, 16 data lines

(D) 13 address, 12 data lines

[GATE 1995 : IIT Kanpur]

Q.42 A computer installation has 1000k of main memory. The jobs arrive and finish in the following sequences.

Job 1 requiring 200k arrives
Job 2 requiring 350k arrives
Job 3 requiring 300k arrives
Job 1 finishes
Job 4 requiring 120k arrives
Job 5 requiring 150k arrives
Job 6 requiring 80k arrives

(a) Draw the memory allocation table using Best Fit and First fit algorithms.

(b) Which algorithm performs better for this sequence?

[GATE 1995 : IIT Kanpur]

Q.43 In a computer system where the 'best-fit' algorithm is used for allocating 'jobs' to 'memory partitions', the following situation was encountered

Partitions sizes in KB	4 K 8 K 20 K 2 K
Jobs sizes in KB	2 K 14 K 3 K 6 K 10 K 20 K 2 K
Time for execution	4 10 2 1 4 1 8 6

When will the 20 K job complete

[GATE 1998 : IIT Delhi]

Q.44 A certain computer system has the segmented paging architecture for virtual memory. The memory is byte addressable. Both virtual and physical address spaces contain 216 bytes each. The virtual address space is divided into 8 non-overlapping equal size segments. The memory management unit (MMU) has a hardware segment table, each entry of which contains the physical address of the page table for the segment. Page tables are stored in the main memory and consists of 2 byte page table entries.

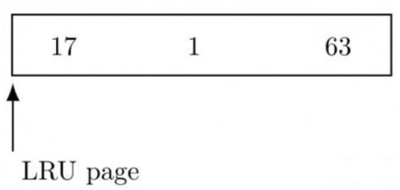
(a) What is the minimum page size in bytes so that the page table for a segment requires at most one page to store it? Assume that the page size can only be a power of 2.

(b) Now suppose that the pages size is 512 bytes. It is proposed to provide a TLB (Transaction look-aside buffer) for speeding up address translation. The proposed TLB will be capable of storing page table entries for 16 recently referenced virtual pages, in a fast cache that will use the direct mapping scheme. What is the number of tag bits that will need to be associated with each cache entry?

(c) Assume that each page table entry contains (besides other information) 1 valid bit, 3 bits for page protection and 1 dirty bit. How many bits are available in page table entry for storing the aging information for the page? Assume that the page size is 512 bytes.

[GATE 1999 : IIT Bombay]

Q.45 A demand paged virtual memory system uses 16 bit virtual address, page size of 256 bytes, and has 1 Kbyte of main memory.LRU page replacement is implemented using the list, whose current status (page number is decimal) is



For each hexadecimal address in the address sequence given below,

00FF, 010D, 10FF, 11B0

dicate

i. the new status of the list
ii. page faults, if any, and
iii. page replacements, if any.

[GATE 1996 : IISc Bangalore]

Q.46 How many 32K x 1 RAM chips are needed to provide a memory capacity of 256Kbytes?

(A) 8 (B) 32
(C) 64 (D) 128

[GATE 2009 : IIT Roorkee]

Q.47 Which of the following is not a form of memory?

(A) Instruction cache

(B) Instruction register

(C) Instruction opcode

(D) Translation look-a-side buffer

[GATE 2002 : IISc Bangalore]

Q.48 In a system with 32 bit virtual addresses and 1 KB page size, use of one- level page tables for virtual to physical address translation is not practical because of

(A) The large amount of internal fragmentation

(B) The large amount of external fragmentation

(C) The large memory overhead in maintaining page tables

(D) The large computation overhead in the translation process

[GATE 2003 : IIT Madras]

Q.49 In a resident OS computer, which of the following systems must reside in the main memory under all situations?

(A) Assembler (B) Linker
(C) Loader (D) Compiler

[GATE 1997 : IIT Madras]

Q.50 A ROM is used to store the table for multiplication of two 8-bit unsigned integers. The size of ROM required is

(A) 256 K Ã - 16 (B) 64 K Ã – 8
(C) 4 K Ã - 16 (D) 64 K Ã - 16

[GATE 1996 : IISc Bangalore]

Q.51 Consider a paging hardware with a TLB. Assume that the entire page table and all the pages are in the physical memory. It takes 10 millisecond to search the TLB and 80 millisecond to access the physical memory. If the TLB hit ratio is 0.6, the effective memory access time (in milliseconds) is __________.

[GATE 2014 : IIT Kharagpur]

Practice Questions

Q.1 The essential content(s) in each entry of a page table is/are.

(A) Virtual page number.

(B) Page frame number.

(C) Both virtual page number and page frame number

(D) Access right information.

[GATE 2009 : IIT Roorkee]

Q.2 In paged memory systems, if the page size is increased, then the internal fragmentation generally:

(A) Becomes less

(B) Becomes more

(C) Remains constant

(D) None of the mentioned

Q.3 Which of the following statements are true?

(A) The main objective for using cache memory is to increase the effective speed of the memory system.

(B) The main objective for using virtual memory is to increase the effective capacity of the memory system.

(C) The size of main memory is larger as compared to cache memory.

(D) Main memory is faster as compared to cache memory.

Q.4 A computer system implements 8 kilobyte pages and a 32- bit physical address space. Each page table entry contains a valid bit, a dirty bit, three permission bits, and the translation. If the maximum size of the page table of a process is 24 megabytes, the length of the virtual address supported by the system is _________ bits.

[GATE 2015 : IIT Kanpur]

Q.5 Dirty bit for a page in a page table

(A) Helps avoid unnecessary writes on paging device

(B) Helps maintain LUR information

(C) Allow only read on a page

(D) None of the above

[GATE 1997 : IIT Madras]

Q.6 Suppose the time to service a page fault is on the average 10 milliseconds, while a memory access takes 1 microsecond. Then a 99.99% hit ratio results in average memory access time of

(A) 1.9999 milliseconds

(B) 1 millisecond

(C) 9.999 microseconds

(D) 1.9999 microseconds

[GATE 2000 : IIT Kharagpur]

Q.7 A paging scheme uses a Translation Look-aside Buffer (TLB). A TLB-access takes 10 ns and a main memory access takes 50 ns. What is the effective access time (in ns) if the TLB hit ratio is 90% and there is no page-fault?

(A) 54 (B) 60

(C) 65 (D) 75

[GATE 2008 : IISc Bangalore]

Q.8 Thrashing

(A) Reduces page I/O

(B) Decreases the degree of multiprogramming

(C) Implies excessive page I/O

(D) Improves the system performance

[GATE 1997 : IIT Madras]

Q.9 Consider a virtual memory system with FIFO page replacement policy. For an arbitrary page access pattern, increasing the number of page frames in main memory will

(A) Always decrease the number of page faults

(B) Always increase the number of page faults

(C) Sometimes increase the number of page faults

(D) Never affect the number of page faults.

[GATE 2001 : IIT Kanpur]

Q.10 In which one of the following page replacement algorithms it is possible for the page fault rate to increase even when the number of allocated frames increases?

(A) LRU (Least Recently Used)

(B) OPT (Optimal Page Replacement)

(C) MRU (Most Recently Used)

(D) FIFO (First In First Out)

[GATE 2016 : IISc Bangalore]

Common Data for Questions 11 & 12

Q.11 A process has been allocated 3 page frames. Assume that none of the pages of the process are available in the memory initially. The process makes the following sequence of page references (reference string): 1, 2, 1, 3, 7, 4, 5, 6, 3, 1

If optimal page replacement policy is used, how many page faults occur for the above reference string?

(A) 7 (B) 8

(C) 9 (D) 10

[GATE 2007 : IIT Kanpur]

Q.12 Least Recently Used (LRU) page replacement policy is a practical approximation to optimal page replacement. For the above reference string, how many more page faults occur with LRU than with the optimal page replacement policy?

(A) 0 (B) 1

(C) 2 (D) 3

[GATE 2007 : IIT Kanpur]

Q.13 A multilevel page table is preferred in comparison to a single level page table for translating virtual address to physical address because

(A) It reduces the memory access time to read or write a memory location.

(B) It helps to reduce the size of page table needed to implement the virtual address space of a process.
(C) It is required by the translation lookaside buffer.
(D) It helps to reduce the number of page faults in page replacement algorithms.

[GATE 2009 : IIT Roorkee]

Q.14 How many address and data lines will be there for a 16 M x 32 memory system?
(A) 24 and 5
(B) 20 and 32
(C) 24 and 32
(D) None of the above

Q.15 The principle of locality justifies the use of
(A) Interrupts
(B) DMA
(C) Polling
(D) Cache Memory

[GATE 1995 : IIT Kanpur]

Q.16 What is the swap space in the disk used for?
(A) Saving temporary html pages
(B) Saving process data
(C) Storing the super-block
(D) Storing device drivers

[GATE 2005 : IIT Bombay]

Q.17 A computer system uses 32-bit virtual address, and 32-bit physical address. The physical memory is byte addressable, and the page size is 4 kbytes. It is decided to use two level page tables to translate from virtual address to physical address. Equal number of bits should be used for indexing first level and second level page table, and the size of each page table entry is 4 bytes.
(A) What is the number of page table entries that can be contained in each page?
(B) How many bits are available for storing protection and other information in each page table entry?

[GATE 2002 : IISc Bangalore]

Q.18 For implementing demand paging, it is required to –
(A) Distinguish between the pages that are in memory and the pages that are on disk
(B) Use a variation of valid-invalid scheme for protection
(C) Implement new MMU functionality
(D) All of the above

[GATE 2002 : IISc Bangalore]

❑❑❑

5 First System and Device Management

Classroom Questions

Q.1 The root directory of a disk should be placed:

(A) At a fixed address in main memory

(B) At a fixed location on the disk

(C) Any where on the disk

(D) At a fixed location on the system disk

[GATE 1993 : IIT Bombay]

Q.2 The data blocks of a very large file in the Unix file system are allocated using

(A) Contiguous allocation

(B) Linked allocation

(C) Indexed allocation

(D) An extension of indexed allocation.

[GATE 2008 : IISc Bangalore]

Q.3 The index node (inode) of a Unix-like file system has 12 direct, one single-indirect and one double-indirect pointer. The disk block size is 4 kB and the disk block addresses 32-bits long. The maximum possible file size is (rounded off to 1 decimal place) __________ GB.

[GATE 2019 : IIT Madras]

Q.5 In a particular Unix OS, each data block is of size 1024 bytes, each node has 10 direct data block addresses and three additional addresses: one for single indirect block, one for double indirect block and one for triple indirect block. Also, each block can contain addresses for 128 blocks. Which one of the following is approximately the maximum size of a file in the file system?

(A) 512 MB (B) 2GB

(C) 8GB (D) 16GB

[GATE 2004 : IIT Delhi]

Q.6 In a computer system, four files of size 11050 bytes, 4990 bytes, 5170 bytes and 12640 bytes need to be stored. For storing these files on disk, we can use either 100 byte disk blocks or 200 byte disk blocks (but can't mix block sizes). For each block used to store a file, 4 bytes of bookkeeping information also needs to be stored on the disk. Thus, the total space used to store a file is the sum of the space taken to store the file and the space taken to store the book keeping information for the blocks allocated for storing the file. A disk block can store either bookkeeping information for a file or data from a file, but not both. What is the total space required for storing the files using 100 byte disk blocks and 200 byte disk blocks respectively?

(A) 35400 and 35800 bytes

(B) 35800 and 35400 bytes

(C) 35600 and 35400 bytes

(D) 35400 and 35600 bytes

[GATE 2005 : IIT Bombay]

Q.6 If the overhead for formatting a disk is 96 bytes for a 4000 byte sector,

(a) Compute the unformatted capacity of the disk for the following parameters:

Number of surfaces: 8
Outer diameter of the disk: 12 cm
Inner diameter of the disk: 4 cm
Inner track space: 0.1 mm
Number of sectors per track: 20

(b) If the disk in (a) is rotating at 360 rpm, determine the effective data transfer rate which is defined as the number of bytes transferred per second between disk and memory.

[GATE 2011 : IIT Madras]

Q.7 A hard disk has 63 sectors per track, 10 platters each with 2 recording surfaces and 1000 cylinders. The address of a sector is given as a triple ⟨c,h,s⟩, where c is the cylinder number, h is the surface number and s is the sector number. Thus, the 0th sector is addresses as ⟨0,0,0⟩, the 1st sector as ⟨0,0,1⟩, and so on

The address of the 1039th sector is

(A) ⟨0,15,31⟩ (B) ⟨0,16,30⟩

(C) ⟨0,16,31⟩ (D) ⟨0,17,31⟩

[GATE 2009 : IIT Roorkee]

Q.8 A hard disk has 63 sectors per track, 10 platters each with 2 recording surfaces and 1000 cylinders. The address of a sector is given as a triple ⟨c,h,s⟩, where c is the cylinder number, h is the surface number and s is the sector number. Thus, the 0th sector is addresses as ⟨0,0,0⟩, the 1st sector as ⟨0,0,1⟩, and so on

The address ⟨400,16, 29⟩ corresponds to sector number:

(A) 505035 (B) 505036

(C) 505037 (D) 505038

[GATE 2009 : IIT Roorkee]

Q.9 An application loads 100 libraries at startup. Loading each library requires exactly one disk access. The seek time of the disk to a random location is given as 10 ms. Rotational speed of disk is 6000 rpm. If all 100 libraries are loaded from random locations on the disk, how long does it take to load all libraries? (The time to transfer data from the disk block once the head has been positioned at the start of the block may be neglected.)

(A) 0.50 s (B) 1.50 s

(C) 1.25 s (D) 1.00 s

[GATE 2011 : IIT Madras]

Q.10 A certain moving arm disk storage, with one head, has the following specifications.

Number of tracks/recording surface = 200

Disk rotation speed = 2400 rpm

Track storage capacity = 62,500 bits

The average latency of this device is P msec and the data transfer rate is Q bits/sec.

Write the value of P and Q

[GATE 1993 : IIT Bombay]

Q.11 A file system with a one-level directory structure is implemented on a disk with disk block size of 4 K bytes. The disk is used as follows:

Disk-block 0: File Allocation Table, consisting of one 8-bit entry per data block, representing the data block address of the next data block in the file

Disk block 1: Directory, with one 32 bit entry per file:

Disk block 2: Data block 1;

Disk block 3: Data block 2; etc.

(A) What is the maximum possible number of files?

(B) What is the maximum possible file size in blocks?

[GATE 1996 : IISc Bangalore]

Q.12 Consider a disk pack with 16 surfaces, 128 tracks per surface and 256 sectors per track. 512 bytes of data are stored in a bit serial manner in a sector. The capacity of the disk pack and the number of bits required to specify a particular sector in the disk are respectively:

(A) 256 Mbyte, 19 bits

(B) 256 Mbyte, 28 bits

(C) 512 Mbyte, 20 bits

(D) 64 Gbyte, 28 bits

[GATE 2007 : IIT Kanpur]

Q.13 A disk has 200 tracks (numbered 0 through 199). At a given time, it was servicing the request of reading data from track 120, and at the previous request, service was for track 90. The pending requests (in order of their arrival) are for track numbers.

30 70 115 130 110 80 20 25.

How many times will the head change its direction for the disk scheduling policies SSTF(Shortest Seek Time First) and FCFS (First Come Fist Serve)

(A) 2 and 3 (B) 3 and 3

(C) 3 and 4 (D) 4 and 4

[GATE 2004 : IIT Delhi]

Q.14 Disk requests come to disk driver for cylinders 10, 22, 20, 2, 40, 06 and 38, in that order at a time when the disk drive is reading from cylinder 20. The seek time is 6 msec per cylinder. Computer the total seek time if the disk arm scheduling algorithm is

(A) First come first served

(B) Closest cylinder next

[GATE 2014 : IIT Kharagpur]

Q.15 Consider a disk queue with requests for I/O to blocks on cylinders 47, 38, 121, 191, 87, 11, 92, 10. The C-LOOK scheduling algorithm is used. The head is initially at cylinder number 63, moving towards larger cylinder numbers on its servicing pass. The cylinders are numbered from 0 to 199. The total head movement (in number of cylinders) incurred while servicing these requests is ________.

[GATE 2016 : IISc Bangalore]

Q.16 Suppose the following disk request sequence (track number) for a disk with 100 tracks is given: 45, 20, 90, 10, 50, 60, 80, 25, 70. Assume that the initial position of the R/W head is on track 50. The additional distance that will be traversed by the R/W head when the Shortest Seek Time First (SSTF) algorithm is used compared to the SCAN (Elevator) algorithm (assuming that SCAN algorithm moves towards 100 when it starts execution) is ________ tracks.

[GATE 2015 : IIT Kanpur]

Q.17 Consider a disk system with 100 cylinders. The requests to access the cylinders occur in following sequence:

4, 34, 10, 7, 19, 73, 2, 15, 6, 20

Assuming that the head is currently at cylinder 50, what is the time taken to satisfy all requests if it takes 1 ms to move form one cylinder to adjacent one and shortest seek time first policy is used?

(A) 95 ms (B) 119 ms

(C) 233 ms (D) 276 ms

[GATE 2009 : IIT Roorkee]

Q.18 Consider a storage disk with 4 platters (numbered as 0, 1, 2 and 3), 200 cylinders (numbered as 0, 1, ... , 199), and 256 sectors per track (numbered as 0, 1, ... 255). The following 6 disk requests of the form [sector number, cylinder number, platter number] are received by the disk controller at the same time:

[120, 72, 2], [180, 134, 1], [60, 20, 0], [212, 86, 3], [56, 116, 2], [118, 16, 1]

Currently head is positioned at sector number 100 of cylinder 80, and is moving towards higher cylinder numbers. The average power dissipation in moving the head over 100 cylinders is 20 milliwatts and for reversing the direction of the head movement once is 15 milliwatts. Power dissipation associated with rotational latency and switching of head between different platters is negligible.

The total power consumption in milliwatts to satisfy all of the above disk requests using the Shortest Seek Time First disk scheduling algorithm is ______ .

[GATE 2018 : IIT Guwahati]

Q.19 Consider a disk with the 100 tracks numbered from 0 to 99 rotating at 3000 rpm. The number of sectors per track is 100. the time to move the head between two successive tracks is 0.2 millisecond.

Consider a set of disk requests to read data from tracks 32, 7, 45, 5 and 10. Assuming that the elevator algorithm is used to schedule disk requests, and the head is initially at track 25 moving up (towards larger track numbers), what is the total seek time for servicing the requests?

[GATE 2001 : IIT Kanpur]

Q.20 The head of a hard disk serves requests following the shortest seek time first (SSTF) policy. The head is initially positioned at track number 180.

Which of the request sets will cause the head to change its direction after servicing every request assuming that the head does not change direction if there is a tie in SSTF and all the

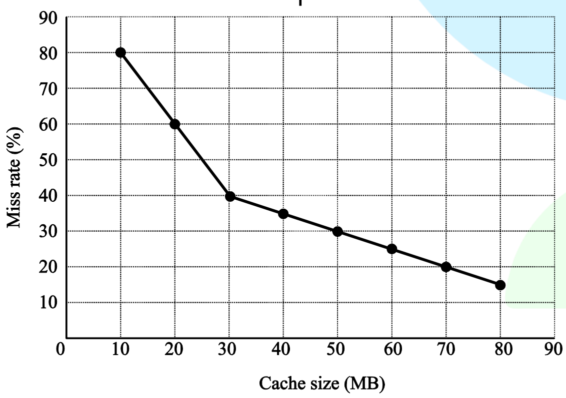
requests arrive before the servicing starts?
(A) 11, 139, 170, 178, 181, 184, 201, 265
(B) 10, 138, 170, 178, 181, 185, 201, 265
(C) 10, 139, 169, 178, 181, 184, 201, 265
(D) 10, 138, 170, 178, 181, 185, 200, 265

[GATE 2007 : IIT Kanpur]

Q.21 What is the maximum cardinality of the request set, so that the head changes its direction after servicing every request if the total number of tracks are 2048 and the head can start from any track?
(A) 9 (B) 10
(C) 11 (D) 12

[GATE 2007 : IIT Kanpur]

Q.22 A file system uses an in- memory cache to cache disk blocks. The miss rate of the cache is shown in the figure. The latency to read a block from the cache is 1 ms and to read a block from the disk is 10 ms. Assume that the cost of checking whether a block exists in the cache is negligible. Available cache sizes are in multiples of 10 MB.



The smallest cache size required to ensure an average read latency of less than 6 ms is_________ MB.

[GATE 2016 : IISc Bangalore]

Q.23 Consider an operating system capable of loading and executing a single sequential user process at a time. The disk head scheduling algorithm used is First Come First Served (FCFS). If FCFS is replaced by Shortest Seek Time First (SSTF), claimed by the vendor to give 50% better benchmark results, what is the expected improvement in the I/O performance of user programs?
(A) 50% (B) 40%
(C) 25% (D) 0%

[GATE 2004 : IIT Delhi]

Q.24 Which of the following disk scheduling strategies is likely to give the best throughput?
(A) Farthest cylinder next
(B) Nearest cylinder next
(C) First come first served
(D) Elevator algorithm

[GATE 1999 : IIT Bombay]

Q.25 Using a larger block size in a fixed block size file system leads to
(A) Better disk throughput but poorer disk space utilization
(B) Better disk throughput and better disk space utilization
(C) Poorer disk throughput but better disk space utilization
(D) Poorer disk throughput and poorer disk space utilization

[GATE 2003 : IIT Madras]

Q.26 When an interrupt occurs, an Operating System
(A) Ignores the interrupt
(B) Always changes state of interrupted process to 'blocked' and schedules another process.
(C) Always resumes execution of interrupted process after processing the interrupts.
(D) May change state of interrupted processor to "blocked and schedule another process.

[GATE 1997 : IIT Madras]

Q.27 Which of the following devices should get higher priority in assigning interrupts?
(A) Hard disk (B) Printer
(C) Keyboard (D) Floppy disk

[GATE 1998 : IIT Delhi]

Q.28 Listed below are some operating system abstraction (in the left column) and the hardware components (Right column). Which matching pair is correct?

List - I	**List – II**
(a) Thread	1. Interrupt
(b) Virtual Address space	2. Memory
(c) File System	3. CPU
(d) Signal	4. Disk

(A) A-2, B-4, C-3, D-1

(B) A-1, B-2, C-3, D-4

(C) A-3, B-2, C-4, D-1

(D) A-4, B-1, C-2, D-3

[GATE 1999 : IIT Bombay]

Q.29 Which of the following requires a device driver?

(A) Register (B) Cache

(C) Main memory (D) Disk

[GATE 2001 : IIT Kanpur]

Q.30 A file system with 300 GB disk uses a file descriptor with 8 direct block addresses, 1 indirect block address and 1 doubly indirect block address. The size of each disk block is 128 Bytes and the size of each disk block address is 8 Bytes. The maximum possible file size in this file system is

(A) 3 KB

(B) 35 KB

(C) 280 KB

(D) Dependent on the size of the disk

[GATE 2012 : IIT Delhi]

Practice Questions

Q.1 In a file allocation system, which of the following allocation scheme(s) can be used if no external fragmentation is allowed?

I. Contiguous

II. Linked

III. Indexed

(A) I and III only (B) II only

(C) III only (D) II and III only

Q.2 In the index allocation scheme of blocks to a file, the maximum possible size of the file depends on

(A) The size of the blocks, and the size of the address of the blocks.

(B) The number of blocks used for the index, and the size of the blocks.

(C) The size of the blocks, the number of blocks used for the index, and the size of the address of the blocks.

(D) None of the above.

Q.3 The correct matching for the following

List- I	**List- II**
(a) Disk scheduling	(1) Round robin
(b) Batch processing	(2) SCAN
(c) Time sharing	(3) LIFO
(d) Interrupt processing	(4) FIFO

(A) a-3, b-4, c-2, d-1

(B) a-4, b-3, c-2, d-1

(C) a-2, b-4, c-1, d-3

(D) a-3, b-4, c-3, d-2

Q.4 Consider a hard disk with 16 recording surfaces (0-15) having 16384 cylinders (0-16383) and each cylinder contains 64 sectors (0-63). Data storage capacity in each sector is 512 bytes. Data are organized cylinder-wise and the addressing format is. A file of size 42797 KB is stored in the disk and the starting disk location of the file is <1200, 9, 40>. What is the cylinder number of the last sector of the file, if it is stored in a contiguous manner?

(A) 1281 (B) 1282

(C) 1283 (D) 1284

Q.5 A certain moving arm disc-storage has the following a specification:

Number of track per surface=4004

Track the storage capacity=130030 bytes

Disk speed =3600 rpm

Average seek time=30 m secs.

Estimate the average latency the disc storage capacity and that data transfer rate.

Q.6 Suppose a disk has 201 cylinders, numbered from 0 to 200. At some time the disk arm is at cylinder 100, and there is a queue of disk access requests for cylinders 30, 85, 90, 100, 105, 110, 135 and 145. If Shortest- Seek Time First (SSTF) is being used for scheduling the disk access, the request for cylinder 90 is serviced after servicing ________ number of requests.

Q.7 Which of the following secondary storage techomology does not support random access?

(A) Tape

(B) Magnetic Disk

(C) Flash Disk

(D) None of the above

Q.8 Which of the following is/are represented by an i-node ?

(A) Files (B) Directories

(C) Pipes (D) None

Q.9 Which of the following are true for disk fragmentation?

(A) On increasing block size the maximum size of the file that can be stored increases.

(B) Contiguous allocation strategy suffers from external fragmentation.

(C) Linked list allocation strategy suffers from external fragmentation.

(D) On increasing block size internal fragmentation increases.

❑❑❑